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Comparing the predictive values of TIMI, HEART, and GRACE for major cardiovascular events in patients admitted to the emergency department with acute chest pain: A systematic review and indirect meta-analysis

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ABSTRACT

Background: The study aimed to compare the predictive values of the Thrombolysis in Myocardial Infarction (TIMI); History, ECG, Age, Risk factors and Troponin (HEART), and Global Registry in Acute Coronary Events (GRACE) scoring systems on major cardiovascular outcomes in acute chest pain (ACP) patients admitted to the emergency department (ED).

Methods: We systematically searched the electronic databases of PubMed, EmBase, and the Cochrane library from their inception to June 2020. The compared parameters included sensitivity, specificity, positive and negative likelihood ratios (PLR and NLR), diagnostic odds ratio (DOR), and the area under the receiver operating characteristic curves (AUC).

Results: The pooled sensitivity and specificity for TIMI, HEART, and GRACE were 0.95 and 0.36, 0.96 and 0.50, and 0.78 and 0.56, respectively. The pooled PLR and NLR for TIMI, HEART, and GRACE were 1.49 and 0.13, 1.94 and 0.08, and 1.77 and 0.40, respectively. The pooled DOR for TIMI, HEART, and GRACE were 9.18, 17.92, and 4.00, respectively. The AUC for TIMI, HEART, and GRACE were 0.80, 0.80, and 0.70, respectively. Finally, the results of indirect comparison suggested the predictive values of TIMI and HEART were superior than those of GRACE for predicting major cardiovascular events (MACEs), while there were no significant differences between TIMI and HEART for predicting MACEs.

Conclusions: TIMI and HEART were superior to GRACE for predicting MACE risk in ACP patients admitted to the ED.

Key words: TIMI; HEART; GRACE; cardiovascular events; meta-analysis

Strengths and limitations of this study

(1) The analysis based on prospective studies and used consistent cutoff values

(2) The pooled results were stable owing to large sample size

(3) The indirect comparisons among TIMI, HEART, and GRACE scoring systems were

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provided

(4) The analysis based on crude data, and the predictive values could be affected by covariates

(5) Substantial heterogeneity was not fully explained

INTRODUCTION

Acute chest pain (ACP) is a common presenting symptom accounting for a significant proportion of attendance and burden in the emergency department (ED) [1]. ACP patients require effective risk stratification, which can initiate treatment in high-risk patients to achieve better prognosis. The early identification of cardiovascular disease (CVD) in ACP patients is important as CVD is the most common cause of chest pain [2]. Nevertheless, patients diagnosed with CVD account for only a small proportion of ACP patients with electrocardiograms (ECGs) on presentation [3,4]. Moreover, patients diagnosed with acute coronary syndrome (ACS) should not be discharged, whereas non-ACS patients are unnecessarily admitted to hospitals due to the heavy burden on resource constraints [5]. Therefore, accurate risk stratification for ACP patients is essential to improve hospital efficacy by administering acute interventions to high-risk patients, avoiding unnecessary tests, and minimizing admissions for low-risk patients. Currently, the Thrombolysis in Myocardial Infarction (TIMI); History, ECG, Age, Risk factors and Troponin (HEART); and Global Registry in Acute Coronary Events (GRACE) scores are widely used for the risk stratification of ACP patients; however, the predictive values using these methods on major cardiovascular events (MACEs) have not been elucidated. The TIMI scoring system consists of items including age >65.0 years, coronary artery stenosis >50.0%, the use of aspirin within 7 days, incidence of chest pain >1 spell within 24 hours, ECG changes, and elevated serum cardiac biomarkers [6-10]. The HEART score is a simple tool based on clinical experience and medical literature, which includes the following components: history, ECG, age, risk factors, and troponin levels [11]. Moreover, the GRACE score, as an objective prediction model, has already been validated for the risk stratification of ACP

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patients and consists of age, vital signs, kidney function, ECG, and troponin levels [12,13]. However, the predictive values of risk stratification measured by the TIMI, HEART, and GRACE scoring systems on MACEs remain both limited and inconclusive. Therefore, the current study was conducted based on prospective cohort studies to provide comprehensive results regarding the risk stratification assessed by the TIMI, HEART, and GRACE scoring systems on MACEs in ACP patients admitted to the ED. Furthermore, the predictive values of risk stratification assessed by the TIMI, HEART, and GRACE scoring systems on MACEs were compared through an indirect analytic approach.

RESULTS

Literature Search

The details regarding the literature search and study selection of eligible studies are presented in Figure 1. A total of 2,794 articles were identified through the electronic search from PubMed, EmBase, and the Cochrane library, and 1,981 were excluded due to duplicate topics. Moreover, the remaining 813 studies were identified through title and abstract review, and 732 were excluded due to irrelevant topics. A total of 81 full texts were retrieved for further evaluation, and 48 studies were excluded due to the following reasons: used other cutoff values (n=21), retrospective study design (n=14), and insufficient data (n=13). An additional 135 potential studies identified from the reference lists of retrieved studies were excluded as they were already included in the results of the electronic search. Subsequently, 33 prospective cohort studies that recruited 40,262 ACP patients were selected for final quantitative meta-analysis [13-45].

Study Characteristics

The baseline characteristics of included studies are summarized in Table 1. The retrieved studies were published from 2005 to 2020, and 255-4,333 ACP patients were included in each study. Nine studies were conducted in Eastern countries, and the remaining 24 studies were conducted in Western countries. The mean age of enrolled

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patients ranged from 48.0 to 69.0 years, and the percentage of males ranged from 40.0 to 68.8%. Risk stratification by the TIMI score was available in 25 studies, 16 studies used the HEART score, and 16 studies employed GRACE. The study quality of the included studies was assessed by QUADAS-2 and presented in Table 2.

TIMI

The predictive value of risk stratification by the TIMI score on MACEs in ACP patients was available in 25 studies. The pooled sensitivity and specificity of the TIMI score for predicting MACEs were 0.95 (95% CI: 0.91-0.98; I^2 = 98.10%) and 0.36 (95% CI: 0.24-0.50; I^2 = 99.64%), respectively (Supplemental 1). Moreover, the pooled PLR and NLR of the TIMI score for predicting MACEs were 1.49 (95% CI: 1.25-1.79; I^2 = 99.21%) and 0.13 (95% CI: 0.07-0.21; I^2 = 95.38%), respectively (Supplemental 2). The pooled DOR of the TIMI score for predicting MACEs was 9.18 (95% CI: 6.22-13.55; P<0.001) with significant heterogeneity across the included studies (I^2 = 87.6%; P<0.001) (Supplemental 3). The AUC of the TIMI score for predicting MACEs was 0.80 (95% CI: 0.76-0.83; Figure 2). No significant publication bias for the TIMI score was detected (P=0.17; Supplemental 4).

HEART

The predictive value of risk stratification by the HEART score on MACEs in ACP patients was available in 16 studies. The pooled sensitivity and specificity of the HEART score for predicting MACEs were 0.96 (95% CI: 0.91-0.98; I^2 = 94.87%) and 0.50 (95% CI: 0.41-0.60; I^2 = 98.84%), respectively (Supplemental 1). The pooled PLR and NLR of the HEART score for predicting MACEs were 1.94 (95% CI: 1.61-2.35; I^2 = 98.01%) and 0.08 (95% CI: 0.03-0.17; I^2 = 94.65%), respectively (Supplemental 2). The pooled DOR for the HEART score was 17.92 (95% CI: 9.40-34.18; P<0.001) with significant heterogeneity across the included studies (I^2 = 88.9%; P<0.001). The AUC of the HEART score for predicting MACEs was 0.80 (95% CI: 0.77-0.84; Figure 3). There was no significant publication bias for the HEART score (P=0.98; Supplemental 4).

GRACE

The predictive value of risk stratification by the GRACE score on MACEs in ACP

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patients was available in 16 studies. The pooled sensitivity and specificity of the GRACE score for predicting MACEs were 0.78 (95% CI: 0.64-0.87; I^2 = 96.78%) and 0.56 (95% CI: 0.46-0.66; I^2 = 99.39%), respectively (Supplemental 1). The pooled PLR and NLR of the GRACE score for predicting MACEs were 1.77 (95% CI: 1.51-2.08; I^2 = 96.34%) and 0.12 (95% CI: 0.06-0.26; I^2 = 94.07%), respectively (Supplemental 2). The DOR of the GRACE score for predicting MACEs was 4.00 (95% CI: 2.78-5.74; P<0.001) with significant heterogeneity across the included studies (I^2 = 88.7%; P < 0.001). The AUC of the GRACE score for predicting MACEs was 0.70 (95% CI: 0.66-0.74; Figure 4). No significant publication bias for the GRACE score was observed (P=0.36; Supplemental 4).

Indirect Comparisons

Indirectindirect comparisons of the diagnostic parameters (sensitivity, specificity, PLR, NLR, DOR, and AUC) among the TIMI, HEART, and GRACE scoring systems for predicting MACEs are summarized in Table 3. First, the sensitivity of TIMI (ratio: 1.22; 95% CI: 1.04-1.43) and HEART (ratio: 1.23; 95% CI: 1.05-1.44) was significantly higher than that of GRACE for predicting MACEs. Second, the specificity of TIMI was significantly lower than that of GRACE for predicting MACEs (ratio: 0.64; 95% CI: 0.43-0.97). Third, the PLR of TIMI was lower than that of HEART for predicting MACEs (ratio: 0.77; 95% CI: 0.59-1.00). Fourth, TIMI (ratio: 0.32; 95% CI: 0.17-0.64) and HEART (ratio: 0.20; 95% CI: 0.08-0.52) were associated with a lower NLR than GRACE for predicting MACEs. Fifth, the DOR of TIMI (ratio: 2.29; 95% CI: 1.35-3.91) and HEART (ratio: 4.48; 95% CI: 2.14-9.39) were significantly higher than that of GRACE for predicting MACEs. Finally, the AUC of TIMI (ratio: 1.14; 95% CI: 1.06-1.23) and HEART (ratio: 1.14; 95% CI: 1.06-1.23) were significantly higher than that of GRACE for predicting MACEs.

Sensitivity and subgroup Analysis

The results of sensitivity analyses found that the predictive values of TIMI, HEART, and GRACE for predicting MACEs were stable and unaltered by excluding any particular study (data not shown). Moreover, subgroup analysis for the predictive parameters were summarized in Table 4. The predictive value of TIMI was superior to

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that of GRACE in most subsets, while there was no significant difference for predicting MACEs if the mean age of patients was <60.0 years or the study was of high quality. Additionally, the predictive value of HEART was higher than that of GRACE in all subsets. Finally, there were no significant differences for the predictive value between TIMI and HEART.

Discussion

This study was the first meta-analysis to conduct indirect comparisons of the predictive values of risk stratification assessed by the TIMI, HEART, and GRACE scores on MACEs in ACP patients. The current study recruited a total of 40,262 ACP patients from 33 prospective cohort studies and across a wide range of patient characteristics. The findings of this study suggest that the TIMI, HEART, and GRACE scoring systems have relatively appropriate predictive values for the risk of MACEs in ACP patients admitted to the ED. Moreover, an indirect analysis indicated that the predictive value of TIMI and HEART was superior to that of GRACE for predicting MACEs, while there were no significant differences between TIMI and HEART for predicting MACEs. Several systematic reviews and meta-analyses have illustrated the predictive values of the TIMI, HEART, and GRACE scoring systems on MACEs in ACP patients [46-48]. A meta-analysis conducted by Hess et al. included 8 prospective studies and found that the TIMI score provided effective risk stratification for predicting MACEs in potential ACS patients, whereas it should not be used as the sole means for determining patient disposition [46]. Van Den Berg et al. conducted a meta-analysis of 2 prospective and 10 retrospective cohort studies and suggested that the HEART score could be used to identify MACEs in patients with a suspected diagnosis of ACS [47]. Roche et al. included 11 studies and found that using 100 as the cutoff value of the GRACE score could predict the discharge of nearly 70% of presentations, while the predictive value for subsequent MACE risk was not obtained [48]. However, the above studies only reported the diagnostic value of a single scoring system for predicting MACEs in ACS patients. Therefore, the current meta-analysis of prospective studies was carried out to

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evaluate the predictive values of the TIMI, HEART, and GRACE scoring systems on the risk of MACEs in ACP patients and systematically compared the predictive values among these three scoring systems.

The predictive value of the TIMI score for MACEs in ACP patients was statistically significant, whereas several studies reported inconsistent results. Sanchis et al. recruited 646 ACP patients who presented with non-ST-segment deviation and normal troponin concentrations. They found that the TIMI score was not associated with the risk of MACEs when 0 was used as the cutoff value [15]. Graham et al. included 352 Chinese ACP patients who were admitted to the ED and found that a low TIMI score could not rule out cardiac causes of chest pain [24]. Holly et al. included 552 ACP patients admitted to an ED observation unit and suggested that the TIMI score was not associated with the risk of MACE at 30 days when 0 was used as the cutoff value [25]. Leung et al. indicated a modified TIMI score of 0 could not rule out 30-day MACEs in ACP patients admitted to the ED [34]. The potential reasons for this could be the TIMI score was designed for risk stratification in patients with non-ST-segment elevation ACS, which is mainly based on appropriate ECG changes or elevations of biomarkers of necrosis. Moreover, the presentation characteristics in ACP patients were not entered into the TIMI score. Finally, the prevalence of MACEs was lower than expected (broad 95% CIs; i.e., no statistically significant difference).

The pooled predictive value of the HEART score for predicting MACE in ACP patients was statistically significant. Nearly all included studies reported a similar conclusion, whereas the study conducted by McCord et al. enrolled 661 ACP patients in the ED from Europe, Australia, and the US and suggested that the HEART score after 4 hours from presentation was associated with marginal predictive values for the risk of MACEs [36]. The potential reason for this may be that this study used a modified HEART score, and the original HEART score only considered the initial cTn value, without taking serial sampling into account, which is associated with a lower prevalence of MACEs [49]. In addition, all included studies reported similar conclusions for the predictive value of the GRACE score on the risk of MACEs. We noted that the predictive value of the GRACE score was inferior compared with that of the TIMI and

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HEART scores. The potential reason for this could be that the GRACE score was initially developed for ACS patients but not for ACP patients.

Three strengths of this quantitative meta-analysis should be highlighted: (1) the study was based on prospective cohort studies and used relatively uniform cutoff values, which were associated with lower selective and informative biases; (2) the analysis of this study was based on a large sample size, and the findings in our study were more robust than any individual study; and (3) the predictive values of the TIMI, HEART, and GRACE scoring systems on the risk of MACEs in ACP patients were compared through an indirect analytic approach. The limitations of this study should also be acknowledged. First, the analysis of this meta-analysis was based on crude data, and the adjusted results were not available. Second, substantial heterogeneity was detected across the included studies, which was not fully explained by using sensitivity and subgroup analyses. Third, the analysis was conducted on published articles, which causes inevitable publication biases. Finally, the current study was based on indirect comparisons between the predictive values of the TIMI, HEART, and GRACE scoring systems as direct comparisons were not available.

The findings of this study indicated that risk stratification assessed by the TIMI, HEART, and GRACE scores provide relatively appropriate predictive values for MACEs in ACP patients. The results of indirect comparison analysis indicated that TIMI and HEART had relatively relative better predictive values than GRACE on subsequent MACE risk. Further prospective cohort studies should be conducted to directly compare the predictive values of TIMI, HEART, and GRACE on MACEs in ACP patients.

Methods

Data Sources, Search Strategy, and Selection Criteria

This study was conducted and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Statement issued in 2009 [50]. Any prospective cohort study investigating the predictive value of TIMI, HEART, and GRACE on

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MACEs in ACP patients were eligible for inclusion in this study. No restrictions were placed on published language and status. The electronic databases of PubMed, EmBase, and the Cochrane library were systematically searched for studies from their inception up to June 2020, and the search strategy was performed using the following terms with Medical Subjecting Heading and free words: “acute chest pain” and (“TIMI” or “HEART” or “GRACE”) and (“prospective” or “cohort”). The reference lists of retrieved studies were also searched manually to identify any new eligible studies.

Two authors independently carried out the literature search and study selection, and any conflicts were resolved through group discussion until a consensus was reached. A study was included if they met the following inclusion criteria: (1) study design: the study had a prospective design; (2) patients: ACP patients admitted to the ED; (3) risk stratifying tools: TIMI, HEART, or GRACE; (4) outcomes: the study had to report the incidence of MACEs and provided a clear definition of MACEs; (5) data abstracted: true and false positives or negatives, or data could transform into the above information must be reported; and (6) cutoff value: the cutoff value of TIMI and HEART was 0-3, and the cutoff value of GRACE was 55-110. Retrospective studies were excluded due to various confounding factors. Additionally, studies that used other cutoff values were excluded.

Data Collection and Quality Assessment

Two authors independently abstracted data items and assessed the quality of the included studies, and any disagreement was settled by an additional author reviewing the original article. The collected information from retrieved studies including the first authors’ name, publication year, country, sample size, age at baseline, percentage of males, risk stratifying tools, patients status, MACE definition, follow-up duration, and true and false positives or negatives. Study quality was assessed using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2), which based on patient selection, index test, reference standard, risk of bias, and concerns about applicability [51].

Statistical Analysis

The pooled sensitivity, specificity, positive likelihood ratio (PLR), negative likelihood

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ratio (NLR), and the area under the receiver operating characteristic curves (AUC) for the risk stratification by TIMI, HEART, and GRACE on MACEs were calculated using a bivariate generalized linear mixed model [52,53], while the pooled diagnostic odds ratio (DOR) was calculated using a random-effects model [54]. The I^2 and Q statistic were used for assessing the heterogeneity across included studies, and P<0.10 was considered as significant heterogeneity [55,56]. The robustness of pooled results was also assessed by using sensitivity analyses [57]. The risk stratification by TIMI, HEART, and GRACE was assessed using an indirect comparison analytic approach, and the ratio among these scoring systems were calculated [58]. Furthermore, subgroup analyses for the predictive value among TIMI, HEART, and GRACE on subsequent MACE risk was also conducted based on country, mean age, percentage of males, follow-up duration, and study quality. The funnel plots and Deeks' asymmetry tests were used to assess publication bias [59,60]. The inspection level for pooled diagnostic parameters was two-sided, and P<0.05 was considered as statistically significant. All statistical analyses were performed using Stata software (version 10.0; Stata Corporation, College Station, TX, USA).

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Authors' contributions

JK: study design and manuscript drafting

YC: study supervision, data analysis and result interpretation

XW: literature search and data collection

ZW: data curation

FC: manuscript revising and final approval

Data availability:

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Competing interests

The authors declare no competing interests.

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Figure legends:

Figure 1. Flow diagram of the literature search and study selection process.

Figure 2. The summary receiver operating characteristic curve of risk stratification assessed by the TIMI score.

Figure 3. The summary receiver operating characteristic curve of risk stratification assessed by the HEART score.

Figure 4. The summary receiver operating characteristic curve of risk stratification assessed by the GRACE score.

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Table 1. Baseline characteristic of studies included in the systematic review and meta-analysis

Study	Country	Sample size	Mean age (years)	Percentage male (%)	Risk stratifying tools	Patients status	MACE definition	Follow-up	NOS score
Tong 2005 [14]	US	957	60.0	52.0	TIMI	Chest pain and a nondiagnostic ECG	Death and MI	30 days	8
Sanchis 2005 [15]	Spain	646	64.0	65.8	TIMI	Acute chest pain	Death, MI, or urgent revascularization	14 days	7
Pollack Jr 2006 [16]	US	3929	51.6	40.0	TIMI	Chest pain in the ED	Death, acute MI, and revascularization	30 days	8
Pelliccia 2006 [17]	Italy	4333	58.4	68.8	TIMI	Acute chest pain	MI	In-hospital	7
Lyon 2007 [18]	UK	954	60.0	62.0	TIMI, GRACE	Undifferentiated chest pain	MI, cardiac arrest, revascularisation, unstable angina with myocardial damage and death	30 days	8
Ramsay 2007 [13]	UK	347	65.2	62.3	TIMI, GRACE	Suspected cardiac pain	Death,non-fatal MI and emergency revascularization	3.0 months	7
Body 2009 [19]	UK	796	58.9	60.4	TIMI	Chest pain in the ED	Death, acute MI or urgent coronary revascularisation	30 days	7
Campbell 2009 [20]	US	3169	53.6	45.0	TIMI	Chest pain in the ED	Death, MI, or revascularization	30 days	8
Hess 2010 [21]	Canada	1017	59.3	60.6	TIMI	ED patients with chest pain and possible ACS	Acute MI, revascularization, or death	30 days	7
Stracke 2010 [22]	Germany	1014	66.0	55.0	GRACE	Chest pain in the ED	Death	In-hospital	7

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Van der Zee 2011 [23]	The Netherlands	524	57.7	60.5	GRACE	Chest pain in the ED	Death	9.4 years	6
Graham 2013 [24]	China	315	69.0	54.9	TIMI	Chest pain in the ED	Death, MI, troponin positive ACS and PCI	30 days	6
Holly 2013 [25]	US	552	54.1	46.0	TIMI	Chest pain in the ED	MI, revascularization, or death	30 days	7
Backus 2013 [26]	The Netherlands	2388	60.6	57.5	HEART	Chest pain in the ED	Acute MI, PCI, CABG, coronary angiography revealing procedurally correctable stenosis managed conservatively, and death	6 weeks	8
Cullen 2013 [27]	Australia	948	54.0	59.9	TIMI, and GRACE	Chest pain in the ED	Cardiac death, acute MI and unstable angina	30 days	7
Graham 2014 [28]	China	925	68.0	51.7	TIMI	Chest pain in the ED	Death, readmission with MI, ACS not diagnosed at initial ED presentation and coronary revascularization	30 days	7
Visser 2015 [29]	The Netherlands	255	64.0	56.0	HEART	Chest pain in the ED	MI, or PCI, or CABG, or coronary angiography revealing significant stenosis or death	6 weeks	8
Boubaker 2015 [30]	Tunisia	3125	57.7	58.3	TIMI, GRACE	Chest pain in the ED	All cause mortality, acute coronary syndrome, and coronary non-ED planned revascularization	30 days	8
Wang 2016 [31]	China	986	54.0	55.0	TIMI, HEART, GRACE	Chest pain in the ED	Death, MI, and/or the need for revascularization by CABG or PCI	6 months	7
Chen 2016 [32]	China	833	65.1	55.3	TIMI, HEART, GRACE	Chest pain in the ED	Coronary revascularization, ventricular arrhythmia needing intervention and	30 days	7

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							high-degree artioventricular block needing intervention		
Sakamoto 2016 [33]	Singapore	604	60.8	69.2	TIMI, HEART, GRACE	Chest pain in the ED	Death, acute MI, PCI, CABG	30 days	7
Leung 2017 [34]	China	602	66.0	48.8	TIMI, HEART	Chest pain in the ED	Death, cardiac arrest, MI, and cardiogenic shock	30 days	8
Poldervaart 2017 [35]	The Netherlands	1748	62.0	54.0	TIMI, HEART, GRACE	Chest pain in the ED	UA, MI, PCI, CABG, stenosis managed conservatively, and death	3.0 months	8
McCord 2017 [36]	Europe, Australia, and the US	661	58.3	58.2	HEART	Chest pain in the ED	Death or acute MI	30 days	8
Reaney 2018 [37]	UK	1000	62.5	57.6	TIMI, HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, cardiac death, cardiogenic shock and life-threatening arrhythmias requiring emergency intervention.	30 days	8
Greenslade 2018 [38]	Australia	1760	60.4	59.3	TIMI	Chest pain in the ED	Cardiac death, cardiac arrest, cardiogenic shock, acute MI, UA, emergency or urgent revascularisation, high-level atrioventricular block and ventricular arrhythmias	30 days	7
Moumneh 2018 [39]	France	641	53.3	53.4	HEART	Non-traumatic chest pain patients	MI, coronary angioplasty, coronary bypass, and sudden unexplained death	6 weeks	7
Ishak 2018 [40]	The Netherlands	1127	63.8	57.7	HEART	Chest pain in the ED	Acute MI, PCI, CABG or death	30 days	8
Wong 2018 [41]	China	1081	48.0	52.3	TIMI, HEART, GRACE	Undifferentiated chest pain	Acute MI, PCI, CABG, and death	30 days	7

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Al-Zaiti 2019 [42]	USA	750	59.0	58.0	TIMI, HEART, GRACE	Chest pain in the ED	Resuscitated or un-resuscitated sudden cardiac arrest, all-cause death, post-discharge re-infarction requiring cardiac revascularization	30 days	8
Huang 2020 [43]	China	509	59.8	53.4	HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, cardiac death, cardiogenic shock or life-threatening arrhythmias requiring emergency intervention or resulting in mortality	30 days	6
Torralba 2020 [44]	Colombia	519	64.3	56.1	TIMI, HEART, GRACE	Chest pain in the ED	Death from any cause, MI, and surgical or percutaneous myocardial revascularization	30 days	7
Shin 2020 [45]	Korea	1247	62.0	60.8	TIMI, HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, or death from cardiac causes	30 days	7

*ACS: acute coronary syndrome; CABG: coronary artery bypass grafting; ECG: electrocardiogram; ED: emergency department; MI: myocardial infarction; PCI: percutaneous coronary intervention; UA: unstable angina

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Table 2. QUADAS-2 Scoring of Included Studies

AUTHORS	PATIENT SELECTION		INDEX TEST		REFERENCE STANDARD		RISK OF BIAS		APPLICA-BILITY
	Consecutive patient enrollment?	Appropriate exclusion criteria?	Inter- or intra-rater reliability?	Training prior to study?	Always same reference standard?	Selection bias?	Bias from index test?	Bias from reference test?	Risk of inappropriate reference test?
Tong 2005 [14]	Low	Low	Inter	Inter	Low	Low	Low	Low	Low
Sanchis 2005 [15]	Low	Low	High	Unclear	Low	Low	Low	Low	Low
Pollack Jr 2006 [16]	Low	Low	Inter	Inter	Low	Low	Low	Low	Low
Pelliccia 2006 [17]	Low	Low	Unclear	Inter	Low	Low	Low	Low	Low
Lyon 2007 [18]	Low	Low	Unclear	Unclear	Low	Low	Low	Low	Low
Ramsay 2007 [13]	Low	Low	Unclear	Unclear	Low	Inter	Low	Low	Low
Body 2009 [19]	Low	Low	Unclear	Unclear	Low	Inter	Low	Low	Low
Campbell 2009 [20]	Low	Low	Unclear	Unclear	Low	Low	Low	Low	Low
Hess 2010 [21]	Low	Low	Unclear	Unclear	Low	Inter	Low	Low	Low
Stracke 2010 [22]	Low	Low	Unclear	Unclear	Low	Inter	Low	Low	Low
Van der Zee 2011 [23]	Low	Low	High	Unclear	Low	Inter	Low	Low	Low
Graham 2013 [24]	Low	Low	High	Unclear	Low	Inter	Low	Low	Low
Holly 2013 [25]	Low	Low	Inter	Unclear	Low	Inter	Low	Low	Low
Backus 2013 [26]	Low	Low	Unclear	Unclear	Low	Low	Low	Low	Low
Cullen 2013 [27]	Low	Low	Inter	Unclear	Low	Low	Low	Low	Low
Graham 2014 [28]	Low	Low	Inter	Unclear	Low	Low	Low	Low	Low
Visser 2015 [29]	Low	Low	Inter	Unclear	Low	Inter	Low	Low	Low

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Boubaker 2015 [30]	Low	Low	Unclear	Unclear	Low	Low	Low	Low	Low
Wang 2016 [31]	Low	Low	Inter	Unclear	Low	Low	Low	Low	Low
Chen 2016 [32]	Low	Low	Inter	Unclear	Low	Inter	Low	Low	Low
Sakamoto 2016 [33]	Low	Low	Inter	Unclear	Low	Inter	Low	Low	Low
Leung 2017 [34]	Low	Low	Unclear	Unclear	Low	Low	Low	Low	Low
Heldervaart 2017 [35]	Low	Low	Unclear	Unclear	Low	Low	Low	Low	Low
McCord 2017 [36]	Low	Low	Unclear	Unclear	Low	Low	Low	Low	Low
Reaney 2018 [37]	Low	Low	Unclear	Unclear	Low	Low	Low	Low	Low
Greenslade 2018 [38]	Low	Low	Inter	Unclear	Low	Inter	Low	Low	Low
Moumneh 2018 [39]	Low	Low	Inter	Unclear	Low	Inter	Low	Low	Low
Ishak 2018 [40]	Low	Low	Unclear	Unclear	Low	Low	Low	Low	Low
Wong 2018 [41]	Low	Low	Inter	Unclear	Low	Inter	Low	Low	Low
Al-Zaiti 2019 [42]	Low	Low	Unclear	Unclear	Low	Low	Low	Low	Low
Huang 2020 [43]	Low	Low	Inter	Unclear	Low	Inter	Low	Low	Low
Torralba 2020 [44]	Low	Low	Inter	Unclear	Low	Low	Low	Low	Low
Shin 2020 [45]	Low	Low	Inter	Unclear	Low	Low	Low	Low	Low

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Table 3. Systematically comparisons the sensitivity, specificity, positive and negative likelihood ratio, diagnostic odds ratio, and the area under the receiver operating characteristic curve of risk stratifying measured by TIMI, HEART, and GRACE.

Parameters	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity and 95%CI	0.95 (0.91-0.98)	0.96 (0.91-0.98)	0.78 (0.64-0.87)	0.99 (0.94-1.04)	1.22 (1.04-1.43)	1.23 (1.05-1.44)
Specificity and 95%CI	0.36 (0.24-0.50)	0.50 (0.41-0.60)	0.56 (0.46-0.66)	0.72 (0.48-1.09)	0.64 (0.43-0.97)	0.89 (0.69-1.16)
PLR and 95%CI	1.49 (1.25-1.79)	1.94 (1.61-2.35)	1.77 (1.51-2.08)	0.77 (0.59-1.00)	0.84 (0.66-1.07)	1.10 (0.86-1.40)
NLR and 95%CI	0.13 (0.07-0.21)	0.08 (0.03-0.17)	0.40 (0.27-0.59)	0.46 (0.15-1.39)	0.32 (0.17-0.64)	0.20 (0.08-0.52)
DOR and 95%CI	9.18 (6.22-13.55)	17.92 (9.40-34.18)	4.00 (2.78-5.74)	0.51 (0.24-1.09)	2.29 (1.35-3.91)	4.48 (2.14-9.39)
AUC and 95%CI	0.80 (0.76-0.83)	0.80 (0.77-0.84)	0.70 (0.66-0.74)	1.00 (0.94-1.06)	1.14 (1.06-1.23)	1.14 (1.06-1.23)

PLR: positive likelihood ratio; NLR: negative likelihood ratio; DOR: diagnostic odds ratio; AUC: the area under the receiver operating characteristic curve; CI: confidence interval

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Table 4. Subgroup analyses

Parameters	Factors	Groups	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity	Country	Western	0.97 (0.93-0.99)	0.97 (0.93-0.99)	0.88 (0.72-0.95)	1.00 (0.96-1.05)	1.10 (0.96-1.27)	1.10 (0.96-1.27)
		Eastern	0.86 (0.69-0.95)	0.94 (0.79-0.98)	0.60 (0.49-0.70)	0.91 (0.75-1.11)	1.43 (1.13-1.82)	1.57 (1.27-1.93)
	Mean age (years)	≥ 60.0	0.97 (0.92-0.99)	0.97 (0.94-0.99)	0.83 (0.67-0.92)	1.00 (0.96-1.05)	1.17 (0.99-1.38)	1.17 (1.00-1.37)
		< 60.0	0.93 (0.81-0.97)	0.91 (0.70-0.98)	0.68 (0.44-0.85)	1.02 (0.84-1.24)	1.37 (0.97-1.92)	1.34 (0.92-1.94)
	Percentage male (%)	≥ 60.0	0.98 (0.92-0.99)	-	0.73 (0.50-0.88)	-	1.34 (1.01-1.79)	-
		< 60.0	0.92 (0.85-0.96)	0.96 (0.91-0.99)	0.80 (0.62-0.90)	0.96 (0.89-1.03)	1.15 (0.95-1.40)	1.20 (0.99-1.45)
	Follow-up duration	≤ 30.0 days	0.96 (0.91-0.98)	0.97 (0.92-0.99)	0.76 (0.61-0.87)	0.99 (0.94-1.04)	1.26 (1.05-1.51)	1.28 (1.06-1.53)
		> 30.0 days	-	0.93 (0.78-0.98)	-	-	-	-
	Study quality	High	0.94 (0.83-0.98)	0.97 (0.91-0.99)	0.77 (0.50-0.92)	0.97 (0.88-1.06)	1.22 (0.89-1.67)	1.26 (0.93-1.71)
		Moderate	0.96 (0.90-0.98)	0.95 (0.85-0.98)	0.78 (0.61-0.89)	1.01 (0.93-1.10)	1.23 (1.01-1.49)	1.22 (1.00-1.49)
Specificity	Country	Western	0.33 (0.20-0.50)	0.56 (0.47-0.64)	0.50 (0.34-0.66)	0.59 (0.36-0.96)	0.66 (0.37-1.16)	1.12 (0.78-1.61)
		Eastern	0.46 (0.27-0.67)	0.44 (0.27-0.62)	0.64 (0.55-0.73)	1.05 (0.56-1.94)	0.72 (0.45-1.16)	0.69 (0.44-1.07)
	Mean age (years)	≥ 60.0	0.35 (0.19-0.55)	0.45 (0.34-0.56)	0.51 (0.37-0.64)	0.78 (0.43-1.40)	0.69 (0.38-1.25)	0.88 (0.61-1.28)
		< 60.0	0.38 (0.23-0.56)	0.60 (0.44-0.74)	0.63 (0.49-0.76)	0.63 (0.38-1.06)	0.60 (0.37-0.99)	0.95 (0.68-1.34)

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	Percentage male (%)	≥ 60.0	0.21 (0.10-0.37)	-	0.56 (0.45-0.66)	-	0.38 (0.19-0.74)	-
		< 60.0	0.45 (0.30-0.61)	0.52 (0.42-0.63)	0.56 (0.42-0.70)	0.87 (0.58-1.30)	0.80 (0.52-1.24)	0.93 (0.67-1.29)
	Follow-up duration	≤ 30.0 days	0.35 (0.23-0.50)	0.46 (0.35-0.57)	0.56 (0.45-0.67)	0.76 (0.48-1.20)	0.63 (0.40-0.97)	0.82 (0.60-1.13)
		> 30.0 days	-	0.60 (0.43-0.75)	-	-	-	-
	Study quality	High	0.41 (0.24-0.62)	0.47 (0.35-0.60)	0.61 (0.36-0.81)	0.87 (0.51-1.51)	0.67 (0.36-1.25)	0.77 (0.47-1.25)
		Moderate	0.33 (0.19-0.51)	0.54 (0.39-0.68)	0.54 (0.44-0.64)	0.61 (0.35-1.08)	0.61 (0.36-1.04)	1.00 (0.72-1.40)
PLR	Country	Western	1.45 (1.16-1.80)	2.20 (1.79-2.70)	1.75 (1.37-2.23)	0.66 (0.49-0.89)	0.83 (0.60-1.15)	1.26 (0.91-1.73)
		Eastern	1.61 (1.24-2.08)	1.66 (1.30-2.13)	1.67 (1.37-2.03)	0.97 (0.68-1.39)	0.96 (0.70-1.33)	0.99 (0.72-1.36)
	Mean age (years)	≥ 60.0	1.49 (1.13-1.96)	1.76 (1.44-2.16)	1.67 (1.34-2.09)	0.85 (0.60-1.19)	0.89 (0.63-1.27)	1.05 (0.78-1.42)
		< 60.0	1.50 (1.23-1.83)	2.26 (1.61-3.18)	1.86 (1.61-2.15)	0.66 (0.45-0.98)	0.81 (0.63-1.03)	1.22 (0.84-1.02)
	Percentage male (%)	≥ 60.0	1.23 (1.06-1.43)	-	1.67 (1.44-1.93)	-	0.74 (0.60-0.91)	-
		< 60.0	1.68 (1.29-2.19)	2.02 (1.64-2.50)	1.81 (1.45-2.28)	0.83 (0.59-1.17)	0.93 (0.66-1.31)	1.12 (0.82-1.52)
	Follow-up duration	≤ 30.0 days	1.47 (1.22-1.79)	1.79 (1.47-2.19)	1.74 (1.45-2.09)	0.82 (0.62-1.08)	0.84 (0.65-1.10)	1.03 (0.78-1.35)
		> 30.0 days	-	2.34 (1.63-3.35)	-	-	-	-
	Study quality	High	1.61 (1.20-2.17)	1.84 (1.45-2.34)	1.97 (1.32-2.95)	0.88 (0.60-1.28)	0.82 (0.50-1.35)	0.93 (0.58-1.49)
		Moderate	1.44 (1.15-1.79)	2.04 (1.53-2.72)	1.70 (1.46-1.97)	0.71 (0.49-1.01)	0.85 (0.65-1.11)	1.20 (0.87-1.66)

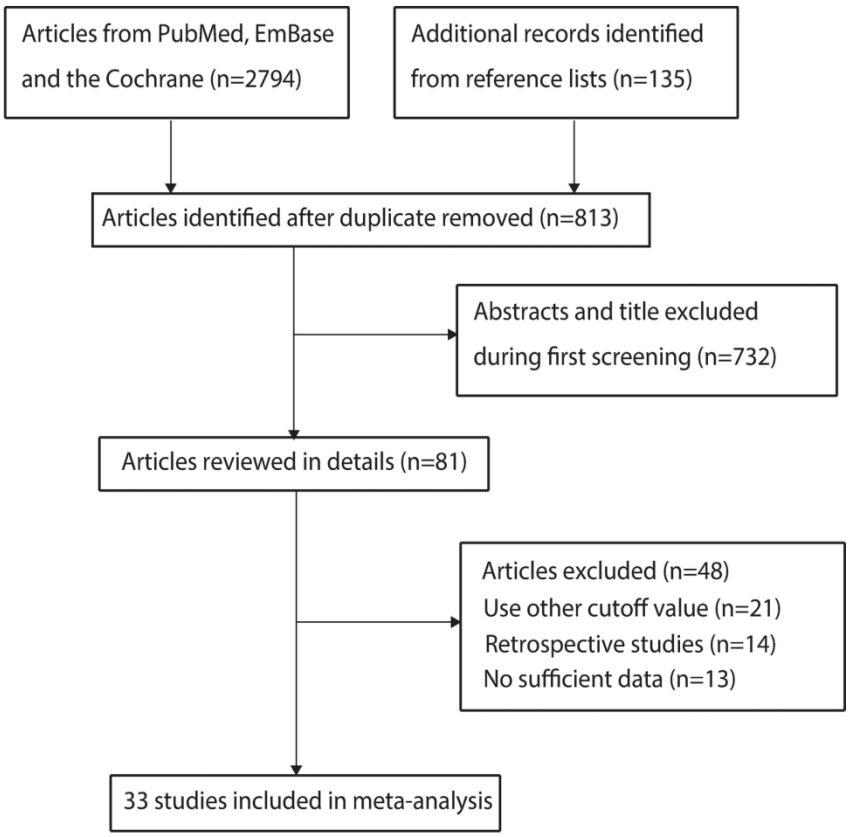
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NLR	Country	Western	0.10 (0.05-0.18)	0.05 (0.02-0.14)	0.25 (0.13-0.49)	2.00 (0.62-6.41)	0.40 (0.16-1.01)	0.20 (0.06-0.65)
		Eastern	0.30 (0.18-0.48)	0.14 (0.06-0.37)	0.63 (0.51-0.77)	2.14 (0.76-6.02)	0.48 (0.28-0.81)	0.22 (0.09-0.56)
	Mean age (years)	≥ 60.0	0.08 (0.03-0.20)	0.06 (0.02-0.14)	0.34 (0.19-0.60)	1.33 (0.34-5.19)	0.24 (0.08-0.71)	0.18 (0.06-0.55)
		< 60.0	0.19 (0.11-0.35)	0.15 (0.04-0.51)	0.50 (0.31-0.81)	1.27 (0.31-5.13)	0.38 (0.18-0.81)	0.30 (0.08-1.17)
	Percentage male (%)	≥ 60.0	0.11 (0.05-0.24)	-	0.48 (0.27-0.87)	-	0.23 (0.09-0.61)	-
		< 60.0	0.17 (0.10-0.28)	0.07 (0.03-0.17)	0.36 (0.22-0.60)	2.43 (0.89-6.66)	0.47 (0.23-0.97)	0.19 (0.07-0.53)
	Follow-up duration	≤ 30.0 days	0.12 (0.07-0.22)	0.07 (0.02-0.18)	0.42 (0.27-0.65)	1.71 (0.50-5.92)	0.29 (0.14-0.59)	0.17 (0.05-0.54)
		> 30.0 days	-	0.12 (0.04-0.35)	-	-	-	-
	Study quality	High	0.13 (0.05-0.33)	0.06 (0.02-0.20)	0.38 (0.20-0.71)	2.17 (0.49-9.60)	0.34 (0.11-1.07)	0.16 (0.04-0.59)
		Moderate	0.12 (0.06-0.24)	0.10 (0.04-0.26)	0.40 (0.24-0.69)	1.20 (0.37-3.85)	0.30 (0.13-0.72)	0.25 (0.09-0.73)
DOR	Country	Western	12.68 (7.19-22.38)	30.41 (10.41-88.82)	6.32 (3.35-11.92)	0.42 (0.12-1.40)	2.01 (0.86-4.70)	4.81 (1.38-16.72)
		Eastern	4.56 (3.67-5.65)	10.05 (5.30-19.06)	2.59 (1.83-3.67)	0.45 (0.23-0.89)	1.76 (1.17-2.65)	3.88 (1.87-8.04)
	Mean age (years)	≥ 60.0	12.86 (6.11-27.07)	23.01 (10.84-48.85)	4.55 (2.50-8.28)	0.56 (0.19-1.61)	2.83 (1.09-7.35)	5.06 (1.93-13.23)
		< 60.0	6.24 (4.46-8.75)	11.52 (3.44-38.55)	3.41 (2.25-5.16)	0.54 (0.15-1.90)	1.83 (1.07-3.12)	3.38 (0.94-12.12)
	Percentage male (%)	≥ 60.0	8.06 (4.08-15.91)	12.32 (1.74-87.48)	3.08 (1.76-5.37)	0.65 (0.08-5.20)	2.62 (1.09-6.31)	4.00 (0.52-30.66)
		< 60.0	9.16 (5.69-14.75)	19.66 (9.31-41.51)	4.51 (2.82-7.22)	0.47 (0.19-1.13)	2.03 (1.04-3.97)	4.36 (1.80-10.54)

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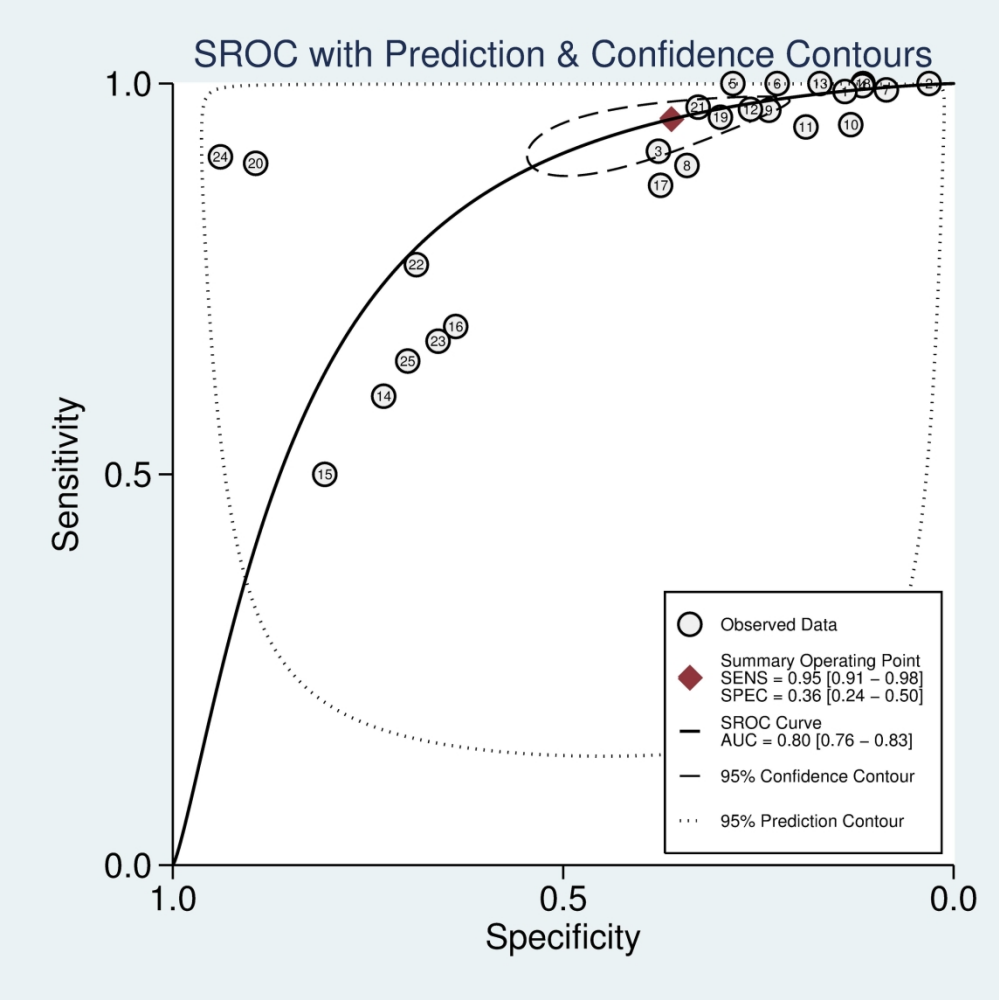
	Follow-up duration	≤ 30.0 days	9.45 (6.19-14.41)	19.21 (8.02-45.99)	3.70 (2.46-5.55)	0.49 (0.19-1.30)	2.55 (1.42-4.59)	5.19 (1.98-13.61)
		> 30.0 days	8.42 (5.14-13.80)	15.90 (8.10-31.21)	5.53 (2.34-13.03)	0.53 (0.23-1.22)	1.52 (0.57-4.10)	2.88 (0.96-8.57)
	Study quality	High	7.96 (4.90-12.91)	21.21 (7.10-63.34)	5.17 (2.51-10.64)	0.38 (0.11-1.24)	1.54 (0.65-3.67)	4.10 (1.11-15.22)
		Moderate	9.44 (5.17-17.23)	16.25 (6.88-38.41)	3.33 (2.22-5.01)	0.58 (0.20-1.66)	2.83 (1.37-5.86)	4.88 (1.88-12.63)
AUC	Country	Western	0.85 (0.82-0.88)	0.82 (0.78-0.85)	0.74 (0.70-0.78)	1.04 (0.98-1.10)	1.15 (1.08-1.23)	1.11 (1.03-1.19)
		Eastern	0.73 (0.69-0.77)	0.74 (0.70-0.78)	0.66 (0.62-0.70)	0.99 (0.91-1.07)	1.11 (1.02-1.20)	1.12 (1.03-1.22)
	Mean age (years)	≥ 60.0	0.85 (0.82-0.88)	0.83 (0.80-0.86)	0.70 (0.66-0.74)	1.02 (0.97-1.08)	1.21 (1.14-1.30)	1.19 (1.11-1.27)
		< 60.0	0.73 (0.69-0.77)	0.79 (0.75-0.82)	0.70 (0.65-0.73)	0.92 (0.86-0.99)	1.04 (0.96-1.13)	1.13 (1.05-1.21)
	Percentage male (%)	≥ 60.0	0.74 (0.70-0.77)	-	0.65 (0.60-0.69)	-	1.14 (1.05-1.24)	-
		< 60.0	0.82 (0.79-0.85)	0.81 (0.78-0.85)	0.72 (0.68-0.76)	1.01 (0.96-1.07)	1.14 (1.07-1.22)	1.13 (1.05-1.21)
	Follow-up duration	≤ 30.0 days	0.81 (0.77-0.84)	0.79 (0.75-0.82)	0.69 (0.65-0.73)	1.03 (0.96-1.09)	1.17 (1.09-1.26)	1.14 (1.06-1.23)
		> 30.0 days	-	0.83 (0.79-0.86)	-	-	-	-
	Study quality	High	0.79 (0.75-0.83)	0.80 (0.76-0.83)	0.74 (0.70-0.78)	0.99 (0.92-1.06)	1.07 (0.99-1.15)	1.08 (1.01-1.16)
		Moderate	0.80 (0.76-0.83)	0.81 (0.77-0.84)	0.67 (0.62-0.71)	0.99 (0.93-1.05)	1.19 (1.10-1.29)	1.21 (1.05-1.14)

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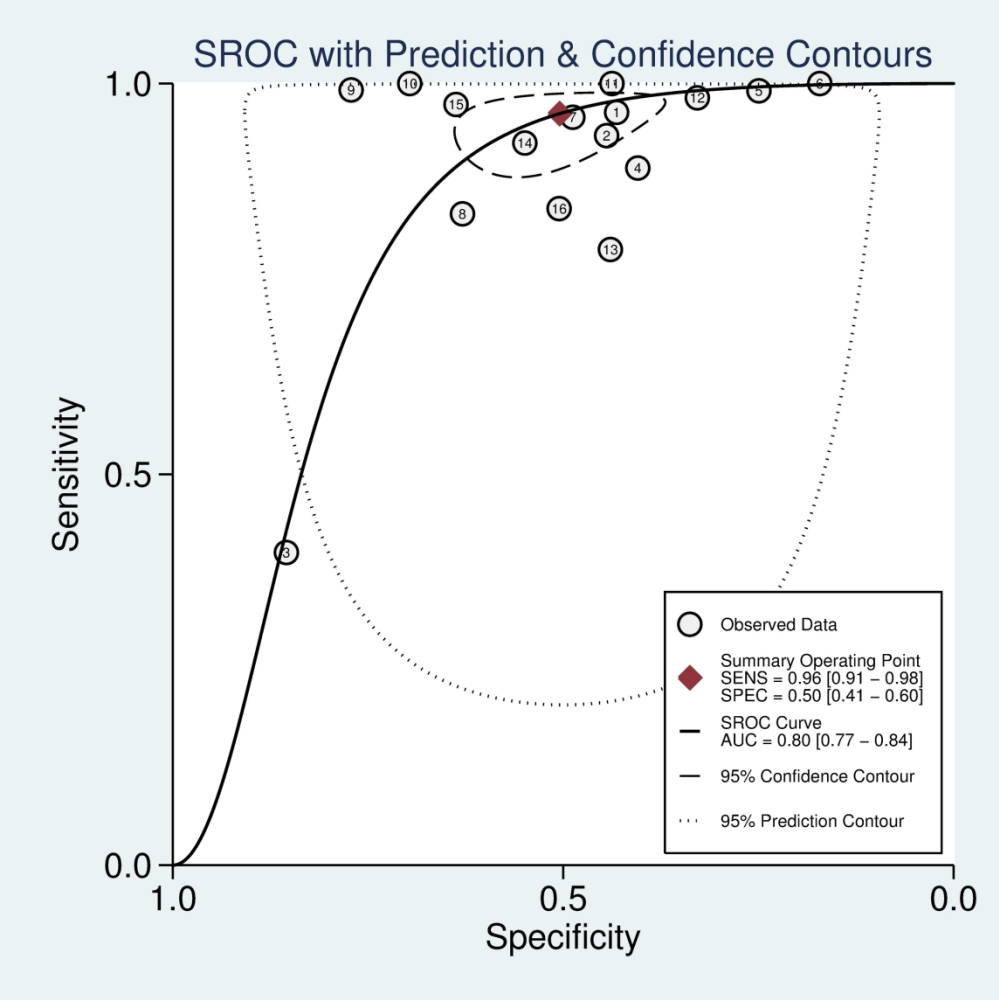
Flow diagram of the literature search and study selection process

154x146mm (300 x 300 DPI)



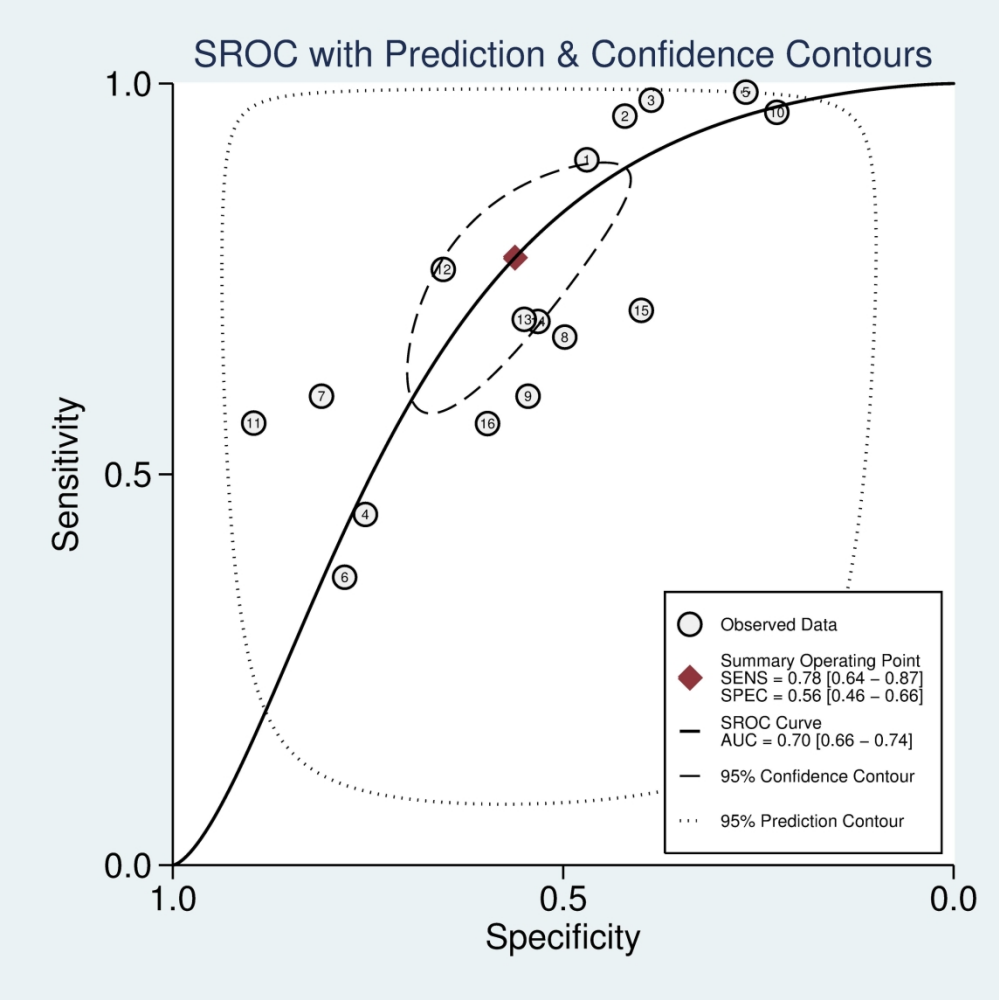
The summary receiver operating characteristic curve of risk stratification assessed by the TIMI score

203x203mm (300 x 300 DPI)



The summary receiver operating characteristic curve of risk stratification assessed by the HEART score

203x203mm (300 x 300 DPI)



The summary receiver operating characteristic curve of risk stratification assessed by the GRACE score

203x203mm (300 x 300 DPI)

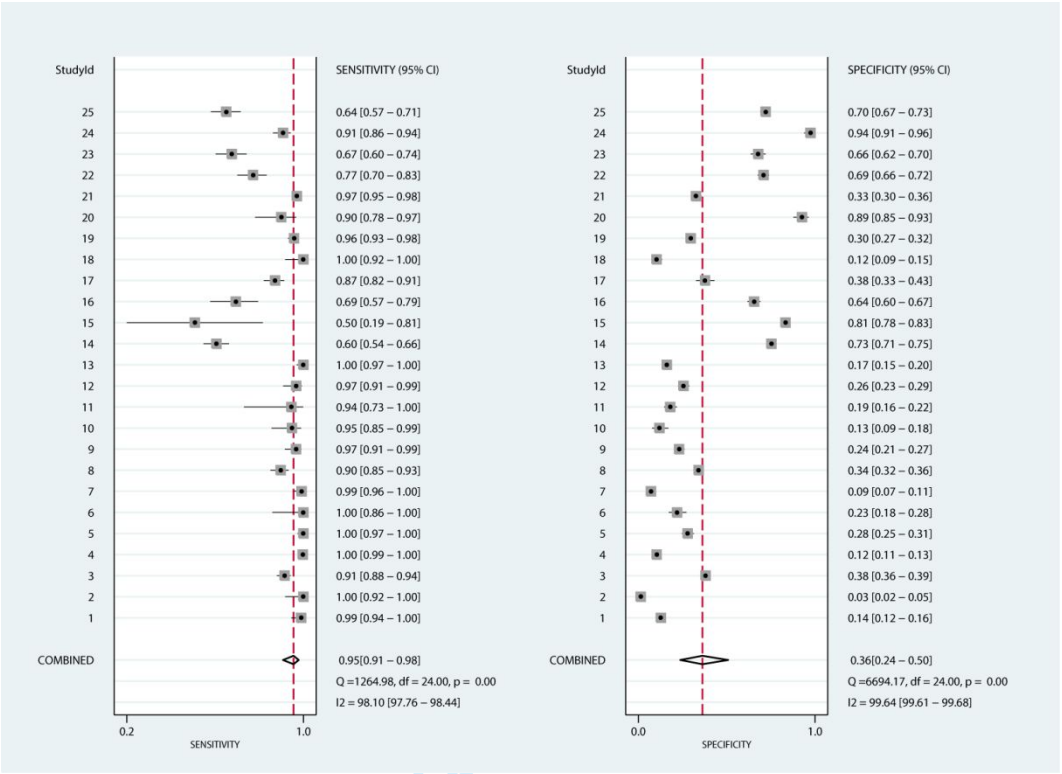


Figure S1. Sensitivity and specificity for TIMI

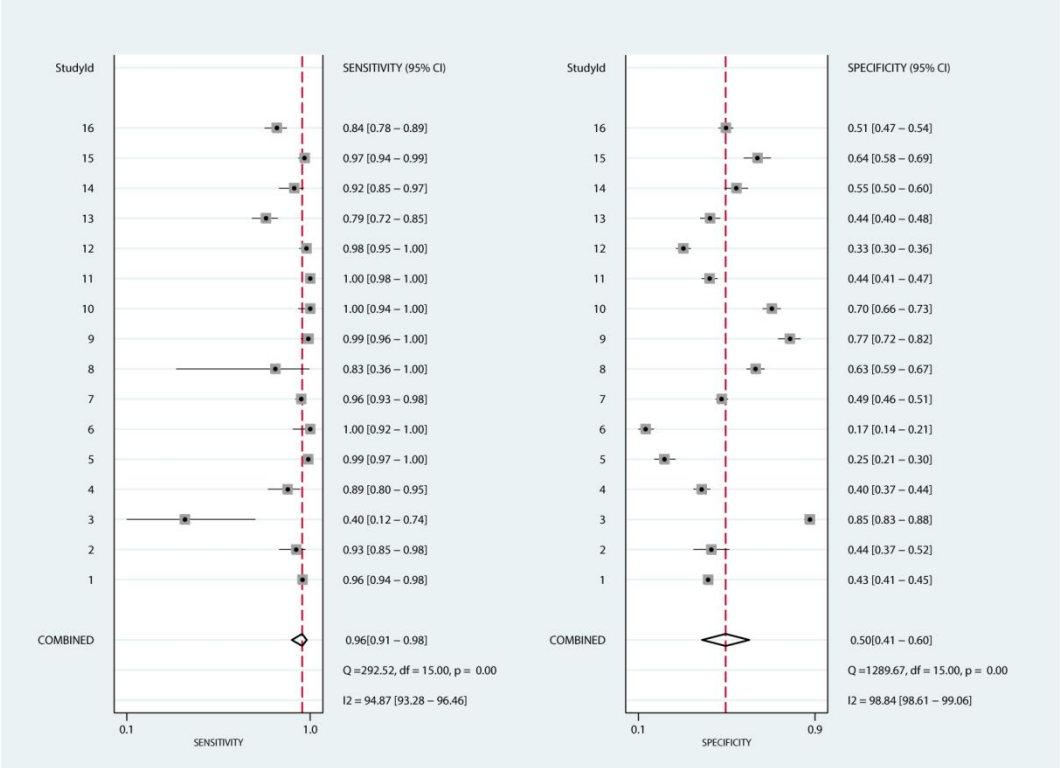


Figure S2. Sensitivity and specificity for HEART

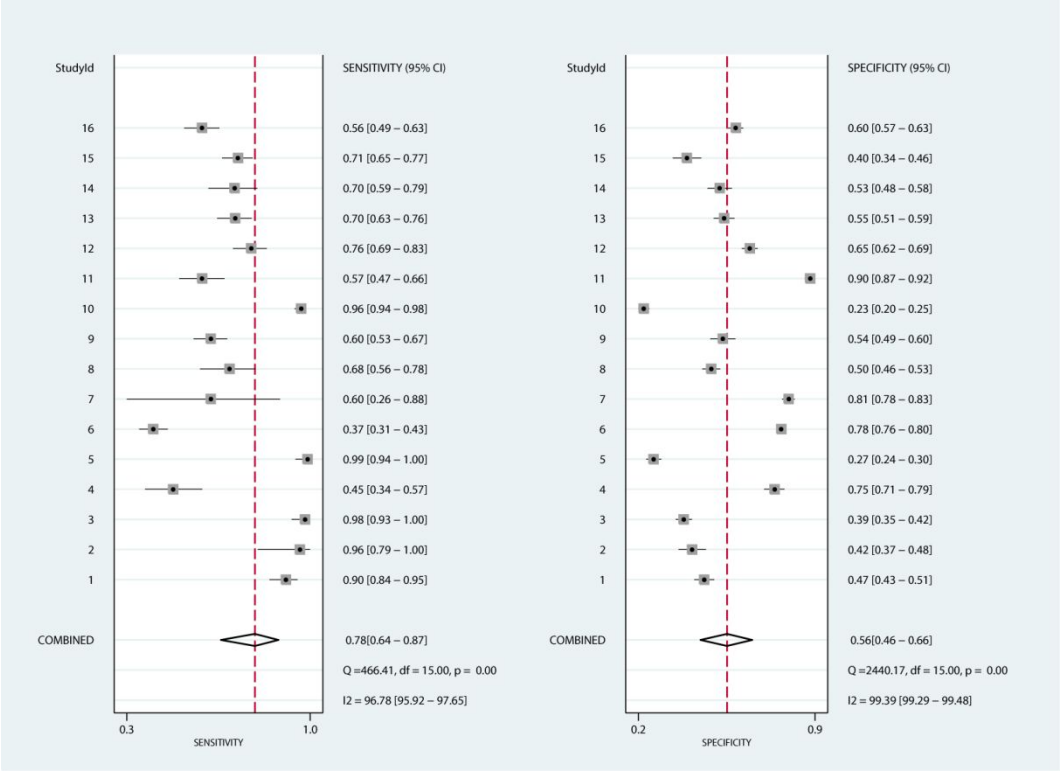


Figure S3. Sensitivity and specificity for GRACE

only

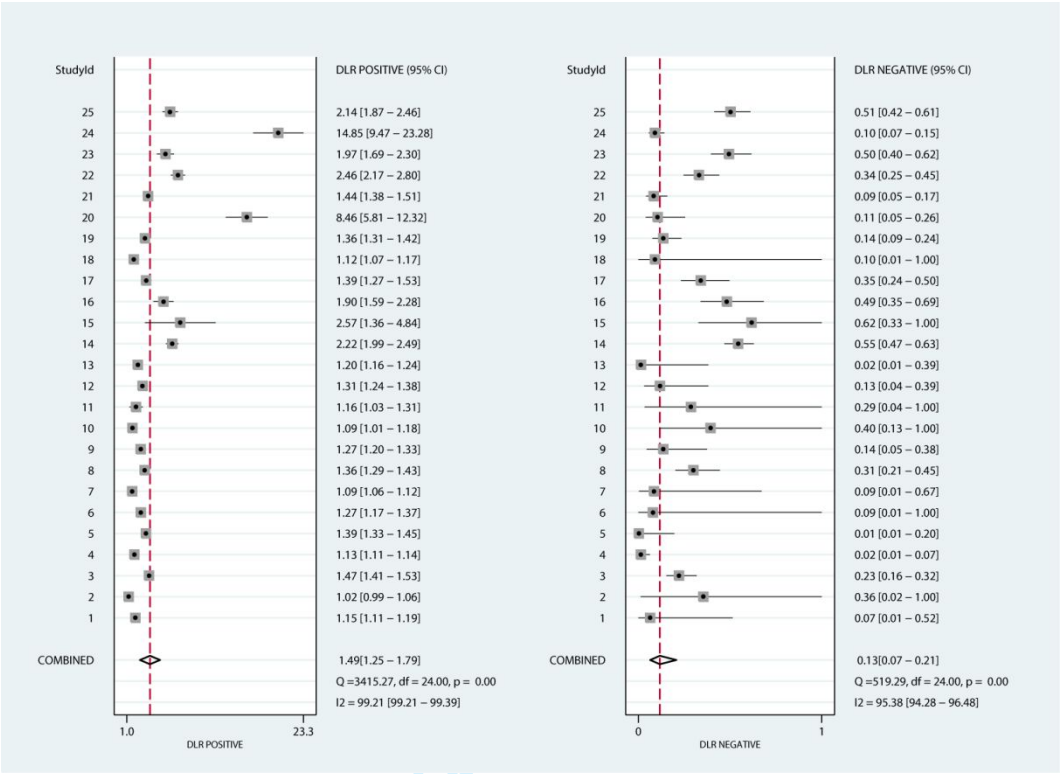


Figure S1. PLR and NLR for TIMI

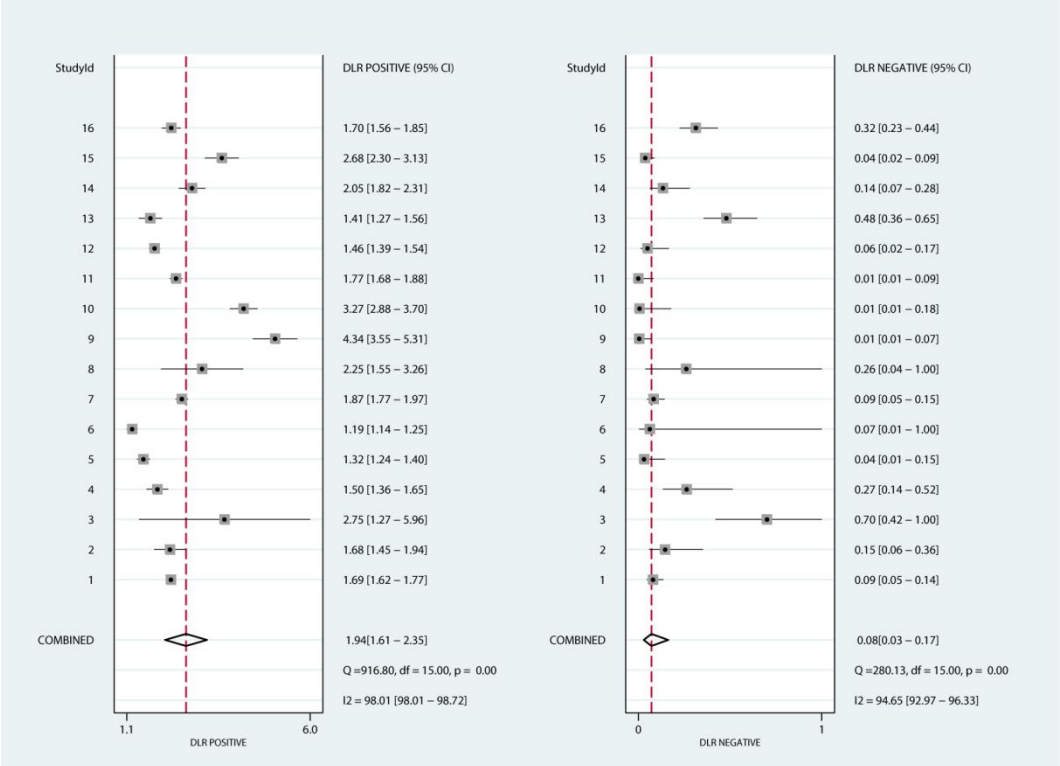


Figure S2. PLR and NLR for HEART

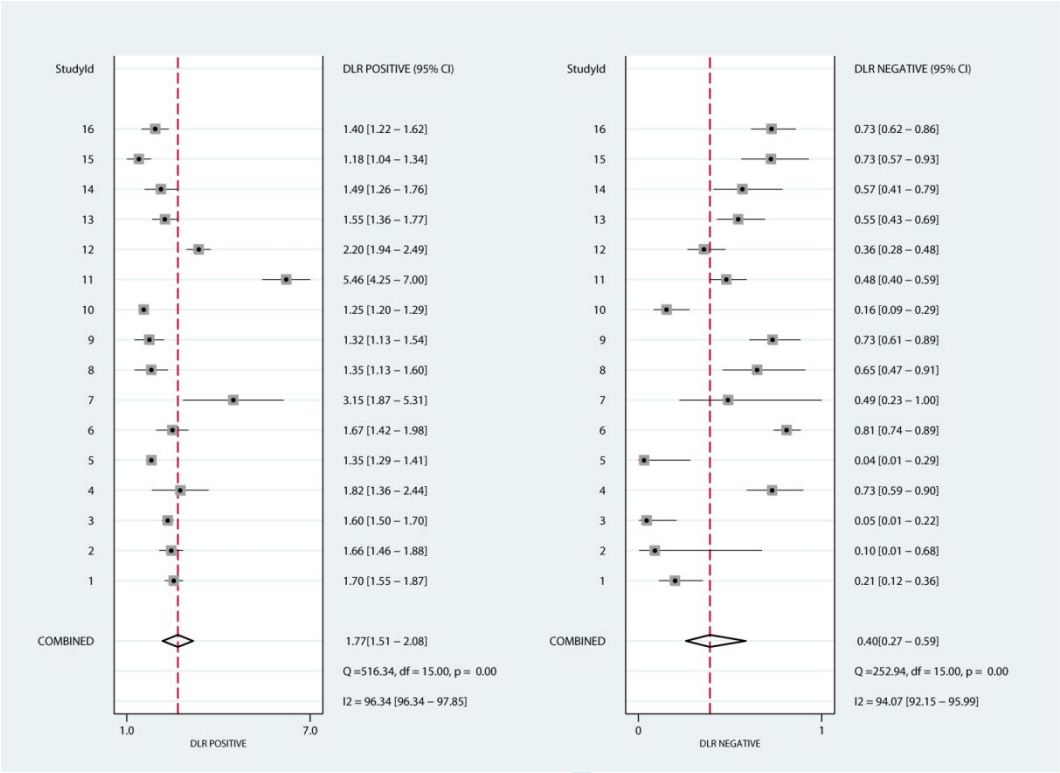


Figure S3. PLR and NLR for GRACE

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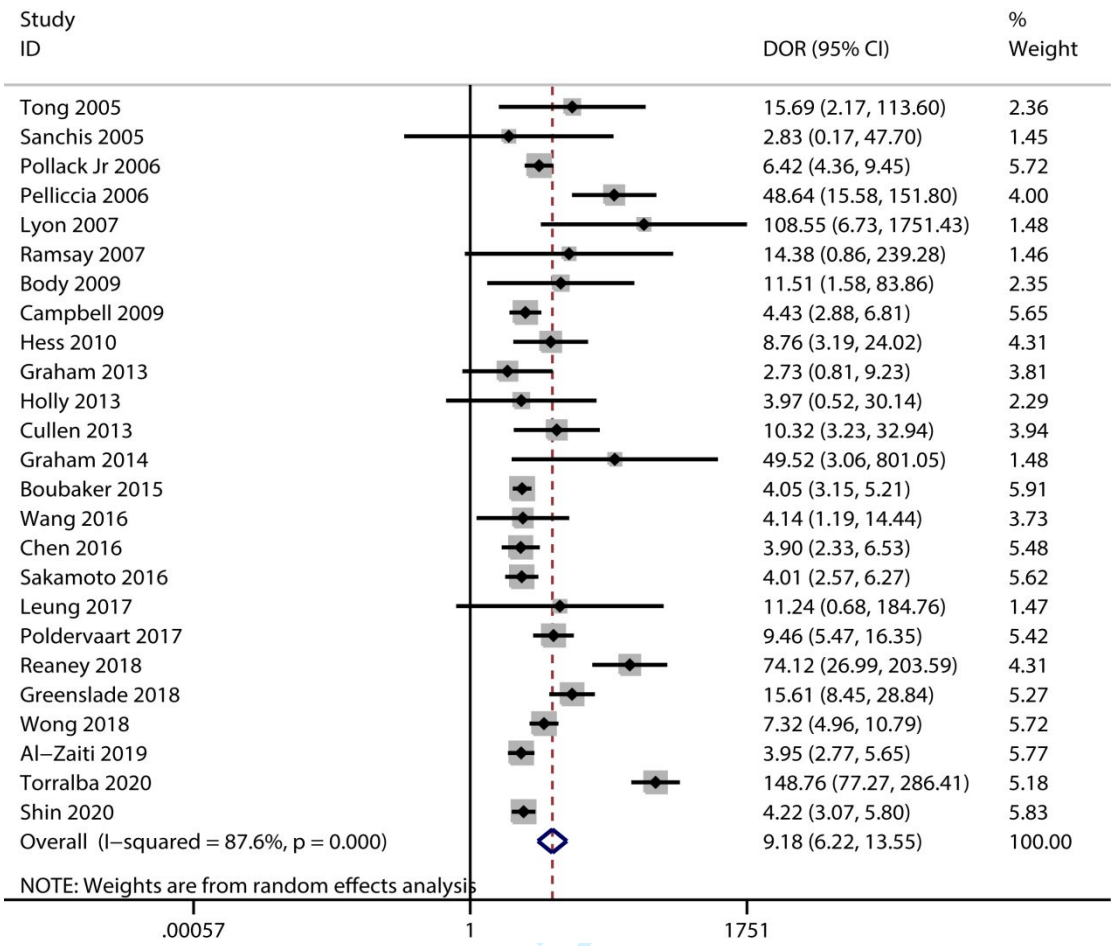


Figure S1. The summary DOR for TIMI

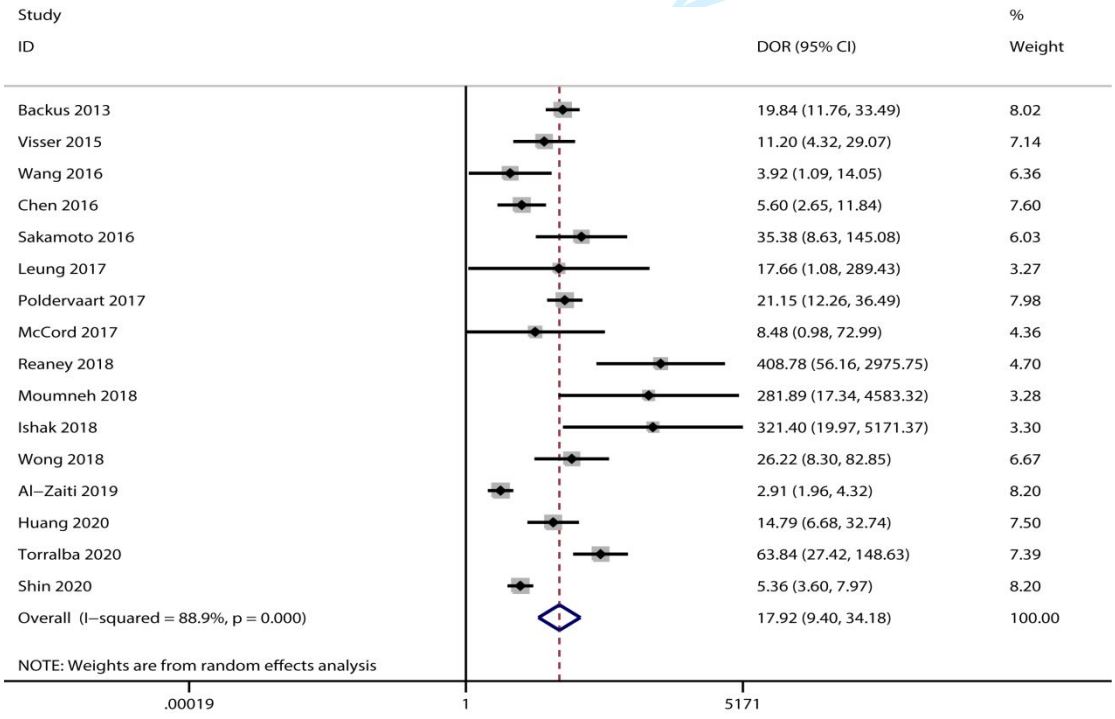


Figure S2. The summary DOR for HEART

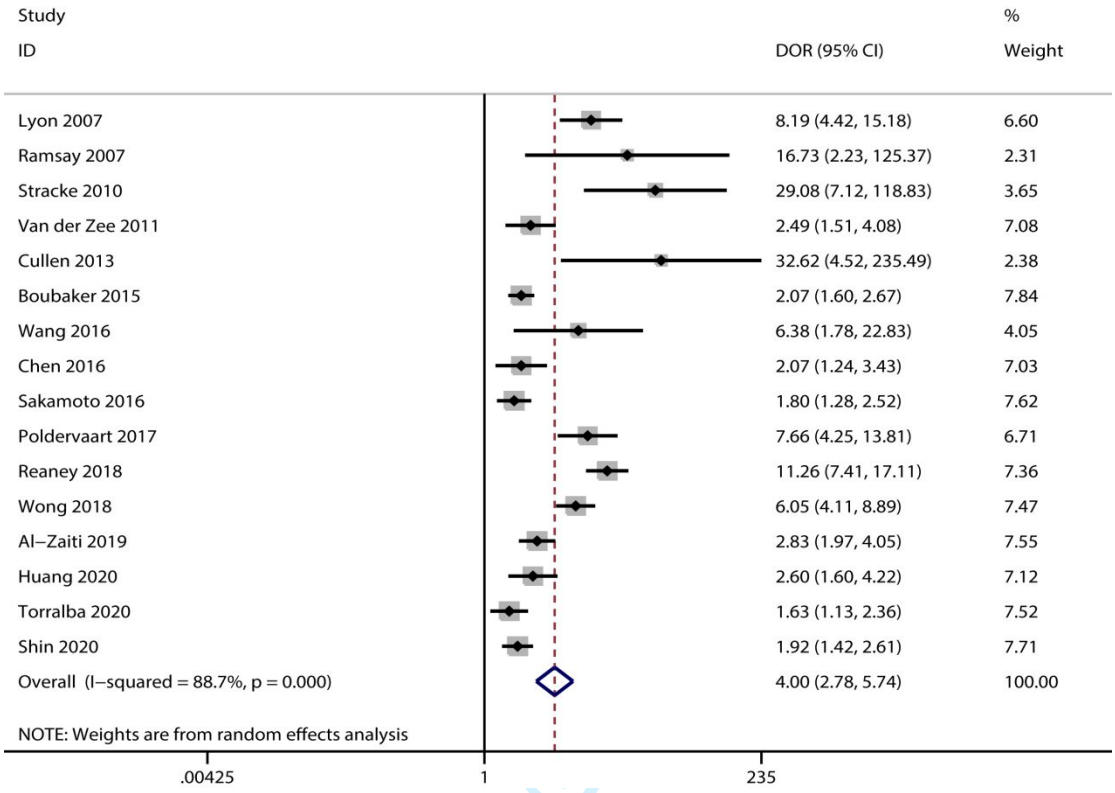


Figure S3. The summary DOR for GRACE

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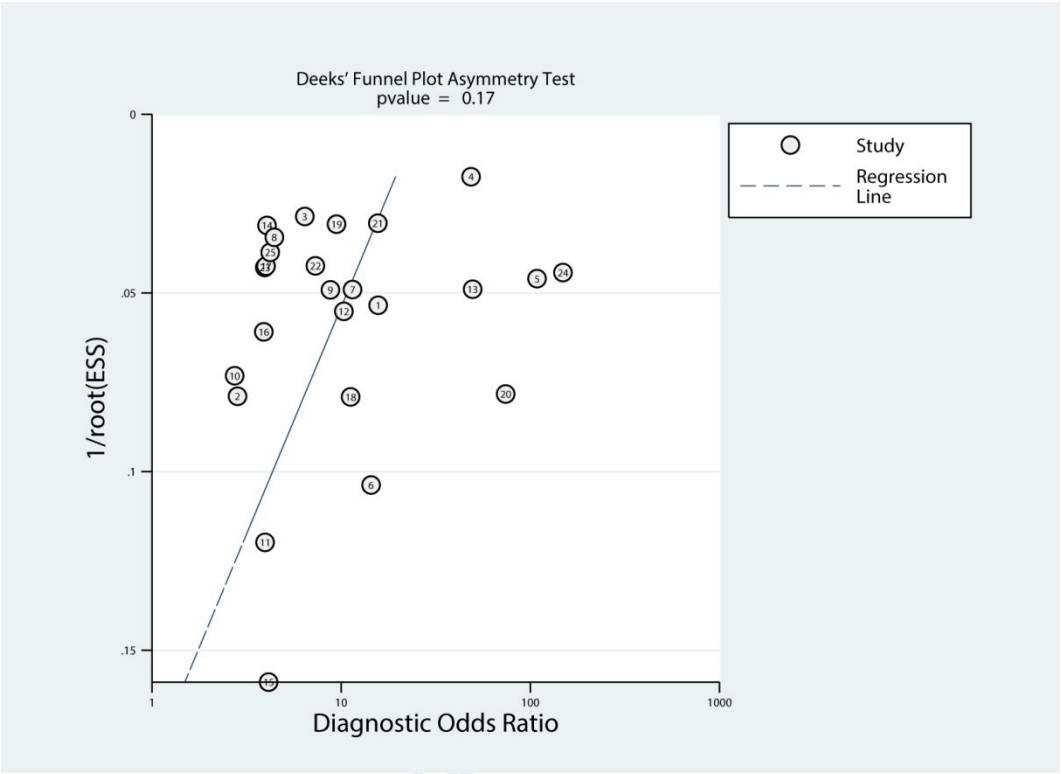


Figure S1. Publication bias for TIMI

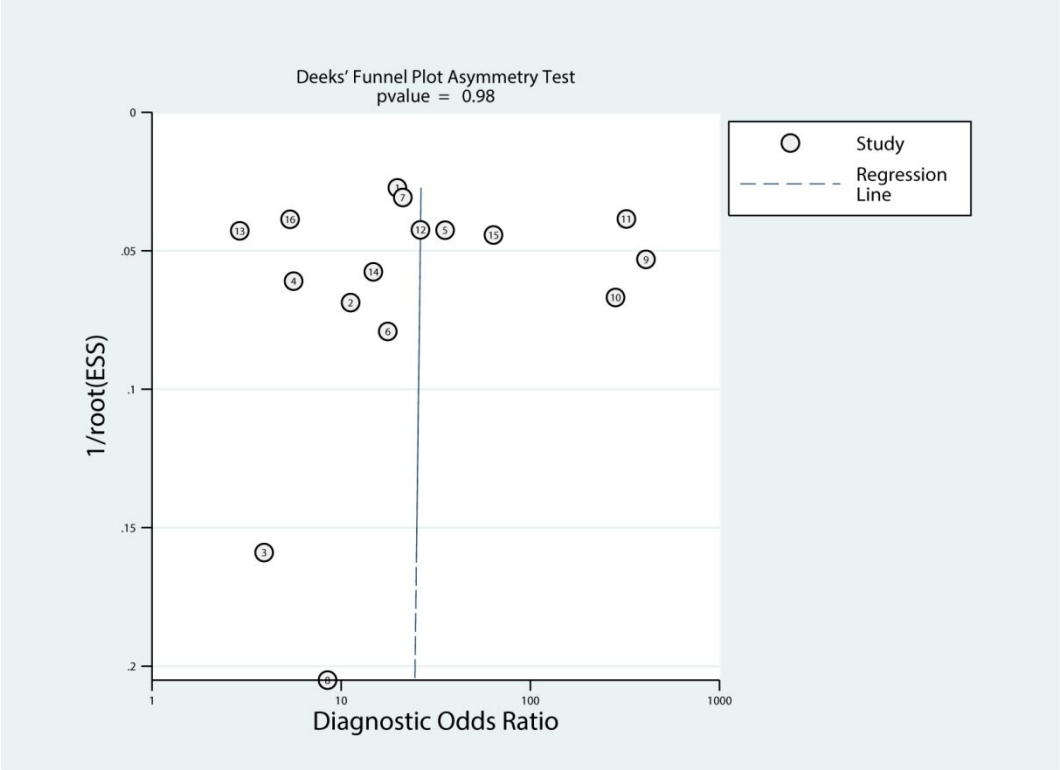


Figure S2. Publication bias for HEART

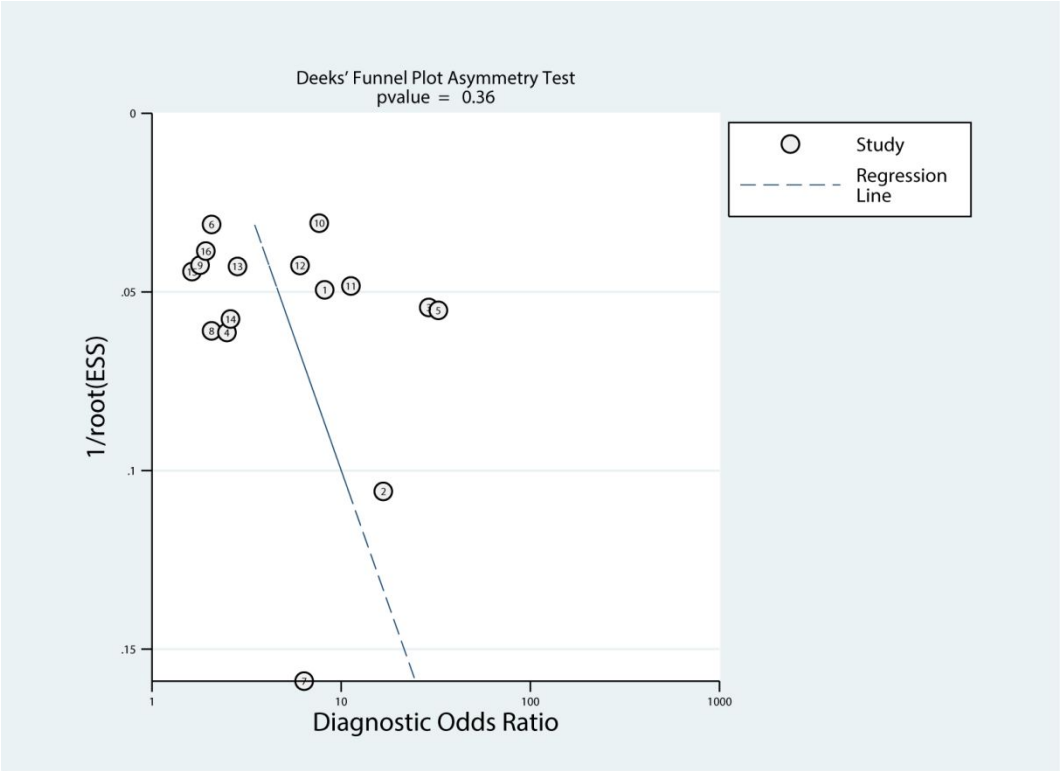


Figure S3. Publication bias for GRACE

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N/A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	10
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	9
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	9-10
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	10
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	10
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	10
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	10-11
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	11

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	11
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	11
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	5-6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	5-6
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	6-7
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7-9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	9
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	11

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

Indirect comparison of TIMI, HEART, and GRACE for predicting major cardiovascular events in patients admitted to the emergency department with acute chest pain: A systematic review and meta-analysis

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Primary Subject Heading:	Cardiovascular medicine
Secondary Subject Heading:	Emergency medicine
Keywords:	Adult cardiology < CARDIOLOGY, ACCIDENT & EMERGENCY MEDICINE, Coronary heart disease < CARDIOLOGY

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Indirect comparison of TIMI, HEART, and GRACE for predicting major cardiovascular events in patients admitted to the emergency department with acute chest pain: A systematic review and meta-analysis

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Word counts: 2,872

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ABSTRACT

Background: The study aimed to compare the predictive values of the Thrombolysis in Myocardial Infarction (TIMI); History, Electrocardiography, Age, Risk factors and Troponin (HEART); and Global Registry in Acute Coronary Events (GRACE) scoring systems for major adverse cardiovascular events (MACE) in acute chest pain (ACP) patients admitted to the emergency department (ED).

Methods: We systematically searched PubMed, Embase, and the Cochrane Library from their inception until June 2020; we compared the following parameters: sensitivity, specificity, positive and negative likelihood ratios (PLR and NLR), diagnostic odds ratio (DOR), and area under receiver operating characteristic curves (AUC).

Results: The pooled sensitivity and specificity for TIMI, HEART, and GRACE were 0.95 and 0.36, 0.96 and 0.50, and 0.78 and 0.56, respectively. The pooled PLR and NLR for TIMI, HEART, and GRACE were 1.49 and 0.13, 1.94 and 0.08, and 1.77 and 0.40, respectively. The pooled DOR for TIMI, HEART, and GRACE was 9.18, 17.92, and 4.00, respectively. The AUC for TIMI, HEART, and GRACE was 0.80, 0.80, and 0.70, respectively. Finally, the results of indirect comparison suggested the superiority of values of TIMI and HEART to those of GRACE for predicting MACEs, while there were no significant differences between TIMI and HEART for predicting MACEs.

Conclusions: TIMI and HEART were superior to GRACE for predicting MACE risk in ACP patients admitted to the ED.

Keywords: TIMI; HEART; GRACE; cardiovascular events; meta-analysis

Strengths and limitations of this study

(1) The analysis is based on prospective studies and used consistent cutoff values

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(2) The pooled results were stable owing to a large sample size

(3) The indirect comparisons among TIMI, HEART, and GRACE scoring systems were provided

(4) The analysis is based on crude data; the predictive values could be affected by covariates

(5) Substantial heterogeneity was not fully explained

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INTRODUCTION

Acute chest pain (ACP) is a common symptom accounting for a significant proportion

of attendance and burden in the emergency department (ED) [1]. ACP patients require effective risk stratification to ensure timely initiation of proper treatment in high-risk cases to achieve better prognoses. The early identification of cardiovascular disease (CVD) in ACP patients is important although CVD accounts for only a small proportion of ACP patients with electrocardiograms (ECGs) on presentation [2-4]. Moreover, patients diagnosed with acute coronary syndrome (ACS) should remain hospitalized, whereas non-ACS patients are unnecessarily admitted to hospitals due to the heavy burden on resource constraints [5]. Therefore, accurate risk stratification for ACP patients is essential to improve hospital efficacy by administering timely interventions to high-risk patients, avoiding unnecessary tests, and minimizing admissions for low-risk patients.

Currently, the Thrombolysis in Myocardial Infarction (TIMI); History, ECG, Age, Risk factors and Troponin (HEART); and Global Registry in Acute Coronary Events (GRACE) scores are widely used for the risk stratification of ACP patients; however, the predictive values using these methods on major cardiovascular events (MACEs) have not been elucidated. The TIMI scoring system, established in 2000 for evaluating patients with unstable angina or non-ST-segment elevation myocardial infarction, consists of factors including age >65.0 years, coronary artery stenosis >50.0%, aspirin use within 7 days, incidence of chest pain >1 episode within 24 hours, ECG changes, and elevated serum cardiac biomarkers [6-10]. The HEART score, developed in 2008, aims to improve the accuracy of diagnosing ACS for patients with undifferentiated chest pain, which includes the following components: history, ECG, age, risk factors, and troponin levels [11]. The GRACE score was developed in 2001 for adults with symptoms of ACS; it comprises the following factors: age, vital signs, kidney function, ECG, and troponin levels [12,13]. However, the predictive values of risk stratification measured by the TIMI, HEART, and GRACE scoring systems on MACEs have not been fully compared. Therefore, this study was conducted based on prospective cohort studies to provide comprehensive results regarding the risk stratification assessed by the TIMI, HEART, and GRACE scoring systems on MACEs in ACP patients admitted to the ED. Furthermore, the predictive values of risk stratification assessed by the TIMI,

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HEART, and GRACE scoring systems on MACEs were compared through an indirect analytic approach.

Methods

Data Sources, Search Strategy, and Selection Criteria

This study was conducted and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Statement issued in 2009 [14]. Any prospective cohort studies investigating the predictive value of TIMI, HEART, and GRACE on MACEs in ACP patients were eligible for inclusion in this study. No restrictions were placed on publishing language and status. The electronic databases of PubMed, Embase, and the Cochrane library were systematically searched for studies from their inception up to June 2020, and the search strategy was performed using the following terms with Medical Subject Heading and free words: ("TIMI" or "HEART" or "GRACE") and "emergency department" and "chest pain" and ("prospective" or "cohort"). The PubMed search strategy details are summarized in the Supplementary file. The reference lists of retrieved studies were also searched manually to find new eligible studies.

Two authors independently performed the literature search and study selection; any conflicts were resolved through group discussion until a consensus was reached. A study was included if they met the following inclusion criteria: (1) study design: the study had a prospective design; (2) patients: ACP patients admitted to the ED; (3) risk stratifying tools: TIMI, HEART, or GRACE; (4) outcomes: the study had to report the incidence of MACEs and provided clear definitions of MACEs; (5) data abstracted: true and false positives or negatives, or data could transform into the above information must be reported; and (6) cut-off value: the cut-off value of TIMI and HEART was 0-3, and the cut-off value of GRACE was 55-110. Retrospective studies were excluded due to various confounding factors. Additionally, studies that used other cut-off values were excluded.

Data Collection and Quality Assessment

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Two authors independently abstracted data items and assessed the quality of the included studies, and any disagreement was settled by an additional author reviewing the original article. The collected information from retrieved studies including the first author's name, publication year, country, sample size, age at baseline, percentage of males, risk stratifying tools, patients' status, MACE definition, follow-up duration, and true and false positives/negatives. Study quality was assessed using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2), which is based on patient selection, index test, reference standard, risk of bias, and concerns about applicability [15].

Statistical Analysis

The pooled sensitivity, specificity, positive likelihood ratio (PLR), negative likelihood ratio (NLR), and the area under the receiver operating characteristic curves (AUC) for the risk stratification by TIMI, HEART, and GRACE on MACEs were calculated using a bivariate generalized linear mixed model [16,17], while the pooled diagnostic odds ratio (DOR) was calculated using a random-effects model [18]. The I^2 and Q statistic were used for assessing the heterogeneity across included studies, and P<0.10 was considered as significant heterogeneity [19,20]. The robustness of pooled results was also assessed by sensitivity analyses, which were also performed for studies using all three scoring systems and the endpoint MACE [21]. TIMI, HEART, and GRACE-based risk stratification were assessed using an indirect comparison analysis, and the ratios among these scoring systems were calculated [22]. Furthermore, subgroup analyses for the predictive values of TIMI, HEART, and GRACE on subsequent MACE risk were also estimated based on country, mean age, percentage of males, follow-up duration, and study quality. The funnel plots and Deeks' asymmetry tests were used to assess publication bias [23,24]. The inspection level for pooled diagnostic parameters was two-sided, and P<0.05 was considered statistically significant. All statistical analyses were performed using Stata software (version 10.0; Stata Corporation, College Station, TX, USA).

Patient and Public Involvement

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There was no patient or public involvement in the design or conduct of this study.

RESULTS

Literature Search

The details regarding the literature search and study selection of eligible studies are presented in Figure 1. A total of 2,794 articles were identified through the electronic search from PubMed, Embase, and the Cochrane library, and 1,981 were excluded because of term duplications. Subsequently, the remaining 813 studies were selected through title and abstract review; 732 were excluded based on irrelevance. A total of 81 full texts were retrieved for further evaluation, and 48 studies were excluded due to the following reasons: used other cut-off values (n=21), retrospective study design (n=14), and insufficient data (n=13). An additional 135 potential studies identified from the reference lists of retrieved studies were excluded because of duplication with the electronic search. Subsequently, 33 prospective cohort studies that recruited 40,262 ACP patients were selected for final quantitative meta-analysis [13,25-56].

Study Characteristics

The baseline characteristics of included studies are summarized in Table 1. The retrieved studies were published from 2005 to 2020, and 255-4,333 ACP patients were included in each study. Nine studies were conducted in Eastern countries, and the remaining 24 studies were conducted in Western countries. The mean age of enrolled patients ranged from 48.0 to 69.0 years, and the percentage of males ranged from 40.0 to 68.8%. Risk stratification by the TIMI score was available in 25 studies and published between 2005 to 2020, 16 studies used the HEART score and published between 2013 to 2020, and 16 studies employed GRACE and published between 2007 to 2020. The study quality of the included studies was assessed by QUADAS-2 and presented in Figure 2.

TIMI

The predictive value of risk stratification by the TIMI score on MACEs in ACP patients was available in 25 studies. The pooled sensitivity and specificity of the TIMI score for

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predicting MACEs were 0.95 (95% CI: 0.91-0.98; I^2 = 98.10%) and 0.36 (95% CI: 0.24-0.50; I^2 = 99.64%), respectively (Supplementary file). Moreover, the pooled PLR and NLR of the TIMI score for predicting MACEs were 1.49 (95% CI: 1.25-1.79; I^2 = 99.21%) and 0.13 (95% CI: 0.07-0.21; I^2 = 95.38%), respectively (Supplementary file). The pooled DOR of the TIMI score for predicting MACEs was 9.18 (95% CI: 6.22-13.55; P<0.001) with significant heterogeneity across the included studies (I^2 = 87.6%; P<0.001) (Supplementary file). The AUC of the TIMI score for predicting MACEs was 0.80 (95% CI: 0.76-0.83; Figure 3). No significant publication bias for the TIMI score was detected (P=0.17; Supplementary file).

HEART

The predictive value of risk stratification by the HEART score on MACEs in ACP patients was available in 16 studies. The pooled sensitivity and specificity of the HEART score for predicting MACEs were 0.96 (95% CI: 0.91-0.98; I^2 = 94.87%) and 0.50 (95% CI: 0.41-0.60; I^2 = 98.84%), respectively (Supplementary file). The pooled PLR and NLR of the HEART score for predicting MACEs were 1.94 (95% CI: 1.61-2.35; I^2 = 98.01%) and 0.08 (95% CI: 0.03-0.17; I^2 = 94.65%), respectively (Supplementary file). The pooled DOR for the HEART score was 17.92 (95% CI: 9.40-34.18; P<0.001) with significant heterogeneity across the included studies (I^2 = 88.9%; P<0.001) (Supplementary file). The AUC of the HEART score for predicting MACEs was 0.80 (95% CI: 0.77-0.84; Figure 4). There was no significant publication bias for the HEART score (P=0.98; Supplementary file).

GRACE

The predictive value of risk stratification by the GRACE score on MACEs in ACP patients was available in 16 studies. The pooled sensitivity and specificity of the GRACE score for predicting MACEs were 0.78 (95% CI: 0.64-0.87; I^2 = 96.78%) and 0.56 (95% CI: 0.46-0.66; I^2 = 99.39%), respectively (Supplementary file). The pooled PLR and NLR of the GRACE score for predicting MACEs were 1.77 (95% CI: 1.51-2.08; I^2 = 96.34%) and 0.12 (95% CI: 0.06-0.26; I^2 = 94.07%), respectively (Supplementary file). The DOR of the GRACE score for predicting MACEs was 4.00 (95% CI: 2.78-5.74; P<0.001) with significant heterogeneity across the included studies

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(I^2 = 88.7%; P < 0.001) (Supplementary file). The AUC of the GRACE score for predicting MACEs was 0.70 (95% CI: 0.66-0.74; Figure 5). No significant publication bias for the GRACE score was observed (P=0.36; Supplementary file).

Indirect Comparisons

Indirect comparisons of the diagnostic parameters (sensitivity, specificity, PLR, NLR, DOR, and AUC) among the TIMI, HEART, and GRACE scoring systems for predicting MACEs are summarized in Table 2. First, the sensitivity of TIMI (ratio: 1.22; 95% CI: 1.04-1.43) and HEART (ratio: 1.23; 95% CI: 1.05-1.44) was significantly higher than that of GRACE for predicting MACEs. Second, the specificity of TIMI was significantly lower than that of GRACE for predicting MACEs (ratio: 0.64; 95% CI: 0.43-0.97). Third, the PLR of TIMI was lower than that of HEART for predicting MACEs (ratio: 0.77; 95% CI: 0.59-1.00). Fourth, TIMI (ratio: 0.32; 95% CI: 0.17-0.64) and HEART (ratio: 0.20; 95% CI: 0.08-0.52) were associated with a lower NLR than GRACE for predicting MACEs. Fifth, the DOR of TIMI (ratio: 2.29; 95% CI: 1.35-3.91) and HEART (ratio: 4.48; 95% CI: 2.14-9.39) was significantly higher than that of GRACE for predicting MACEs. Finally, the AUC of TIMI (ratio: 1.14; 95% CI: 1.06-1.23) and HEART (ratio: 1.14; 95% CI: 1.06-1.23) was significantly higher than that of GRACE for predicting MACEs.

Sensitivity and subgroup Analysis

The results of sensitivity analyses found that the predictive values of TIMI, HEART, and GRACE for predicting MACEs were stable and unaltered by sequential removal of one study from the overall analysis (data not shown). Sensitivity analyses were also performed after removing studies not using all 3 scoring systems (Table 3). We noted that TIMI had lower sensitivity that that of HEART (ratio: 0.86; 95%CI: 0.75-0.99). Moreover, TIMI had lower NLR (ratio: 0.55; 95%CI: 0.31-0.99), higher DOR (ratio: 2.71; 95%CI: 1.17-6.24), and higher AUC (ratio: 1.19; 95%CI: 1.11-1.27), compared to GRACE. Furthermore, HEART had higher sensitivity (ratio: 1.32; 95%CI: 1.11-1.58), DOR (ratio: 4.36; 95%CI: 1.63-11.64), or AUC (ratio: 1.14; 95%CI: 1.06-1.23), and lower NLR (ratio: 0.22; 95%CI: 0.09-0.56) as compared with GRACE. Additionally, Table 4 presents the subgroup analysis results for the predictive values of

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TIMI, HEART, and GRACE. First, TIMI has lower specificity than HEART in the pooled studies from Western countries. Furthermore, TIMI versus HEART showed lower PLR in the pooled studies of Western countries (mean age < 60.0 years). Moreover, the DOR of TIMI was lower than of HEART in the pooled studies of Eastern countries, while TIMI had a lower AUC than HEART for mean age < 60.0 years. Second, TIMI with higher sensitivity than GRACE when pooled studies conducted in Eastern countries, percentage of males ≥60.0%, follow-up duration ≤30.0 days, and studies of moderate quality. Moreover, TIMI versus GRACE showed lower specificity if mean age <60.0 years, percentage of males ≥60.0%, and follow-up duration ≤ 30.0 days. Furthermore, TIMI has lower PLR than GRACE if percentage of males ≥ 60.0%, while TIMI has lower NLR than GRACE in the pooled studies of Eastern countries, irrespective of the mean age or percentage male status, follow-up duration ≤30.0 days, and studies of moderate quality. Moreover, TIMI versus GRACE showed higher DOR and AUC in most subgroups. Third, HEART versus GRACE showed higher sensitivity in the pooled studies of Eastern countries, mean age ≥ 60.0 years, follow-up duration ≤ 30.0 days, and studies of moderate quality. Moreover, HEART has lower NLR than GRACE in most subgroups, except that of mean age < 60.0 years. Furthermore, HEART versus GRACE showed higher DOR and AUC in most subgroups.

Discussion

This study was the first meta-analysis to conduct indirect comparisons of the predictive values of risk stratification assessed by the TIMI, HEART, and GRACE scores on MACEs in ACP patients. The current study included a total of 40,262 ACP patients from 33 prospective cohort studies and across a wide range of patient characteristics. The findings of this study suggest that the TIMI, HEART, and GRACE scoring systems have relatively powerful predictive values for the risk of MACEs in ACP patients admitted to the ED. Moreover, an indirect analysis indicated that the predictive value of TIMI and HEART was superior to that of GRACE for predicting MACEs, while there were no significant differences between TIMI and HEART for predicting MACEs.

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The results of sensitivity analyses for studies using all 3 scoring systems were consistent with results of the overall analysis.

Several systematic reviews and meta-analyses have illustrated the predictive values of the TIMI, HEART, and GRACE scoring systems on MACEs in ACP patients [57-59]. A meta-analysis conducted by Hess et al. included 8 prospective studies and found that the TIMI score provided effective risk stratification for predicting MACEs in potential ACS patients, whereas it should not be used as the sole means for determining patient disposition [57]. Van Den Berg et al. conducted a meta-analysis of 2 prospective and 10 retrospective cohort studies and suggested that the HEART score could be used to identify MACEs in patients with a suspected diagnosis of ACS [58]. Roche et al. included 11 studies and found that using 100 as the cut-off value of the GRACE score could predict the discharge of nearly 70% of presentations, while the predictive value for subsequent MACE risk was not obtained [59]. However, the above studies only reported the diagnostic value of a single scoring system for predicting MACEs in ACS patients. Therefore, we performed the current meta-analysis of prospective studies to evaluate the predictive values of the TIMI, HEART, and GRACE scoring systems on the risk of MACEs in ACP patients and systematically compared the predictive values among them.

The predictive value of the TIMI score for MACEs in ACP patients was statistically significant, whereas several studies reported inconsistent results. Sanchis et al. recruited 646 ACP patients who presented with non-ST-segment deviation and normal troponin concentrations. They found that the TIMI score was not associated with the risk of MACEs when 0 was used as the cutoff value [26]. Graham et al. included 352 Chinese ACP patients who were admitted to the ED and found that a low TIMI score could not rule out cardiac causes of chest pain [35]. Holly et al. included 552 ACP patients admitted to an ED observation unit and suggested that the TIMI score was not associated with the risk of MACE at 30 days when 0 was used as the cut-off value [36]. Leung et al. indicated a modified TIMI score of 0 could not rule out 30-day MACEs in ACP patients admitted to the ED [45]. The potential reasons for this could be the TIMI score was designed for risk stratification in patients with non-ST-segment elevation

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ACS, which is mainly based on appropriate ECG changes or elevations of biomarkers of necrosis. Moreover, the presentation characteristics in ACP patients were not entered into the TIMI score. Finally, the prevalence of MACEs was lower than expected (broad 95% CIs; i.e., no statistically significant difference).

The pooled predictive value of the HEART score for predicting MACE in ACP patients was statistically significant. Nearly all included studies reported a similar conclusion, whereas the study conducted by McCord et al. enrolled 661 ACP patients in the ED from Europe, Australia, and the US and suggested that the HEART score after 4 hours of the presentation was associated with marginal predictive values for the risk of MACEs [47]. The potential reason for this may be that this study used a modified HEART score, and the original HEART score only considered the initial cTn value, without taking serial sampling into account, which is associated with a lower prevalence of MACEs [60]. Furthermore, all included studies reported similar conclusions for the predictive value of the GRACE score on the risk of MACEs. Especially, the GRACE score was initially designed for post-ACS risk stratification, including unstable angina, and non-ST-elevation ACS [11]. The America Heart Association suggested the use of the GRACE score for admission and discharge of ACS patients. Moreover, the risk assessment for patients evaluated outside the hospital should be recommended to use GRACE [61].

We noted that the predictive value of the GRACE score was inferior to that of TIMI and HEART scores. Sensitivity analyses were performed for studies reporting all 3 scoring systems, which included 9 studies [42-44,46,48,52,53,55,56] and the results indicated TIMI and HEART having higher predictive values for MACEs than GRACE. The potential reason for this could be that the GRACE score was initially developed for ACS patients but not for ACP patients, and the potential risk factors for the progression of MACEs were not considered in the GRACE score. Interestingly, we noted no significant difference between TIMI and HEART for predicting MACEs, while several studies reported that the predictive value of HEART for MACEs was superior to that of TIMI [44,46,48,52,53,55,56]. The potential explanation could be the use of HEART score in the absence of exact definitions for medical history across included studies [11].

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Three strengths of this quantitative meta-analysis should be highlighted: (1) the study was based on prospective cohort studies and used relatively uniform cut-off values, which were associated with lower selective and informative biases; (2) the analysis of this study was based on a large sample size, and the findings in our study were more robust than any individual study; and (3) the predictive values of the TIMI, HEART, and GRACE scoring systems on the risk of MACEs in ACP patients were compared through an indirect analytic approach. The limitations of this study should also be acknowledged. First, the analysis of this meta-analysis was based on crude data, and the adjusted results were not available. Second, substantial heterogeneity was detected across the included studies, and the heterogeneity were not fully explained by sensitivity and subgroup analyses, which might be attributed to the various definitions of MACEs, follow-up durations, and individual risk level for MACEs. Third, the analysis was conducted on published articles, which causes inevitable publication biases. Finally, the current study was based on indirect comparisons between the predictive values of the TIMI, HEART, and GRACE scoring systems as direct comparisons were not available.

The findings of this study indicated that risk stratification assessed by the TIMI, HEART, and GRACE scores provide relatively appropriate predictive values for MACEs in ACP patients. The results of indirect comparison analysis indicated that TIMI and HEART had relatively relative better predictive values than GRACE on subsequent MACE risk. Further prospective cohort studies should be conducted to directly compare the predictive values of TIMI, HEART, and GRACE on MACEs in ACP patients.

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Authors' contributions

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JK: study design and manuscript drafting

YC: study supervision, data analysis and result interpretation

XW: literature search and data collection

ZW: data curation

FC: manuscript revising and final approval

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Competing interests

The authors declare no competing interests.

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None.

Ethnic Statement

Not applicable.

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Figure legends:

Figure 1. Flow diagram of the literature search and study selection process.

Figure 2. QUADAS-2 Scoring of Included Studies.

Figure 3. The summary receiver operating characteristic curve of risk stratification assessed by the TIMI score.

Figure 4. The summary receiver operating characteristic curve of risk stratification assessed by the HEART score.

Figure 5. The summary receiver operating characteristic curve of risk stratification assessed by the GRACE score.

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Table 1. Baseline characteristics of studies included in the systematic review and meta-analysis.

Study	Country	Sample size	Mean age (years)	Percentage male (%)	Risk stratifying tools	Patients' status	MACE definition	Follow-up
Tong 2005 [25]	US	957	60.0	52.0	TIMI	Chest pain and a nondiagnostic ECG	Death and MI	30 days
Sanchis 2005 [26]	Spain	646	64.0	65.8	TIMI	Acute chest pain	Death, MI, or urgent revascularization	14 days
Pollack Jr 2006 [27]	US	3929	51.6	40.0	TIMI	Chest pain in the ED	Death, acute MI, and revascularization	30 days
Pelliccia 2006 [28]	Italy	4333	58.4	68.8	TIMI	Acute chest pain	MI	In-hospital
Lyon 2007 [29]	UK	954	60.0	62.0	TIMI, GRACE	Undifferentiated chest pain	MI, cardiac arrest, revascularization, unstable angina with myocardial damage and death	30 days
Ramsay 2007 [13]	UK	347	65.2	62.3	TIMI, GRACE	Suspected cardiac pain	Death, non-fatal MI, and emergency revascularization	3.0 months
Body 2009 [30]	UK	796	58.9	60.4	TIMI	Chest pain in the ED	Death, acute MI or urgent coronary revascularization	30 days
Campbell 2009 [31]	US	3169	53.6	45.0	TIMI	Chest pain in the ED	Death, MI, or revascularization	30 days
Hess 2010 [32]	Canada	1017	59.3	60.6	TIMI	ED patients with chest pain and possible ACS	Acute MI, revascularization, or death	30 days
Stracke 2010 [33]	Germany	1014	66.0	55.0	GRACE	Chest pain in the ED	Death	In-hospital

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Van der Zee 2011 [34]	The Netherlands	524	57.7	60.5	GRACE	Chest pain in the ED	Death	9.4 years
Graham 2013 [35]	China	315	69.0	54.9	TIMI	Chest pain in the ED	Death, MI, troponin positive ACS and PCI	30 days
Holly 2013 [36]	US	552	54.1	46.0	TIMI	Chest pain in the ED	MI, revascularization, or death	30 days
Backus 2013 [37]	The Netherlands	2388	60.6	57.5	HEART	Chest pain in the ED	Acute MI, PCI, CABG, coronary angiography revealing procedurally correctable stenosis managed conservatively, and death	6 weeks
Cullen 2013 [38]	Australia	948	54.0	59.9	TIMI, and GRACE	Chest pain in the ED	Cardiac death, acute MI, and unstable angina	30 days
Graham 2014 [39]	China	925	68.0	51.7	TIMI	Chest pain in the ED	Death, readmission with MI, ACS not diagnosed at initial ED presentation and coronary revascularization	30 days
Visser 2015 [40]	The Netherlands	255	64.0	56.0	HEART	Chest pain in the ED	MI, or PCI, or CABG, or coronary angiography revealing significant stenosis or death	6 weeks
Boubaker 2015 [41]	Tunisia	3125	57.7	58.3	TIMI, GRACE	Chest pain in the ED	All-cause mortality, acute coronary syndrome, and coronary non-ED planned revascularization	30 days
Wang 2016 [42]	China	986	54.0	55.0	TIMI, HEART, GRACE	Chest pain in the ED	Death, MI, and/or the need for revascularization by CABG or PCI	6 months
Chen 2016 [43]	China	833	65.1	55.3	TIMI, HEART, GRACE	Chest pain in the ED	Coronary revascularization, ventricular arrhythmia needing intervention and	30 days

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							high-degree atrioventricular block needing intervention	
Sakamoto 2016 [44]	Singapore	604	60.8	69.2	TIMI, HEART, GRACE	Chest pain in the ED	Death, acute MI, PCI, CABG	30 days
Leung 2017 [45]	China	602	66.0	48.8	TIMI, HEART	Chest pain in the ED	Death, cardiac arrest, MI, and cardiogenic shock	30 days
Poldervaart 2017 [46]	The Netherlands	1748	62.0	54.0	TIMI, HEART, GRACE	Chest pain in the ED	UA, MI, PCI, CABG, stenosis managed conservatively, and death	3.0 months
McCord 2017 [47]	Europe, Australia, and the US	661	58.3	58.2	HEART	Chest pain in the ED	Death or acute MI	30 days
Reaney 2018 [48]	UK	1000	62.5	57.6	TIMI, HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, cardiac death, cardiogenic shock, and life-threatening arrhythmias requiring emergency intervention.	30 days
Greenslade 2018 [49]	Australia	1760	60.4	59.3	TIMI	Chest pain in the ED	Cardiac death, cardiac arrest, cardiogenic shock, acute MI, UA, emergency or urgent revascularization, high-level atrioventricular block, and ventricular arrhythmias	30 days
Moumneh 2018 [50]	France	641	53.3	53.4	HEART	Non-traumatic chest pain patients	MI, coronary angioplasty, coronary bypass, and sudden unexplained death	6 weeks
Ishak 2018 [51]	The Netherlands	1127	63.8	57.7	HEART	Chest pain in the ED	Acute MI, PCI, CABG, or death	30 days
Wong 2018 [52]	China	1081	48.0	52.3	TIMI, HEART, GRACE	Undifferentiated chest pain	Acute MI, PCI, CABG, and death	30 days

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Al-Zaiti 2019 [53]	USA	750	59.0	58.0	TIMI, HEART, GRACE	Chest pain in the ED	Resuscitated or un-resuscitated sudden cardiac arrest, all-cause death, post-discharge re-infarction requiring cardiac revascularization	30 days
Huang 2020 [54]	China	509	59.8	53.4	HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, cardiac death, cardiogenic shock, or life-threatening arrhythmias requiring emergency intervention or resulting in mortality	30 days
Torralba 2020 [55]	Colombia	519	64.3	56.1	TIMI, HEART, GRACE	Chest pain in the ED	Death from any cause, MI, and surgical or percutaneous myocardial revascularization	30 days
Shin 2020 [56]	Korea	1247	62.0	60.8	TIMI, HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, or death from cardiac causes	30 days

*ACS: acute coronary syndrome; CABG: coronary artery bypass grafting; ECG: electrocardiogram; ED: emergency department; MI: myocardial infarction; PCI: percutaneous coronary intervention; UA: unstable angina.

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Table 2. Systematic comparisons of the sensitivity, specificity, positive and negative likelihood ratio, diagnostic odds ratio, and the area under the receiver operating characteristic curve of risk stratifying measured by TIMI, HEART, and GRACE.

Parameters	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity and 95%CI	0.95 (0.91-0.98)	0.96 (0.91-0.98)	0.78 (0.64-0.87)	0.99 (0.94-1.04)	1.22 (1.04-1.43)	1.23 (1.05-1.44)
Specificity and 95%CI	0.36 (0.24-0.50)	0.50 (0.41-0.60)	0.56 (0.46-0.66)	0.72 (0.48-1.09)	0.64 (0.43-0.97)	0.89 (0.69-1.16)
PLR and 95%CI	1.49 (1.25-1.79)	1.94 (1.61-2.35)	1.77 (1.51-2.08)	0.77 (0.59-1.00)	0.84 (0.66-1.07)	1.10 (0.86-1.40)
NLR and 95%CI	0.13 (0.07-0.21)	0.08 (0.03-0.17)	0.40 (0.27-0.59)	0.46 (0.15-1.39)	0.32 (0.17-0.64)	0.20 (0.08-0.52)
DOR and 95%CI	9.18 (6.22-13.55)	17.92 (9.40-34.18)	4.00 (2.78-5.74)	0.51 (0.24-1.09)	2.29 (1.35-3.91)	4.48 (2.14-9.39)
AUC and 95%CI	0.80 (0.76-0.83)	0.80 (0.77-0.84)	0.70 (0.66-0.74)	1.00 (0.94-1.06)	1.14 (1.06-1.23)	1.14 (1.06-1.23)

PLR: positive likelihood ratio; NLR: negative likelihood ratio; DOR: diagnostic odds ratio; AUC: the area under the receiver operating characteristic curve.

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Table 3. Sensitivity analysis for direct comparisons of the sensitivity, specificity, positive and negative likelihood ratio, diagnostic odds ratio, and the area under the receiver operating characteristic curve of risk stratifying measured by TIMI, HEART, and GRACE.

Parameters	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity and 95%CI	0.81 (0.70-0.89)	0.94 (0.84-0.98)	0.71 (0.58-0.81)	0.86 (0.75-0.99)	1.14 (0.94-1.39)	1.32 (1.11-1.58)
Specificity and 95%CI	0.70 (0.54-0.83)	0.53 (0.39-0.66)	0.59 (0.44-0.73)	1.32 (0.94-1.86)	1.19 (0.85-1.65)	0.90 (0.62-1.29)
PLR and 95%CI	2.72 (1.67-4.41)	1.99 (1.51-2.63)	1.74 (1.33-2.28)	1.37 (0.78-2.39)	1.56 (0.90-2.72)	1.14 (0.78-1.68)
NLR and 95%CI	0.27 (0.16-0.44)	0.11 (0.05-0.28)	0.49 (0.36-0.66)	2.45 (0.90-6.66)	0.55 (0.31-0.99)	0.22 (0.09-0.56)
DOR and 95%CI	9.45 (4.77-18.73)	15.21 (6.45-35.88)	3.49 (2.16-5.63)	0.62 (0.21-1.86)	2.71 (1.17-6.24)	4.36 (1.63-11.64)
AUC and 95%CI	0.83 (0.80-0.86)	0.80 (0.76-0.83)	0.70 (0.66-0.74)	1.04 (0.98-1.10)	1.19 (1.11-1.27)	1.14 (1.06-1.23)

PLR: positive likelihood ratio; NLR: negative likelihood ratio; DOR: diagnostic odds ratio; AUC: the area under the receiver operating characteristic curve.

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Table 4. Subgroup analyses.

Parameters	Factors	Groups	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity	Country	Western	0.97 (0.93-0.99)	0.97 (0.93-0.99)	0.88 (0.72-0.95)	1.00 (0.96-1.05)	1.10 (0.96-1.27)	1.10 (0.96-1.27)
		Eastern	0.86 (0.69-0.95)	0.94 (0.79-0.98)	0.60 (0.49-0.70)	0.91 (0.75-1.11)	1.43 (1.13-1.82)	1.57 (1.27-1.93)
	Mean age (years)	⩾ 60.0	0.97 (0.92-0.99)	0.97 (0.94-0.99)	0.83 (0.67-0.92)	1.00 (0.96-1.05)	1.17 (0.99-1.38)	1.17 (1.00-1.37)
		< 60.0	0.93 (0.81-0.97)	0.91 (0.70-0.98)	0.68 (0.44-0.85)	1.02 (0.84-1.24)	1.37 (0.97-1.92)	1.34 (0.92-1.94)
	Percentage male (%)	⩾ 60.0	0.98 (0.92-0.99)	-	0.73 (0.50-0.88)	-	1.34 (1.01-1.79)	-
		< 60.0	0.92 (0.85-0.96)	0.96 (0.91-0.99)	0.80 (0.62-0.90)	0.96 (0.89-1.03)	1.15 (0.95-1.40)	1.20 (0.99-1.45)
	Follow-up duration	⩽ 30.0 days	0.96 (0.91-0.98)	0.97 (0.92-0.99)	0.76 (0.61-0.87)	0.99 (0.94-1.04)	1.26 (1.05-1.51)	1.28 (1.06-1.53)
		> 30.0 days	-	0.93 (0.78-0.98)	-	-	-	-
	Study quality	High	0.94 (0.83-0.98)	0.97 (0.91-0.99)	0.77 (0.50-0.92)	0.97 (0.88-1.06)	1.22 (0.89-1.67)	1.26 (0.93-1.71)
		Moderate	0.96 (0.90-0.98)	0.95 (0.85-0.98)	0.78 (0.61-0.89)	1.01 (0.93-1.10)	1.23 (1.01-1.49)	1.22 (1.00-1.49)
Specificity	Country	Western	0.33 (0.20-0.50)	0.56 (0.47-0.64)	0.50 (0.34-0.66)	0.59 (0.36-0.96)	0.66 (0.37-1.16)	1.12 (0.78-1.61)
		Eastern	0.46 (0.27-0.67)	0.44 (0.27-0.62)	0.64 (0.55-0.73)	1.05 (0.56-1.94)	0.72 (0.45-1.16)	0.69 (0.44-1.07)
	Mean age	⩾ 60.0	0.35 (0.19-0.55)	0.45 (0.34-0.56)	0.51 (0.37-0.64)	0.78 (0.43-1.40)	0.69 (0.38-1.25)	0.88 (0.61-1.28)

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	(years)	< 60.0	0.38 (0.23-0.56)	0.60 (0.44-0.74)	0.63 (0.49-0.76)	0.63 (0.38-1.06)	0.60 (0.37-0.99)	0.95 (0.68-1.34)
	Percentage male (%)	⩾ 60.0	0.21 (0.10-0.37)	-	0.56 (0.45-0.66)	-	0.38 (0.19-0.74)	-
		< 60.0	0.45 (0.30-0.61)	0.52 (0.42-0.63)	0.56 (0.42-0.70)	0.87 (0.58-1.30)	0.80 (0.52-1.24)	0.93 (0.67-1.29)
	Follow-up duration	⩽ 30.0 days	0.35 (0.23-0.50)	0.46 (0.35-0.57)	0.56 (0.45-0.67)	0.76 (0.48-1.20)	0.63 (0.40-0.97)	0.82 (0.60-1.13)
		> 30.0 days	-	0.60 (0.43-0.75)	-	-	-	-
	Study quality	High	0.41 (0.24-0.62)	0.47 (0.35-0.60)	0.61 (0.36-0.81)	0.87 (0.51-1.51)	0.67 (0.36-1.25)	0.77 (0.47-1.25)
		Moderate	0.33 (0.19-0.51)	0.54 (0.39-0.68)	0.54 (0.44-0.64)	0.61 (0.35-1.08)	0.61 (0.36-1.04)	1.00 (0.72-1.40)
PLR	Country	Western	1.45 (1.16-1.80)	2.20 (1.79-2.70)	1.75 (1.37-2.23)	0.66 (0.49-0.89)	0.83 (0.60-1.15)	1.26 (0.91-1.73)
		Eastern	1.61 (1.24-2.08)	1.66 (1.30-2.13)	1.67 (1.37-2.03)	0.97 (0.68-1.39)	0.96 (0.70-1.33)	0.99 (0.72-1.36)
	Mean age (years)	⩾ 60.0	1.49 (1.13-1.96)	1.76 (1.44-2.16)	1.67 (1.34-2.09)	0.85 (0.60-1.19)	0.89 (0.63-1.27)	1.05 (0.78-1.42)
		< 60.0	1.50 (1.23-1.83)	2.26 (1.61-3.18)	1.86 (1.61-2.15)	0.66 (0.45-0.98)	0.81 (0.63-1.03)	1.22 (0.84-1.02)
	Percentage male (%)	⩾ 60.0	1.23 (1.06-1.43)	-	1.67 (1.44-1.93)	-	0.74 (0.60-0.91)	-
		< 60.0	1.68 (1.29-2.19)	2.02 (1.64-2.50)	1.81 (1.45-2.28)	0.83 (0.59-1.17)	0.93 (0.66-1.31)	1.12 (0.82-1.52)
	Follow-up duration	⩽ 30.0 days	1.47 (1.22-1.79)	1.79 (1.47-2.19)	1.74 (1.45-2.09)	0.82 (0.62-1.08)	0.84 (0.65-1.10)	1.03 (0.78-1.35)
		> 30.0 days	-	2.34 (1.63-3.35)	-	-	-	-
	Study	High	1.61 (1.20-2.17)	1.84 (1.45-2.34)	1.97 (1.32-2.95)	0.88 (0.60-1.28)	0.82 (0.50-1.35)	0.93 (0.58-1.49)

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	quality	Moderate	1.44 (1.15-1.79)	2.04 (1.53-2.72)	1.70 (1.46-1.97)	0.71 (0.49-1.01)	0.85 (0.65-1.11)	1.20 (0.87-1.66)
NLR	Country	Western	0.10 (0.05-0.18)	0.05 (0.02-0.14)	0.25 (0.13-0.49)	2.00 (0.62-6.41)	0.40 (0.16-1.01)	0.20 (0.06-0.65)
		Eastern	0.30 (0.18-0.48)	0.14 (0.06-0.37)	0.63 (0.51-0.77)	2.14 (0.76-6.02)	0.48 (0.28-0.81)	0.22 (0.09-0.56)
	Mean age (years)	⩾ 60.0	0.08 (0.03-0.20)	0.06 (0.02-0.14)	0.34 (0.19-0.60)	1.33 (0.34-5.19)	0.24 (0.08-0.71)	0.18 (0.06-0.55)
		< 60.0	0.19 (0.11-0.35)	0.15 (0.04-0.51)	0.50 (0.31-0.81)	1.27 (0.31-5.13)	0.38 (0.18-0.81)	0.30 (0.08-1.17)
	Percentage male (%)	⩾ 60.0	0.11 (0.05-0.24)	-	0.48 (0.27-0.87)	-	0.23 (0.09-0.61)	-
		< 60.0	0.17 (0.10-0.28)	0.07 (0.03-0.17)	0.36 (0.22-0.60)	2.43 (0.89-6.66)	0.47 (0.23-0.97)	0.19 (0.07-0.53)
	Follow-up duration	⩽ 30.0 days	0.12 (0.07-0.22)	0.07 (0.02-0.18)	0.42 (0.27-0.65)	1.71 (0.50-5.92)	0.29 (0.14-0.59)	0.17 (0.05-0.54)
		> 30.0 days	-	0.12 (0.04-0.35)	-	-	-	-
	Study quality	High	0.13 (0.05-0.33)	0.06 (0.02-0.20)	0.38 (0.20-0.71)	2.17 (0.49-9.60)	0.34 (0.11-1.07)	0.16 (0.04-0.59)
		Moderate	0.12 (0.06-0.24)	0.10 (0.04-0.26)	0.40 (0.24-0.69)	1.20 (0.37-3.85)	0.30 (0.13-0.72)	0.25 (0.09-0.73)
DOR	Country	Western	12.68 (7.19-22.38)	30.41 (10.41-88.82)	6.32 (3.35-11.92)	0.42 (0.12-1.40)	2.01 (0.86-4.70)	4.81 (1.38-16.72)
		Eastern	4.56 (3.67-5.65)	10.05 (5.30-19.06)	2.59 (1.83-3.67)	0.45 (0.23-0.89)	1.76 (1.17-2.65)	3.88 (1.87-8.04)
	Mean age (years)	⩾ 60.0	12.86 (6.11-27.07)	23.01 (10.84-48.85)	4.55 (2.50-8.28)	0.56 (0.19-1.61)	2.83 (1.09-7.35)	5.06 (1.93-13.23)
		< 60.0	6.24 (4.46-8.75)	11.52 (3.44-38.55)	3.41 (2.25-5.16)	0.54 (0.15-1.90)	1.83 (1.07-3.12)	3.38 (0.94-12.12)
	Percentage	⩾ 60.0	8.06 (4.08-15.91)	12.32 (1.74-87.48)	3.08 (1.76-5.37)	0.65 (0.08-5.20)	2.62 (1.09-6.31)	4.00 (0.52-30.66)

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	male (%)	< 60.0	9.16 (5.69-14.75)	19.66 (9.31-41.51)	4.51 (2.82-7.22)	0.47 (0.19-1.13)	2.03 (1.04-3.97)	4.36 (1.80-10.54)
	Follow-up duration	⩽ 30.0 days	9.45 (6.19-14.41)	19.21 (8.02-45.99)	3.70 (2.46-5.55)	0.49 (0.19-1.30)	2.55 (1.42-4.59)	5.19 (1.98-13.61)
		> 30.0 days	8.42 (5.14-13.80)	15.90 (8.10-31.21)	5.53 (2.34-13.03)	0.53 (0.23-1.22)	1.52 (0.57-4.10)	2.88 (0.96-8.57)
	Study quality	High	7.96 (4.90-12.91)	21.21 (7.10-63.34)	5.17 (2.51-10.64)	0.38 (0.11-1.24)	1.54 (0.65-3.67)	4.10 (1.11-15.22)
		Moderate	9.44 (5.17-17.23)	16.25 (6.88-38.41)	3.33 (2.22-5.01)	0.58 (0.20-1.66)	2.83 (1.37-5.86)	4.88 (1.88-12.63)
AUC	Country	Western	0.85 (0.82-0.88)	0.82 (0.78-0.85)	0.74 (0.70-0.78)	1.04 (0.98-1.10)	1.15 (1.08-1.23)	1.11 (1.03-1.19)
		Eastern	0.73 (0.69-0.77)	0.74 (0.70-0.78)	0.66 (0.62-0.70)	0.99 (0.91-1.07)	1.11 (1.02-1.20)	1.12 (1.03-1.22)
	Mean age (years)	⩾ 60.0	0.85 (0.82-0.88)	0.83 (0.80-0.86)	0.70 (0.66-0.74)	1.02 (0.97-1.08)	1.21 (1.14-1.30)	1.19 (1.11-1.27)
		< 60.0	0.73 (0.69-0.77)	0.79 (0.75-0.82)	0.70 (0.65-0.73)	0.92 (0.86-0.99)	1.04 (0.96-1.13)	1.13 (1.05-1.21)
	Percentage male (%)	⩾ 60.0	0.74 (0.70-0.77)	-	0.65 (0.60-0.69)	-	1.14 (1.05-1.24)	-
		< 60.0	0.82 (0.79-0.85)	0.81 (0.78-0.85)	0.72 (0.68-0.76)	1.01 (0.96-1.07)	1.14 (1.07-1.22)	1.13 (1.05-1.21)
	Follow-up duration	⩽ 30.0 days	0.81 (0.77-0.84)	0.79 (0.75-0.82)	0.69 (0.65-0.73)	1.03 (0.96-1.09)	1.17 (1.09-1.26)	1.14 (1.06-1.23)
		> 30.0 days	-	0.83 (0.79-0.86)	-	-	-	-
	Study quality	High	0.79 (0.75-0.83)	0.80 (0.76-0.83)	0.74 (0.70-0.78)	0.99 (0.92-1.06)	1.07 (0.99-1.15)	1.08 (1.01-1.16)
		Moderate	0.80 (0.76-0.83)	0.81 (0.77-0.84)	0.67 (0.62-0.71)	0.99 (0.93-1.05)	1.19 (1.10-1.29)	1.21 (1.05-1.14)

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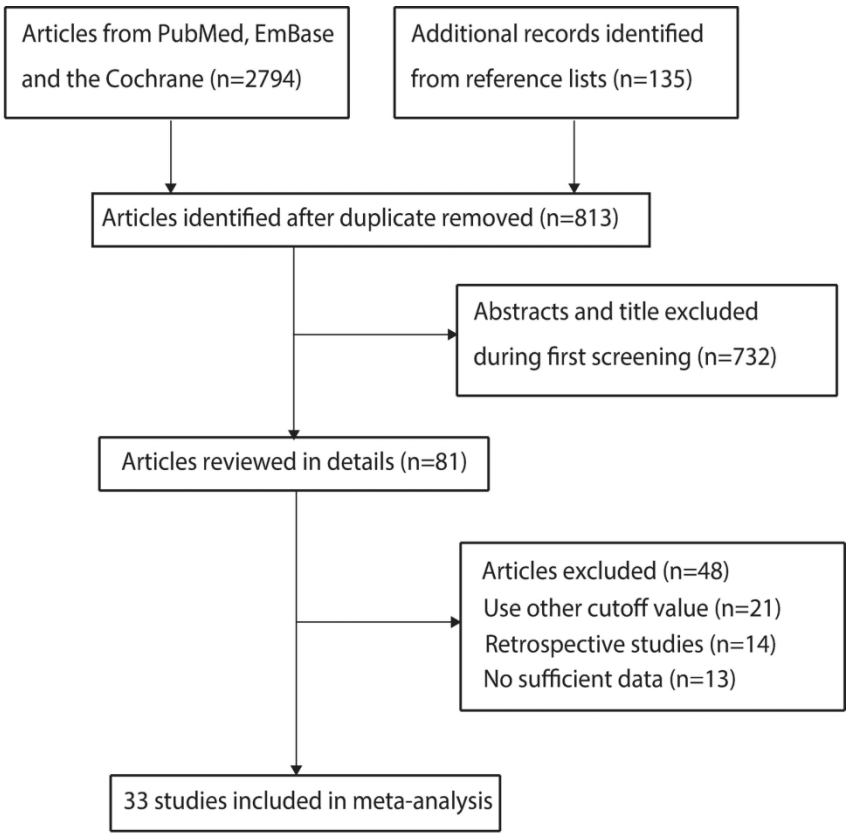


Figure 1. Flow diagram of the literature search and study selection process.

154x146mm (300 x 300 DPI)

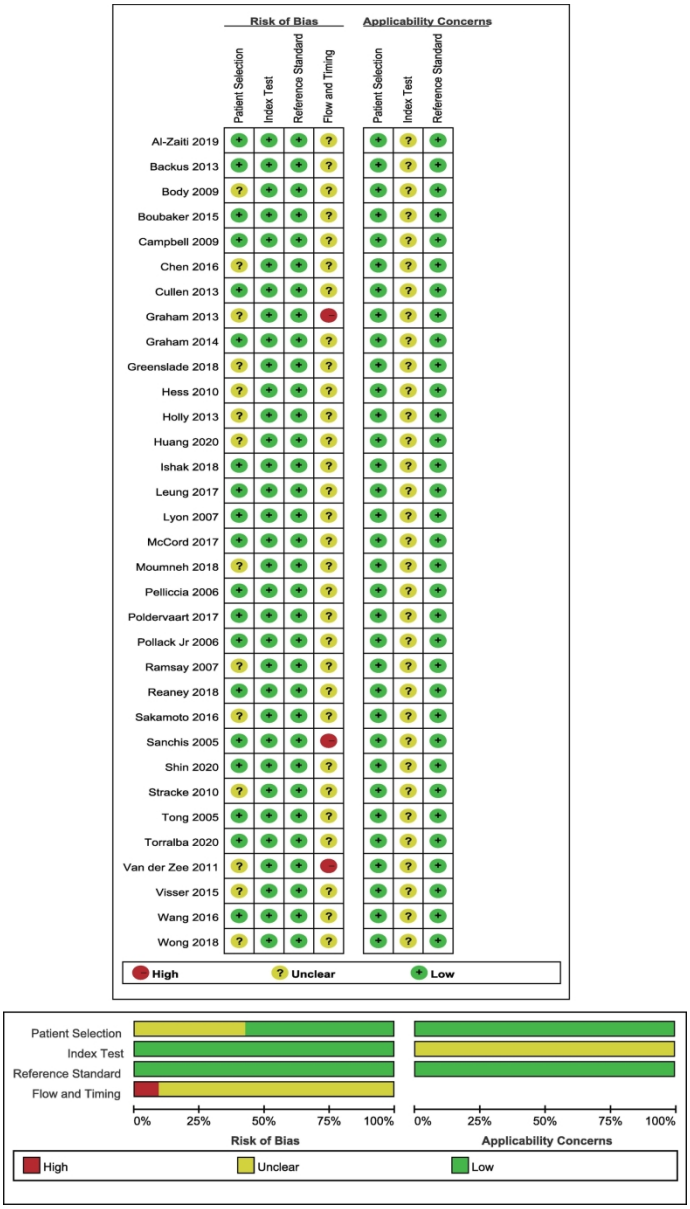


Figure 2. QUADAS-2 Scoring of Included Studies.

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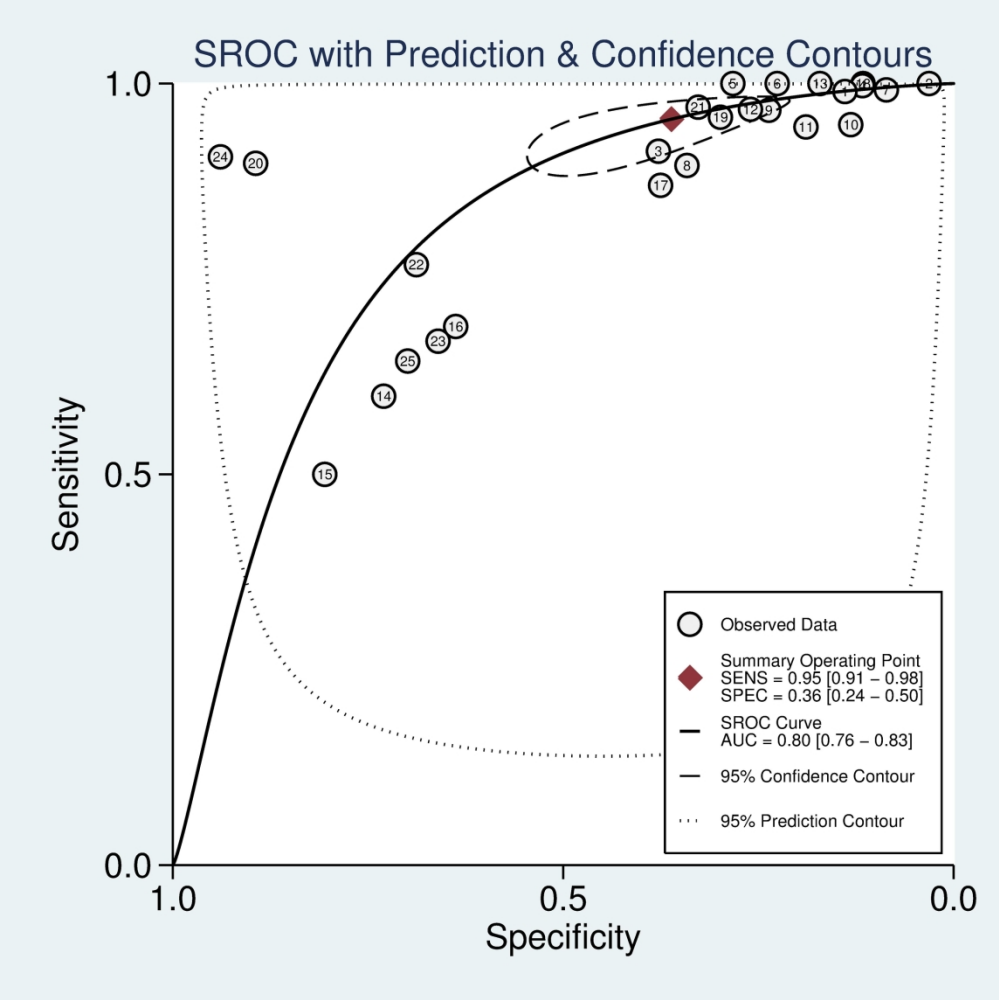


Figure 3. The summary receiver operating characteristic curve of risk stratification assessed by the TIMI score.

203x203mm (300 x 300 DPI)

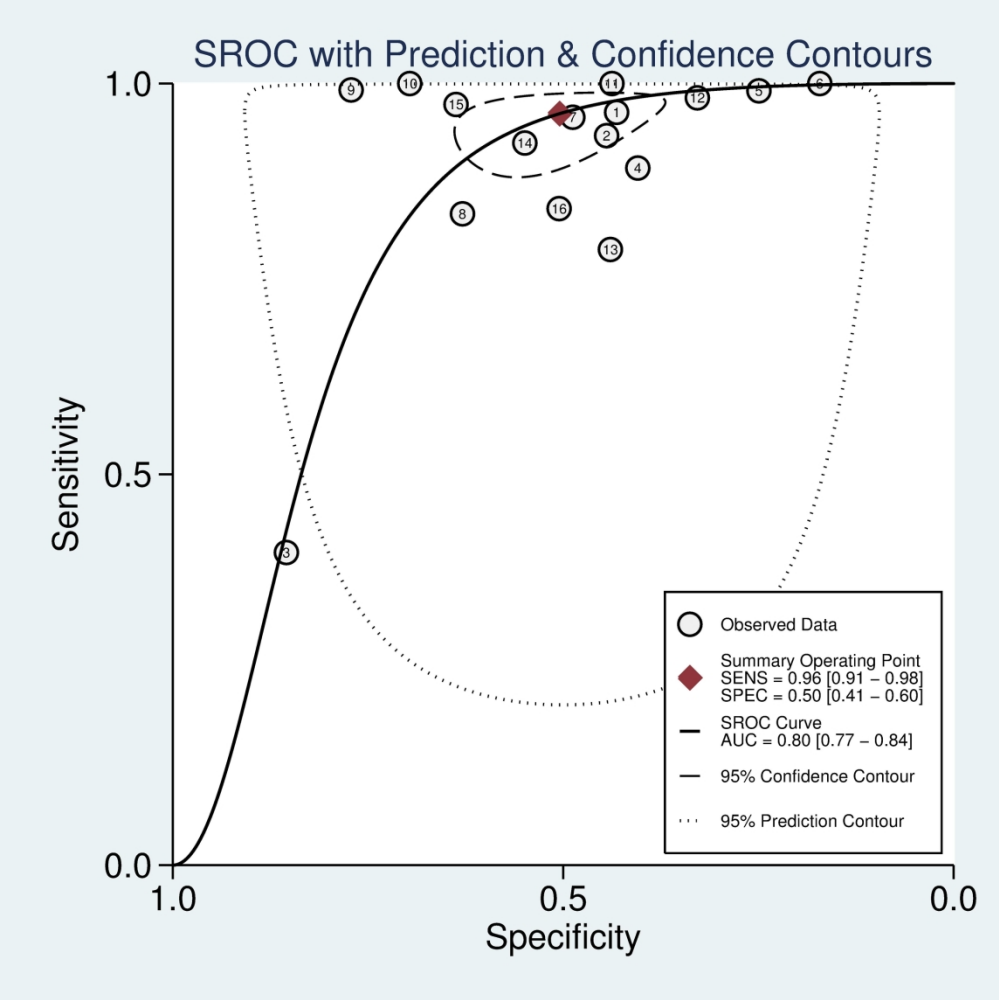


Figure 4. The summary receiver operating characteristic curve of risk stratification assessed by the HEART score.

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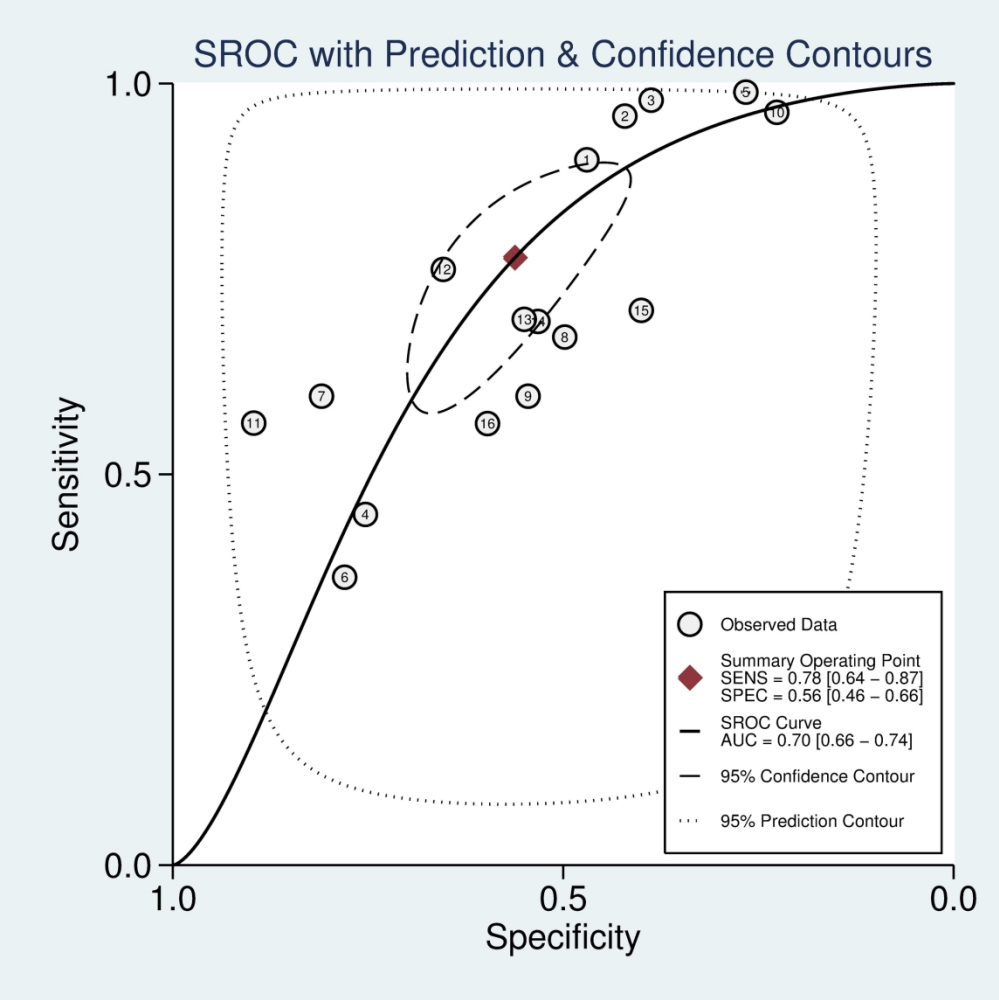


Figure 5. The summary receiver operating characteristic curve of risk stratification assessed by the GRACE score.

203x203mm (300 x 300 DPI)

1. Search strategies in PubMed:

#1 ("Chest Pain"[Mesh] OR "Chest Pain"[Title/Abstract])

#2 ("Emergency Service, Hospital/classification"[Mesh] OR "Emergency Service, Hospital/organization and administration"[Mesh] OR "emergency department"[Title/Abstract])

#3 ("TIMI"[Title/Abstract] OR "HEART"[Title/Abstract] OR "GRACE"[Title/Abstract])

#4 registr*[Title/Abstract] OR Registries[Mesh] OR cohort*[Title/Abstract] OR Follow-Up Stud*[Title/Abstract] OR Follow-Up*[Title/Abstract] OR Longitudinal Stud*[Title/Abstract] OR Prospective*[Title/Abstract] OR Prospective Stud*[Title/Abstract] OR Concurrent Stud*[Title/Abstract] OR Incidence Stud*[Title/Abstract] OR "Cohort Studies"[Mesh]

#5 AND/#1-#4

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2. Sensitivity and Specificity

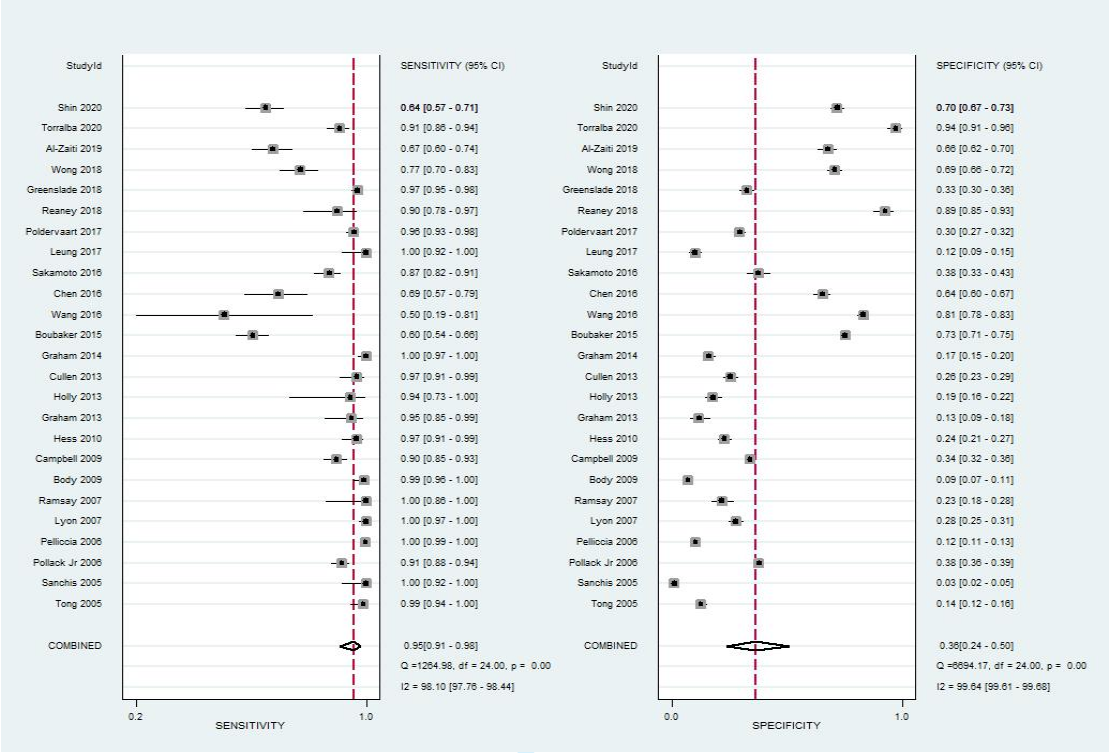


Figure S1. Sensitivity and specificity for TIMI

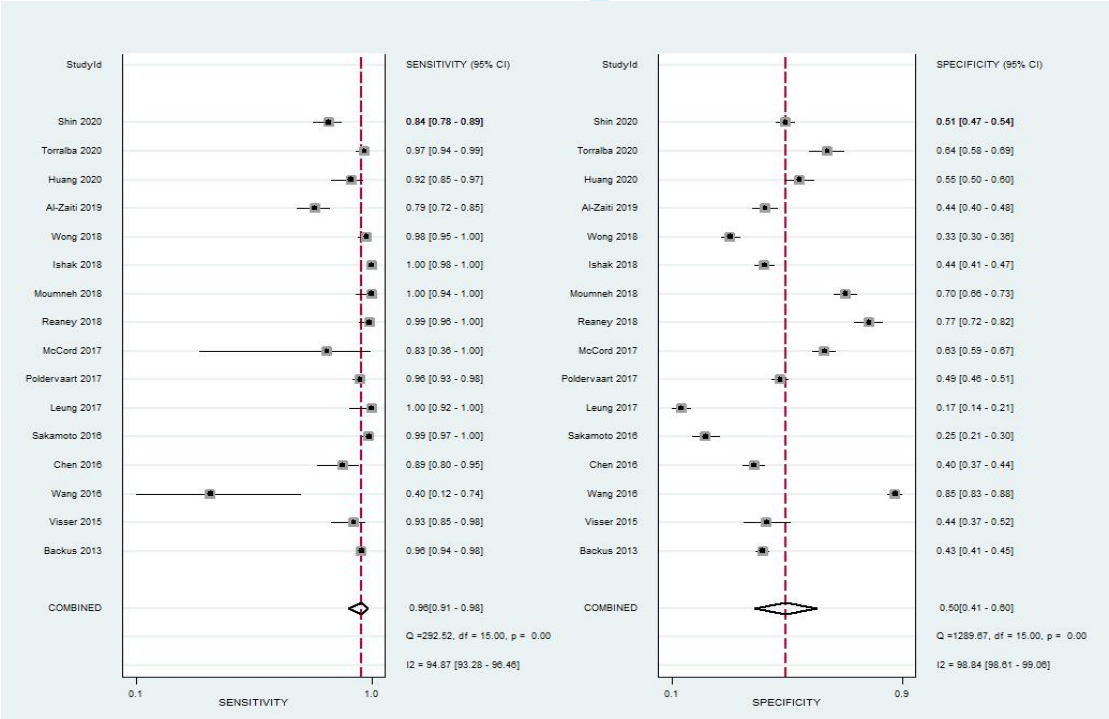


Figure S2. Sensitivity and specificity for HEART

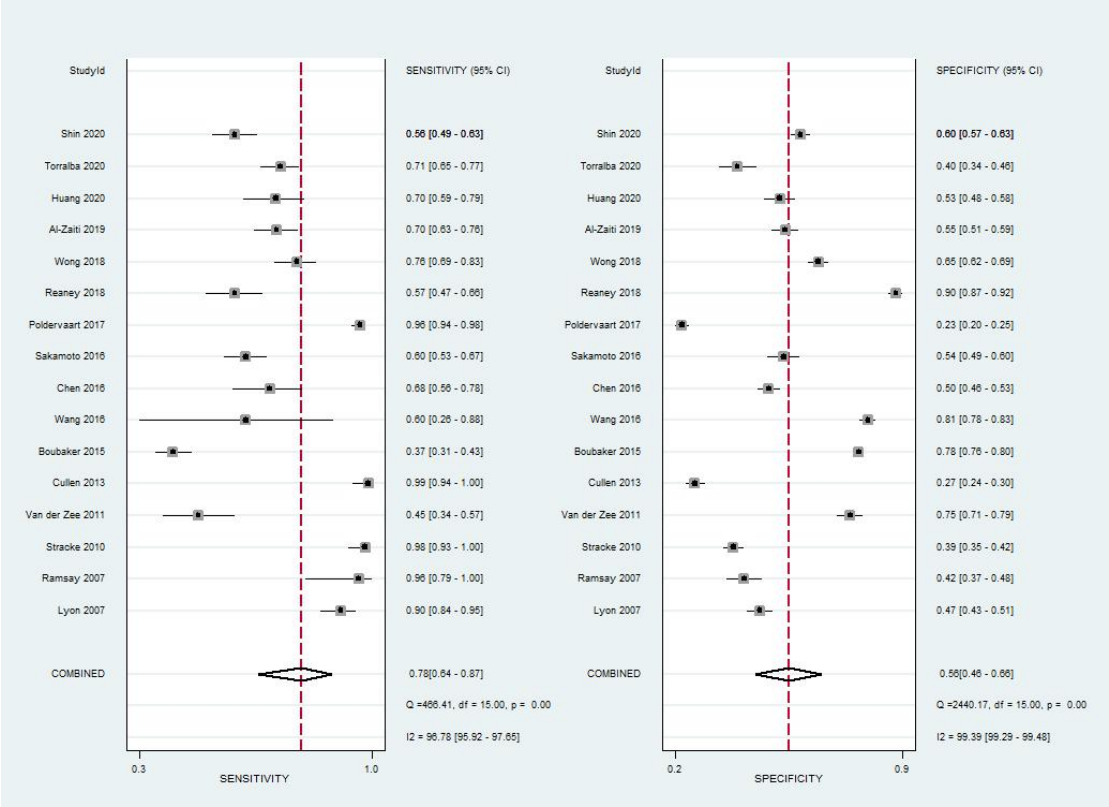


Figure S3. Sensitivity and specificity for GRACE

3. PLR and NLR

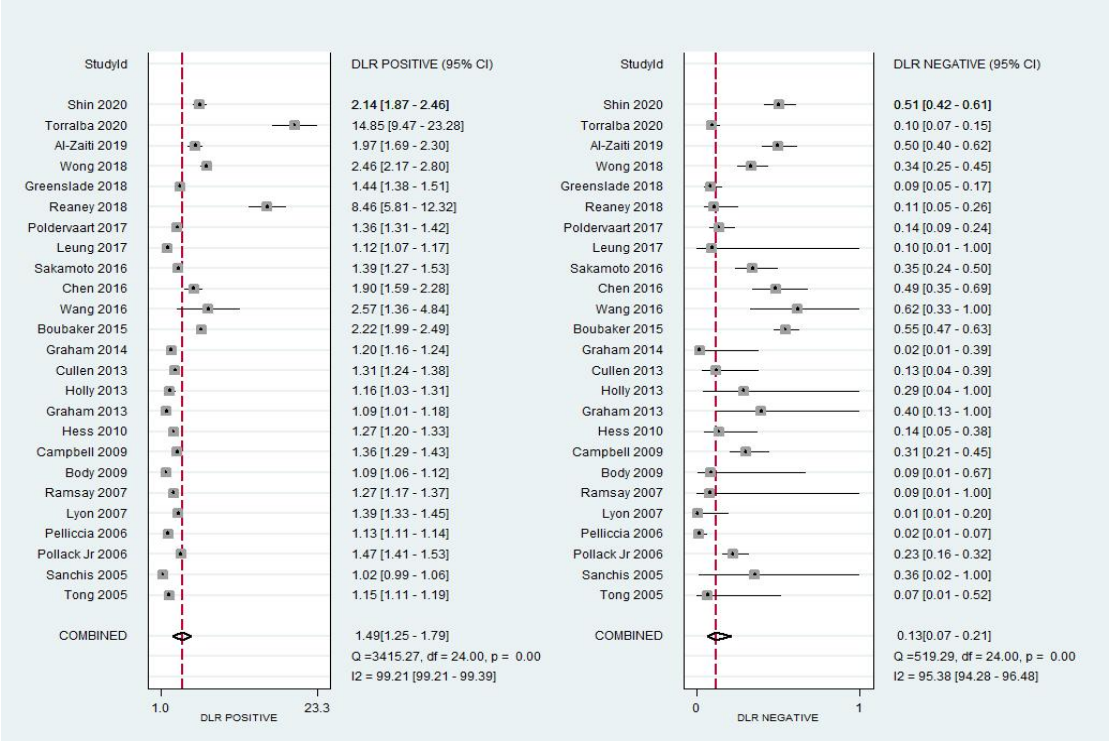


Figure S1. PLR and NLR for TIMI

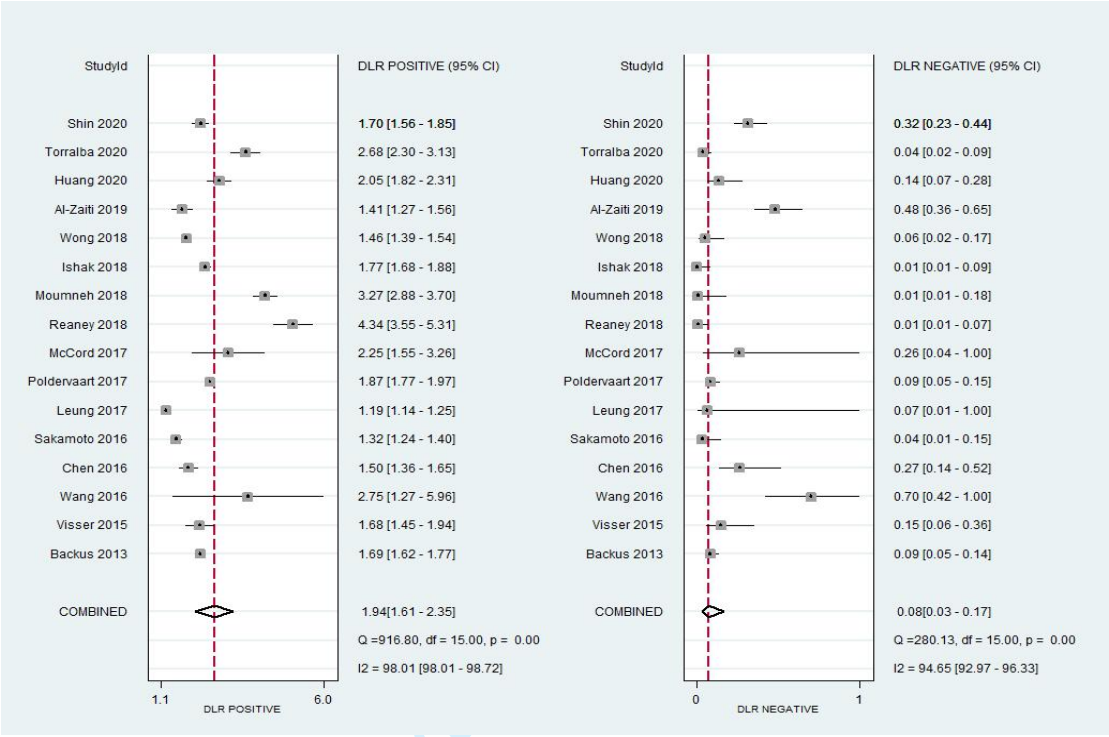


Figure S2. PLR and NLR for HEART

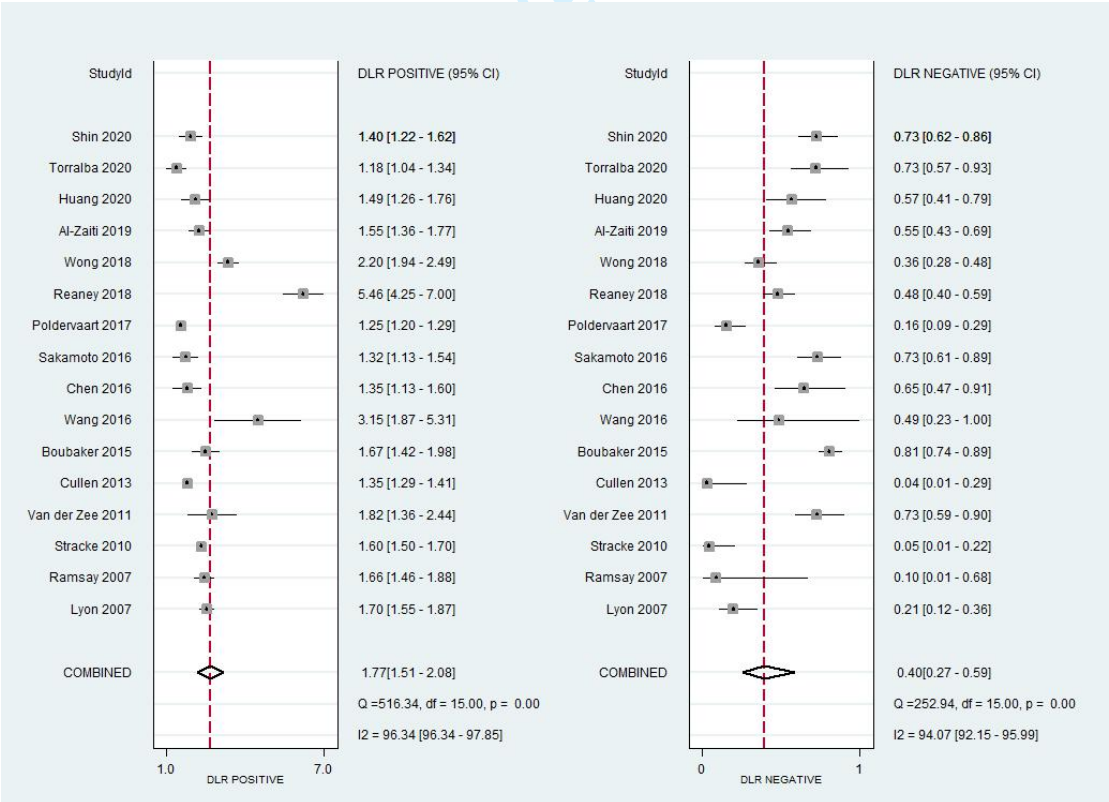


Figure S3. PLR and NLR for GRACE

4. DOR

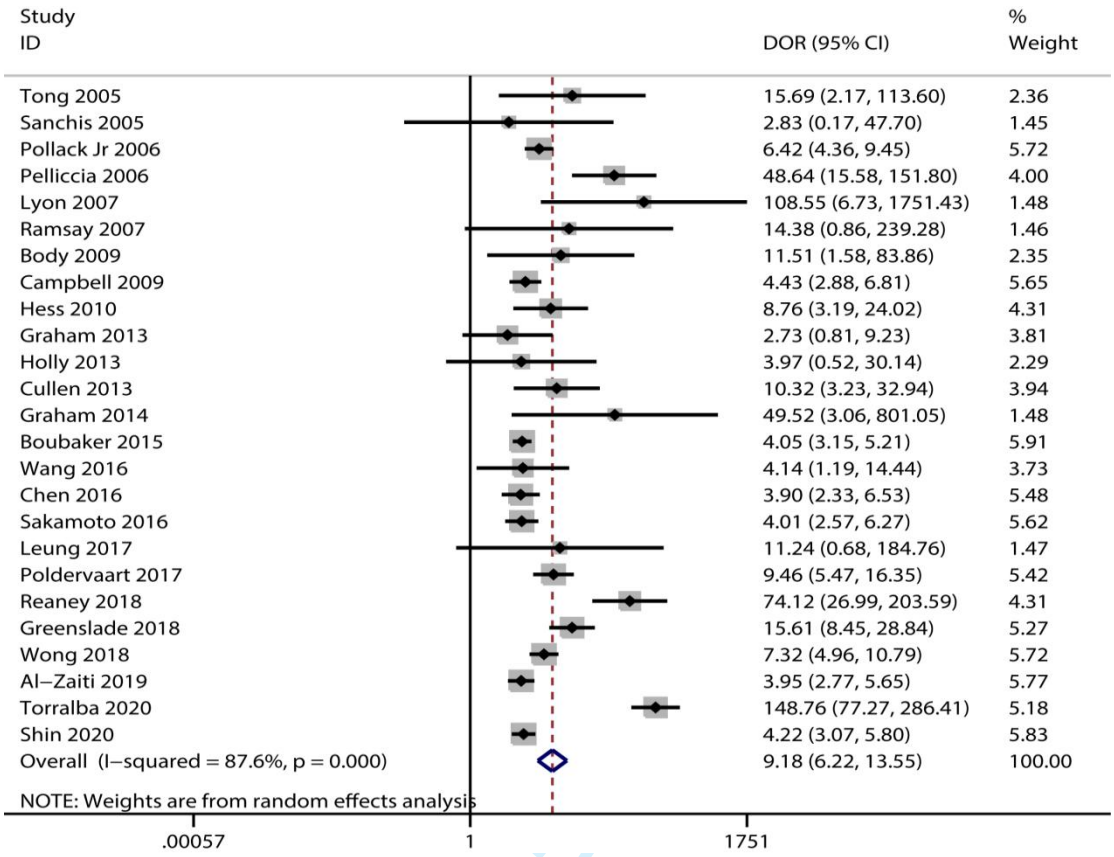


Figure S1. The summary DOR for TIMI

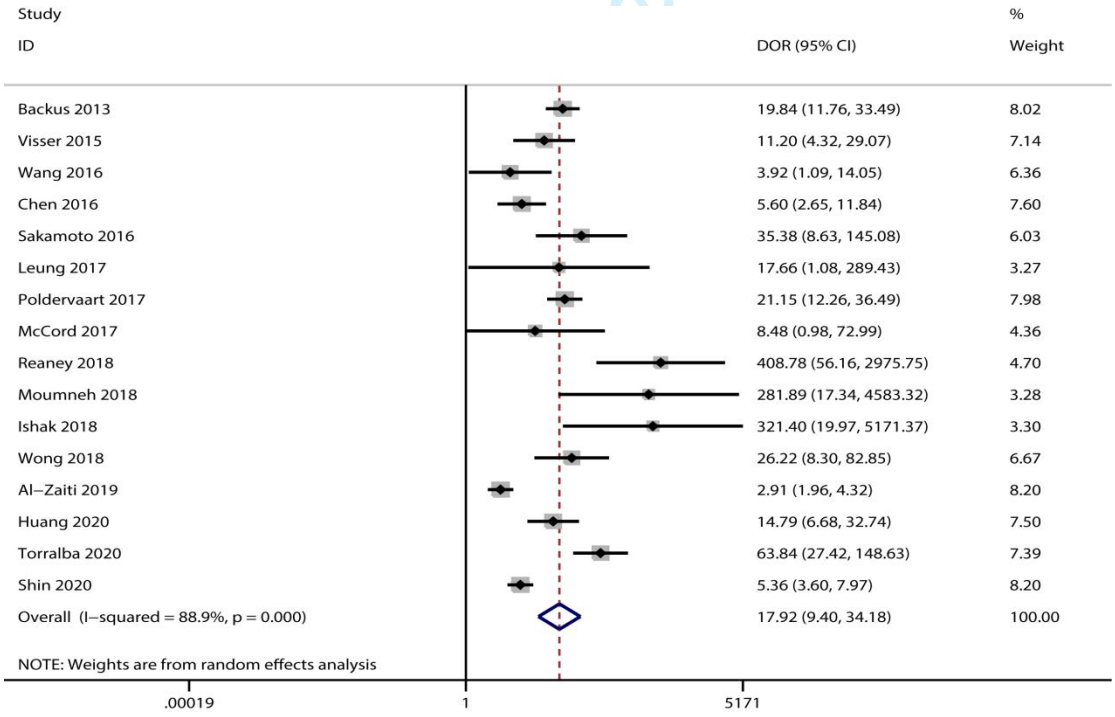


Figure S2. The summary DOR for HEART

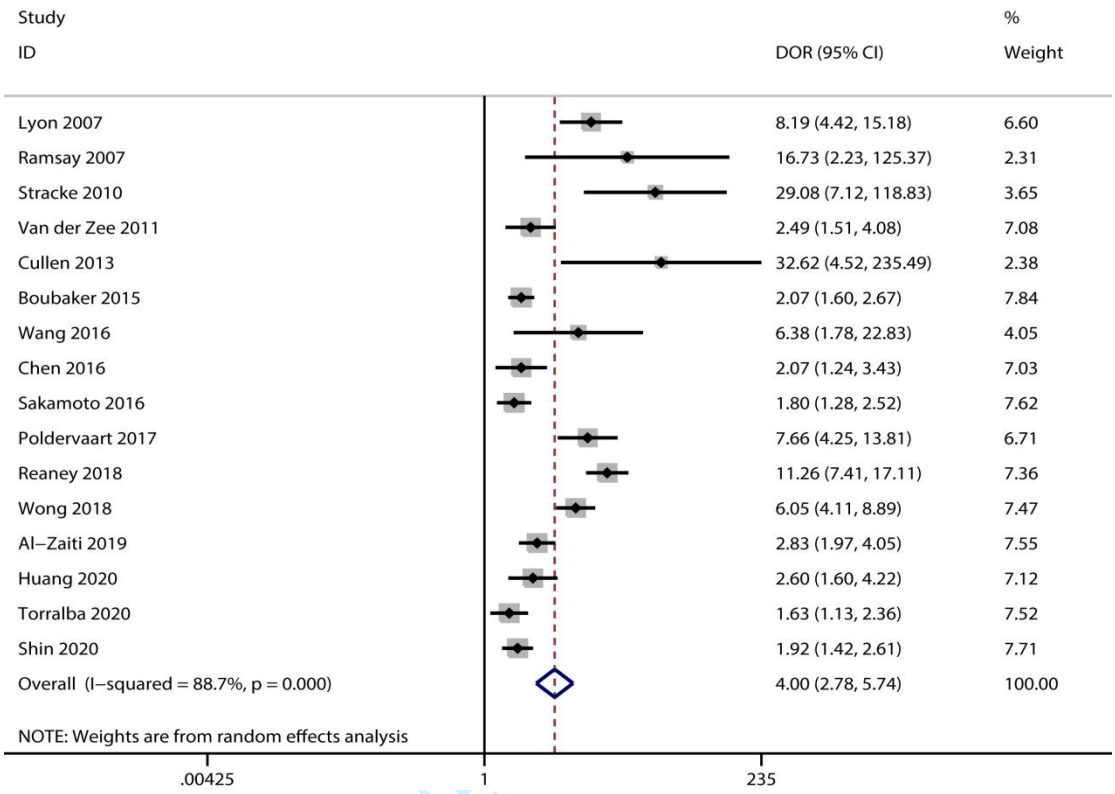


Figure S3. The summary DOR for GRACE

5. Publication bias

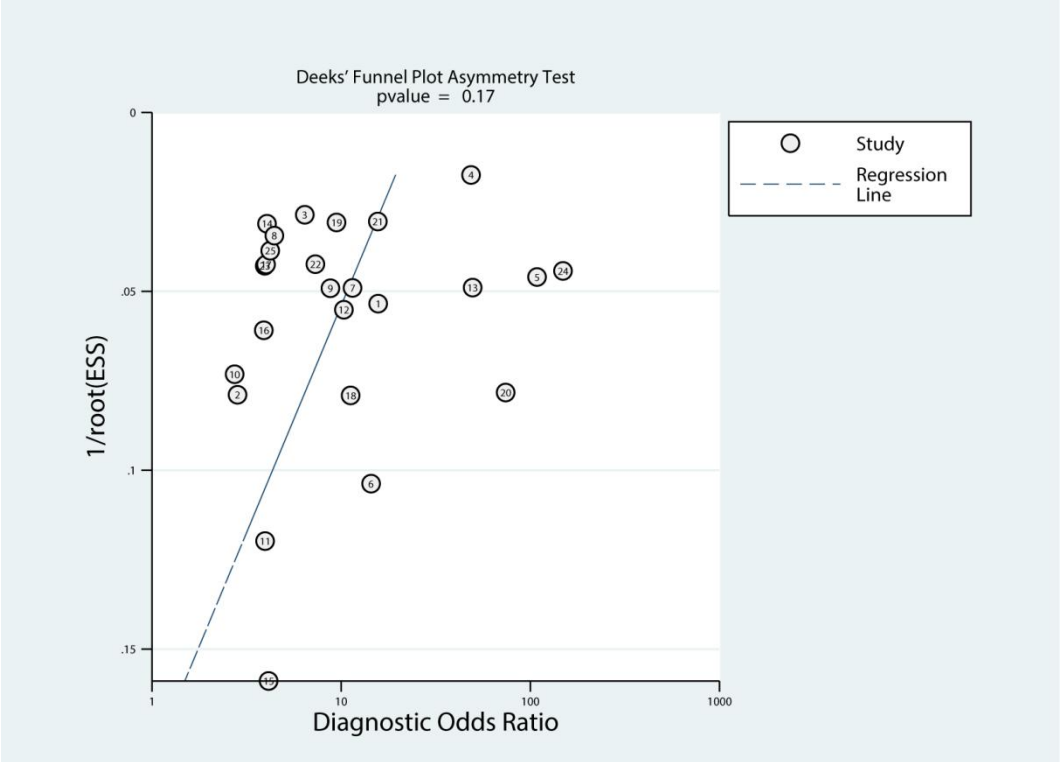


Figure S1. Publication bias for TIMI

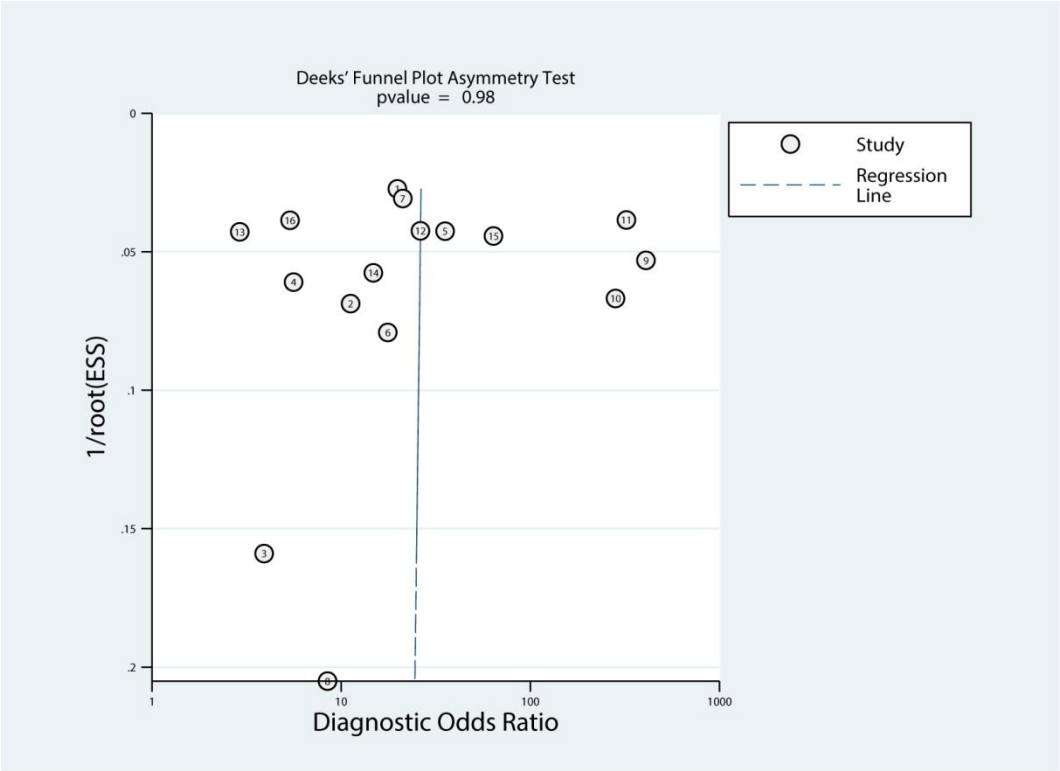


Figure S2. Publication bias for HEART

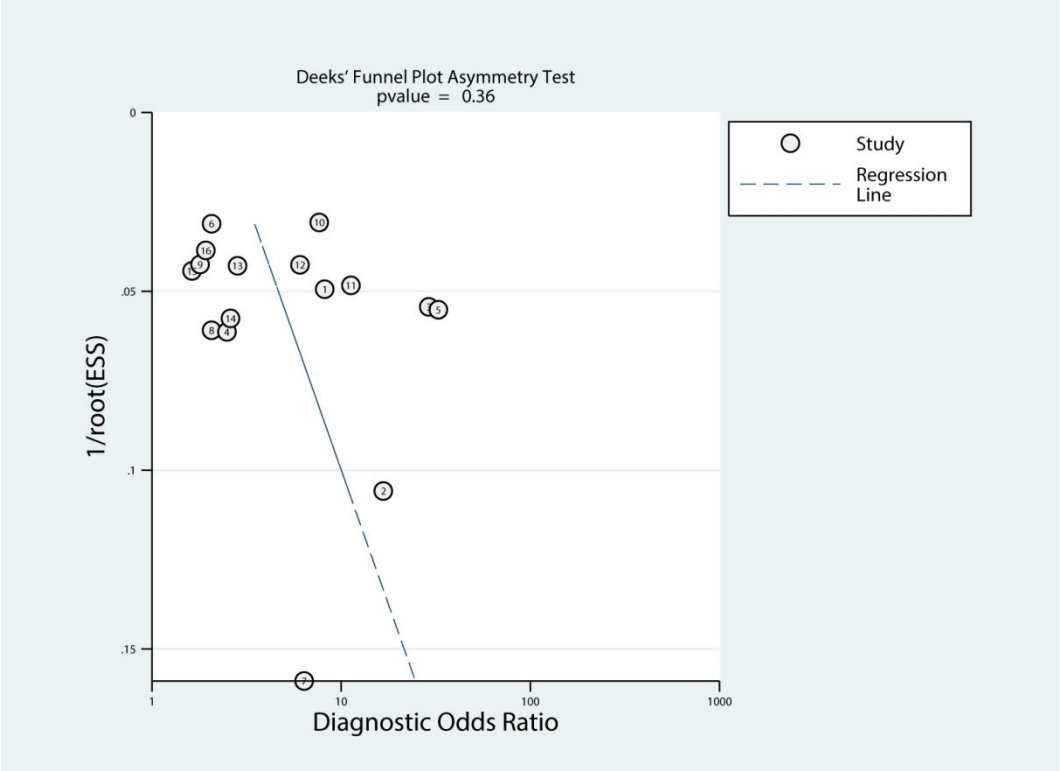


Figure S3. Publication bias for GRACE

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N/A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	9-10
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10-13
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	13
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	14

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Indirect comparison of TIMI, HEART, and GRACE for predicting major cardiovascular events in patients admitted to the emergency department with acute chest pain: A systematic review and meta-analysis

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Primary Subject Heading:	Cardiovascular medicine
Secondary Subject Heading:	Emergency medicine
Keywords:	Adult cardiology < CARDIOLOGY, ACCIDENT & EMERGENCY MEDICINE, Coronary heart disease < CARDIOLOGY

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Indirect comparison of TIMI, HEART, and GRACE for predicting major cardiovascular events in patients admitted to the emergency department with acute chest pain: A systematic review and meta-analysis

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Word count: 3429

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ABSTRACT

Background: The study aimed to compare the predictive values of the Thrombolysis in Myocardial Infarction (TIMI); History, Electrocardiography, Age, Risk factors, and Troponin (HEART); and Global Registry in Acute Coronary Events (GRACE) scoring systems for major adverse cardiovascular events (MACEs) in acute chest pain (ACP) patients admitted to the emergency department (ED).

Methods: We systematically searched PubMed, Embase, and the Cochrane Library from their inception until June 2020; we compared the following parameters: sensitivity, specificity, positive and negative likelihood ratios (PLR and NLR), diagnostic odds ratio (DOR), and area under the receiver operating characteristic curves (AUC).

Results: The pooled sensitivity and specificity for TIMI, HEART, and GRACE were 0.95 and 0.36, 0.96 and 0.50, and 0.78 and 0.56, respectively. The pooled PLR and NLR for TIMI, HEART, and GRACE were 1.49 and 0.13, 1.94 and 0.08, and 1.77 and 0.40, respectively. The pooled DOR for TIMI, HEART, and GRACE was 9.18, 17.92, and 4.00, respectively. The AUC for TIMI, HEART, and GRACE was 0.80, 0.80, and 0.70, respectively. Finally, the results of indirect comparison suggested the superiority of values of TIMI and HEART to those of GRACE for predicting MACEs, while there were no significant differences between TIMI and HEART for predicting MACEs.

Conclusions: TIMI and HEART were superior to GRACE for predicting MACE risk in ACP patients admitted to the ED.

Keywords: TIMI; HEART; GRACE; cardiovascular events; meta-analysis

Strengths and limitations of this study

(1) The analysis is based on prospective studies and used consistent cut-off values.

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(2) The pooled results were stable owing to a large sample size.

(3) The indirect comparisons among TIMI, HEART, and GRACE scoring systems were provided.

(4) The analysis is based on crude data; the predictive values could be affected by covariates.

(5) Substantial heterogeneity was not fully explained.

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INTRODUCTION

Acute chest pain (ACP) is a common symptom accounting for a significant proportion

of attendance and burden in the emergency department (ED) [1]. ACP patients require effective risk stratification to ensure timely initiation of proper treatment in high-risk cases to achieve better prognoses. The early identification of cardiovascular disease (CVD) in ACP patients is important although CVD accounts for only a small proportion of ACP patients with electrocardiograms (ECGs) on presentation [2-4]. Moreover, patients diagnosed with acute coronary syndrome (ACS) should remain hospitalized, whereas non-ACS patients are unnecessarily admitted to hospitals due to the heavy burden on resource constraints [5]. Therefore, accurate risk stratification for ACP patients is essential to improve hospital efficacy by administering timely interventions to high-risk patients, avoiding unnecessary tests, and minimizing admissions for low-risk patients.

Currently, the Thrombolysis in Myocardial Infarction (TIMI); History, ECG, Age, Risk factors, and Troponin (HEART); and Global Registry in Acute Coronary Events (GRACE) scores are widely used for the risk stratification of ACP patients; however, the predictive values using these methods on major cardiovascular events (MACEs) have not been elucidated. The TIMI scoring system was established in 2000 for evaluating patients with unstable angina or non-ST-segment elevation myocardial infarction [6-10]. The HEART score, developed in 2008, aims to improve the accuracy of diagnosing ACS for patients with undifferentiated chest pain [11]. The GRACE score was developed in 2001 for adults with symptoms of ACS; it comprises the following factors: age, vital signs, kidney function, ECG, and troponin levels [12,13]. However, the predictive values of risk stratification measured by the TIMI, HEART, and GRACE scoring systems on MACEs have not been fully compared. Therefore, this study was conducted based on prospective cohort studies to provide comprehensive results regarding the risk stratification assessed by the TIMI, HEART, and GRACE scoring systems on MACEs in ACP patients admitted to the ED. Furthermore, the predictive values of risk stratification assessed by the TIMI, HEART, and GRACE scoring systems on MACEs were compared through an indirect analytic approach.

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Methods

Data Sources, Search Strategy, and Selection Criteria

This study was conducted and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Statement issued in 2009 [14]. Any prospective cohort studies investigating the predictive value of TIMI, HEART, and GRACE on MACEs in ACP patients were eligible for inclusion in this study. No restrictions were placed on publishing language and status. The electronic databases of PubMed, Embase, and the Cochrane library were systematically searched for studies from their inception up to June 2020, and the search strategy was performed using the following terms with Medical Subject Heading and free words: ("TIMI" or "HEART" or "GRACE") and "emergency department" and "chest pain" and ("prospective" or "cohort"). The PubMed search strategy details are summarized in the Supplementary file. The reference lists of retrieved studies were also searched manually to find new eligible studies.

Two authors independently performed the literature search and study selection; any conflicts were resolved through group discussion until a consensus was reached. A study was included if they met the following inclusion criteria: (1) study design: the study had a prospective design; (2) patients: ACP patients admitted to the ED; (3) risk stratifying tools: TIMI, HEART, or GRACE; (4) outcomes: the study had to report the incidence of MACEs and provided clear definitions of MACEs; (5) data abstracted: true and false positives or negatives, or data could transform into the above information must be reported; and (6) cut-off value: the cut-off value of TIMI and HEART was 0-3, and the cut-off value of GRACE was 55-110. Retrospective studies were excluded due to various confounding factors. Additionally, studies that used other cut-off values were excluded.

Data Collection and Quality Assessment

Two authors independently abstracted data items and assessed the quality of the included studies, and any disagreement was settled by an additional author reviewing the original article. The collected information from retrieved studies including the first

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author's name, publication year, country, sample size, age at baseline, percentage of males, risk stratifying tools, patients' status, MACE definition, follow-up duration, and true and false positives/negatives. Study quality was assessed using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2), which is based on patient selection, index test, reference standard, risk of bias, and concerns about applicability [15].

Statistical Analysis

The pooled sensitivity, specificity, positive likelihood ratio (PLR), negative likelihood ratio (NLR), and the area under the receiver operating characteristic curves (AUC) for the risk stratification by TIMI, HEART, and GRACE on MACEs were calculated using a bivariate generalized linear mixed model [16,17], while the pooled diagnostic odds ratio (DOR) was calculated using a random-effects model [18]. The I^2 and Q statistic were used for assessing the heterogeneity across included studies, and P<0.10 was considered as significant heterogeneity [19,20]. The robustness of pooled results was also assessed by sensitivity analyses, which were also performed for studies using all three scoring systems and the endpoint MACE [21]. TIMI-, HEART-, and GRACE-based risk stratification were assessed using an indirect comparison analysis, and the ratios among these scoring systems were calculated [22]. Furthermore, subgroup analyses for the predictive values of TIMI, HEART, and GRACE on subsequent MACE risk were also estimated based on country, mean age, percentage of males, follow-up duration, and study quality. The funnel plots and Deeks' asymmetry tests were used to assess publication bias [23,24]. The inspection level for pooled diagnostic parameters was two-sided, and P<0.05 was considered statistically significant. All statistical analyses were performed using Stata software (version 10.0; Stata Corporation, College Station, TX, USA).

Patient and Public Involvement

There was no patient or public involvement in the design or conduct of this study.

RESULTS

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Literature Search

The details regarding the literature search and study selection of eligible studies are presented in Figure 1. A total of 2,794 articles were identified through the electronic search from PubMed, Embase, and the Cochrane library, and 1,981 were excluded because of term duplications. Subsequently, the remaining 813 studies were selected through title and abstract review; 732 were excluded based on irrelevance. A total of 81 full texts were retrieved for further evaluation, and 48 studies were excluded due to the following reasons: used other cut-off values (n=21), retrospective study design (n=14), and insufficient data (n=13). An additional 135 potential studies identified from the reference lists of retrieved studies were excluded because of duplication with the electronic search. Subsequently, 33 prospective cohort studies that recruited 40,262 ACP patients were selected for final quantitative meta-analysis [13,25-56].

Study Characteristics

The baseline characteristics of included studies are summarized in Table 1. The retrieved studies were published from 2005 to 2020, and 255-4,333 ACP patients were included in each study. Nine studies were conducted in Eastern countries, and the remaining 24 studies were conducted in Western countries. The mean age of enrolled patients ranged from 48.0 to 69.0 years, and the percentage of males ranged from 40.0 to 68.8%. Risk stratification by the TIMI score was available in 25 studies published between 2005 and 2020, 16 studies used the HEART score and were published between 2013 and 2020, and 16 studies employed GRACE and were published between 2007 and 2020. The study quality of the included studies was assessed by QUADAS-2 (Figure 2).

TIMI

The predictive value of risk stratification by the TIMI score on MACEs in ACP patients was available in 25 studies. The pooled sensitivity and specificity of the TIMI score for predicting MACEs were 0.95 (95% CI: 0.91-0.98; I^2 = 98.10%) and 0.36 (95% CI: 0.24-0.50; I^2 = 99.64%), respectively (Supplementary file). Moreover, the pooled PLR and NLR of the TIMI score for predicting MACEs were 1.49 (95% CI: 1.25-1.79; I^2 = 99.21%) and 0.13 (95% CI: 0.07-0.21; I^2 = 95.38%), respectively (Supplementary file).

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The pooled DOR of the TIMI score for predicting MACEs was 9.18 (95% CI: 6.22-13.55; P<0.001) with significant heterogeneity across the included studies (I^2 = 87.6%; P<0.001) (Supplementary file). The AUC of the TIMI score for predicting MACEs was 0.80 (95% CI: 0.76-0.83; Figure 3). No significant publication bias for the TIMI score was detected (P=0.17; Supplementary file).

HEART

The predictive value of risk stratification by the HEART score on MACEs in ACP patients was available in 16 studies. The pooled sensitivity and specificity of the HEART score for predicting MACEs were 0.96 (95% CI: 0.91-0.98; I^2 = 94.87%) and 0.50 (95% CI: 0.41-0.60; I^2 = 98.84%), respectively (Supplementary file). The pooled PLR and NLR of the HEART score for predicting MACEs were 1.94 (95% CI: 1.61-2.35; I^2 = 98.01%) and 0.08 (95% CI: 0.03-0.17; I^2 = 94.65%), respectively (Supplementary file). The pooled DOR for the HEART score was 17.92 (95% CI: 9.40-34.18; P<0.001) with significant heterogeneity across the included studies (I^2 = 88.9%; P<0.001) (Supplementary file). The AUC of the HEART score for predicting MACEs was 0.80 (95% CI: 0.77-0.84; Figure 4). There was no significant publication bias for the HEART score (P=0.98; Supplementary file).

GRACE

The predictive value of risk stratification by the GRACE score on MACEs in ACP patients was available in 16 studies. The pooled sensitivity and specificity of the GRACE score for predicting MACEs were 0.78 (95% CI: 0.64-0.87; I^2 = 96.78%) and 0.56 (95% CI: 0.46-0.66; I^2 = 99.39%), respectively (Supplementary file). The pooled PLR and NLR of the GRACE score for predicting MACEs were 1.77 (95% CI: 1.51-2.08; I^2 = 96.34%) and 0.12 (95% CI: 0.06-0.26; I^2 = 94.07%), respectively (Supplementary file). The DOR of the GRACE score for predicting MACEs was 4.00 (95% CI: 2.78-5.74; P<0.001) with significant heterogeneity across the included studies (I^2 = 88.7%; P<0.001) (Supplementary file). The AUC of the GRACE score for predicting MACEs was 0.70 (95% CI: 0.66-0.74; Figure 5). No significant publication bias for the GRACE score was observed (P=0.36; Supplementary file).

Indirect Comparisons

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Indirect comparisons of the diagnostic parameters (sensitivity, specificity, PLR, NLR, DOR, and AUC) among the TIMI, HEART, and GRACE scoring systems for predicting MACEs are summarized in Table 2. First, the sensitivity of TIMI (ratio: 1.22; 95% CI: 1.04-1.43) and HEART (ratio: 1.23; 95% CI: 1.05-1.44) was significantly higher than that of GRACE for predicting MACEs. Second, the specificity of TIMI was significantly lower than that of GRACE for predicting MACEs (ratio: 0.64; 95% CI: 0.43-0.97). Third, the PLR of TIMI was lower than that of HEART for predicting MACEs (ratio: 0.77; 95% CI: 0.59-1.00). Fourth, TIMI (ratio: 0.32; 95% CI: 0.17-0.64) and HEART (ratio: 0.20; 95% CI: 0.08-0.52) were associated with a lower NLR than GRACE for predicting MACEs. Fifth, the DOR of TIMI (ratio: 2.29; 95% CI: 1.35-3.91) and HEART (ratio: 4.48; 95% CI: 2.14-9.39) was significantly higher than that of GRACE for predicting MACEs. Finally, the AUC of TIMI (ratio: 1.14; 95% CI: 1.06-1.23) and HEART (ratio: 1.14; 95% CI: 1.06-1.23) was significantly higher than that of GRACE for predicting MACEs.

Sensitivity and Subgroup Analysis

The results of sensitivity analyses found that the predictive values of TIMI, HEART, and GRACE for predicting MACEs were stable and unaltered by sequential removal of one study from the overall analysis (data not shown). Sensitivity analyses were also performed after removing studies not using all 3 scoring systems (Table 3). We noted that TIMI had lower sensitivity that that of HEART (ratio: 0.86; 95% CI: 0.75-0.99). Moreover, TIMI had lower NLR (ratio: 0.55; 95% CI: 0.31-0.99), higher DOR (ratio: 2.71; 95% CI: 1.17-6.24), and higher AUC (ratio: 1.19; 95% CI: 1.11-1.27), compared to GRACE. Furthermore, HEART had higher sensitivity (ratio: 1.32; 95% CI: 1.11-1.58), DOR (ratio: 4.36; 95% CI: 1.63-11.64), and AUC (ratio: 1.14; 95% CI: 1.06-1.23) and lower NLR (ratio: 0.22; 95% CI: 0.09-0.56) as compared with GRACE. Additionally, Table 4 presents the subgroup analysis results for the predictive values of TIMI, HEART, and GRACE. First, TIMI has lower specificity than HEART in the pooled studies from Western countries. Furthermore, TIMI versus HEART showed lower PLR in the pooled studies of Western countries (mean age < 60.0 years). Moreover, the DOR of TIMI was lower than of HEART in the pooled studies of Eastern

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countries, while TIMI had a lower AUC than HEART for mean age <60.0 years. Second, TIMI with higher sensitivity than GRACE when pooled studies conducted in Eastern countries, percentage of males ≥60.0%, follow-up duration ≤30.0 days, and studies of moderate quality. Moreover, TIMI versus GRACE showed lower specificity if mean age <60.0 years, percentage of males ≥60.0%, and follow-up duration ≤30.0 days. Furthermore, TIMI has lower PLR than GRACE if percentage of males ≥60.0%, while TIMI has lower NLR than GRACE in the pooled studies of Eastern countries, irrespective of the mean age or percentage male status, follow-up duration ≤30.0 days, and studies of moderate quality. In addition, TIMI versus GRACE showed higher DOR and AUC in most subgroups. Third, HEART versus GRACE showed higher sensitivity in the pooled studies of Eastern countries, mean age ≥60.0 years, follow-up duration ≤30.0 days, and studies of moderate quality. Moreover, HEART has lower NLR than GRACE in most subgroups, except that of mean age <60.0 years. Furthermore, HEART versus GRACE showed higher DOR and AUC in most subgroups.

Discussion

This study was the first meta-analysis to conduct indirect comparisons of the predictive values of risk stratification assessed by the TIMI, HEART, and GRACE scores on MACEs in ACP patients. The current study included a total of 40,262 ACP patients from 33 prospective cohort studies and across a wide range of patient characteristics. The findings of this study suggest that the predictive values of TIMI, HEART, and GRACE scoring systems were better for MACEs in ACP patients admitted to the ED. Moreover, an indirect analysis indicated that the predictive value of TIMI and HEART was superior to that of GRACE for predicting MACEs, while there were no significant differences between TIMI and HEART for predicting MACEs. The results of sensitivity analyses for studies using all 3 scoring systems were consistent with those of the overall analysis. Meanwhile, we noted that the sensitivity of TIMI was lower than HEART for predicting MACEs.

Several systematic reviews and meta-analyses have illustrated the predictive values of

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the TIMI, HEART, and GRACE scoring systems on MACEs in ACP patients [57-59]. Hess et al. included 8 prospective studies and found that the TIMI score provided effective risk stratification for predicting MACEs in potential ACS patients, whereas it should not be used as the sole means for determining patient disposition [57]. Van Den Berg et al. identified 2 prospective and 10 retrospective cohort studies and suggested that the HEART score could be used to identify MACEs in patients with a suspected diagnosis of ACS [58]. Roche et al. included 11 studies and found that using 100 as the cut-off value of the GRACE score could predict the discharge of nearly 70% of presentations, while the predictive value for subsequent MACE risk was not obtained [59]. However, the above studies only reported the diagnostic value of a single scoring system for predicting MACEs in ACS patients. Therefore, we performed the current meta-analysis of prospective studies to evaluate the predictive values of the TIMI, HEART, and GRACE scoring systems on the risk of MACEs in ACP patients and systematically compared the predictive values among them.

The predictive value of the TIMI score for MACEs in ACP patients was statistically significant, whereas several studies reported inconsistent results. Sanchis et al. found that the TIMI score was not associated with the risk of MACEs when 0 was used as the cut-off value [26]. Graham et al. found that a low TIMI score could not rule out cardiac causes of chest pain [35]. Holly et al. suggested that the TIMI score was not associated with the risk of MACE at 30 days when 0 was used as the cut-off value [36]. Leung et al. indicated that a modified TIMI score of 0 could not rule out 30-day MACEs in ACP patients admitted to the ED [45]. The potential reasons for this could be that the TIMI score was designed for risk stratification in patients with non-ST-segment elevation ACS, which is mainly based on appropriate ECG changes or elevations of biomarkers of necrosis. Moreover, the presentation characteristics in ACP patients were not entered into the TIMI score. Finally, the prevalence of MACEs during follow-up in these studies was lower than expected, resulting in broad 95% CIs, i.e., no statistically significant difference [26,35,36,45].

The predictive value of the HEART score for predicting MACEs in ACP patients was statistically significant. Nearly all included studies reported a similar conclusion,

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whereas the study conducted by McCord et al. suggested that the HEART score after 4 hours of the presentation was associated with marginal predictive values for the risk of MACEs [47]. The potential reason for this may be that this study used a modified HEART score, and the original HEART score only considered the initial cTn value, without taking serial sampling into account, which is associated with a lower prevalence of MACEs [60].

The predictive value of the GRACE score for predicting MACE in ACP patients was statistically significant, and all included studies reported similar conclusions for the predictive value of the GRACE score on the risk of MACEs. Especially, the GRACE score was initially designed for post-ACS risk stratification, including unstable angina and non-ST-elevation ACS [11]. The America Heart Association suggested the use of the GRACE score for admission and discharge of ACS patients. Moreover, the risk assessment for patients evaluated outside the hospital should be recommended to use GRACE [61]. Therefore, the predictive value of the GRACE score in low-risk individuals was restricted, which should be addressed in clinical practice.

We noted that the predictive value of the GRACE score was inferior to that of TIMI and HEART scores. Sensitivity analyses were performed for studies reporting all 3 scoring systems, which included 9 studies [42-44,46,48,52,53,55,56], and the results indicated TIMI and HEART having higher predictive values for MACEs than GRACE. The results of the sensitivity analyses were more reliable owing to the analysis of three scoring systems based on direct comparisons. Moreover, the GRACE score was initially developed for ACS patients but not for ACP patients, and the potential risk factors for the progression of MACEs were not considered in the GRACE score. Interestingly, we noted no significant difference between TIMI and HEART for predicting MACEs, while several studies reported that the predictive value of HEART for MACEs was superior to that of TIMI [44,46,48,52,53,55,56]. Subgroup analysis found TIMI with lower AUC compared with HEART if the mean age of patients was <60.0 years. The potential explanation could be the use of HEART score in the absence of exact definitions for medical history across included studies [11], and the predictive value of HEART was more suitable in low-risk individuals.

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Three strengths of this quantitative meta-analysis should be highlighted: (1) the study was based on prospective cohort studies and used relatively uniform cut-off values, which were associated with lower selective and informative biases; (2) the analysis of this study was based on a large sample size, and the findings in our study were more robust than any individual study; and (3) the predictive values of the TIMI, HEART, and GRACE scoring systems on the risk of MACEs in ACP patients were compared through an indirect analytic approach.

Despite the abovementioned findings, the predictive values of the TIMI, HEART, and GRACE could be affected by the definitions of MACEs and the ranges from a single endpoint (death or myocardial infarction) to a composite endpoint. Therefore, MACE definition could affect the predictive value and follow-up durations owing to these factors that attribute to the weight of pooled conclusion. Moreover, this meta-analysis was based on crude data, and the adjusted results were not available. Furthermore, substantial heterogeneity was detected across the included studies, and the heterogeneity was not fully explained by sensitivity and subgroup analyses. In addition, the analysis was conducted on published articles, which causes inevitable publication biases. Finally, the current study was based on indirect comparisons between the predictive values of the TIMI, HEART, and GRACE scoring systems as direct comparisons were not available.

The findings of this study indicated that risk stratification assessed by the TIMI, HEART, and GRACE scores provides relatively appropriate predictive values for MACEs in ACP patients. The results of indirect comparison analysis indicated that TIMI and HEART had relatively relative better predictive values than GRACE on subsequent MACE risk. Further prospective cohort studies should be conducted to directly compare the predictive values of TIMI, HEART, and GRACE on MACEs in ACP patients.

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Authors' Contributions

JK: study design and manuscript drafting

YC: study supervision, data analysis, and result interpretation

XW: literature search and data collection

ZW: data curation

FC: manuscript revising and final approval

Data Availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Competing Interests

The authors declare no competing interests.

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None.

Ethics Approval

Neither ethics approval nor participant consent was required as this study was based solely on the summary results of previously published articles. Individual patient data were not obtained or accessed.

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Figure Legends

Figure 1. Flow diagram of the literature search and study selection process.

Figure 2. QUADAS-2 scoring of included studies.

Figure 3. The summary receiver operating characteristic curve of risk stratification assessed by the TIMI score.

Figure 4. The summary receiver operating characteristic curve of risk stratification assessed by the HEART score.

Figure 5. The summary receiver operating characteristic curve of risk stratification assessed by the GRACE score.

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Table 1. Baseline characteristics of studies included in the systematic review and meta-analysis.

Study	Country	Sample size	Mean age (years)	Percentage male (%)	Risk stratifying tools	Patients' status	MACE definition	Follow-up
Tong 2005 [25]	US	957	60.0	52.0	TIMI	Chest pain and a nondiagnostic ECG	Death and MI	30 days
Sanchis 2005 [26]	Spain	646	64.0	65.8	TIMI	Acute chest pain	Death, MI, or urgent revascularization	14 days
Pollack Jr 2006 [27]	US	3929	51.6	40.0	TIMI	Chest pain in the ED	Death, acute MI, and revascularization	30 days
Pelliccia 2006 [28]	Italy	4333	58.4	68.8	TIMI	Acute chest pain	MI	In-hospital
Lyon 2007 [29]	UK	954	60.0	62.0	TIMI, GRACE	Undifferentiated chest pain	MI, cardiac arrest, revascularization, unstable angina with myocardial damage and death	30 days
Ramsay 2007 [13]	UK	347	65.2	62.3	TIMI, GRACE	Suspected cardiac pain	Death, non-fatal MI, and emergency revascularization	3.0 months
Body 2009 [30]	UK	796	58.9	60.4	TIMI	Chest pain in the ED	Death, acute MI or urgent coronary revascularization	30 days
Campbell 2009 [31]	US	3169	53.6	45.0	TIMI	Chest pain in the ED	Death, MI, or revascularization	30 days
Hess 2010 [32]	Canada	1017	59.3	60.6	TIMI	ED patients with chest pain and possible ACS	Acute MI, revascularization, or death	30 days
Stracke 2010 [33]	Germany	1014	66.0	55.0	GRACE	Chest pain in the ED	Death	In-hospital

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Van der Zee 2011 [34]	The Netherlands	524	57.7	60.5	GRACE	Chest pain in the ED	Death	9.4 years
Graham 2013 [35]	China	315	69.0	54.9	TIMI	Chest pain in the ED	Death, MI, troponin positive ACS and PCI	30 days
Holly 2013 [36]	US	552	54.1	46.0	TIMI	Chest pain in the ED	MI, revascularization, or death	30 days
Backus 2013 [37]	The Netherlands	2388	60.6	57.5	HEART	Chest pain in the ED	Acute MI, PCI, CABG, coronary angiography revealing procedurally correctable stenosis managed conservatively, and death	6 weeks
Cullen 2013 [38]	Australia	948	54.0	59.9	TIMI, and GRACE	Chest pain in the ED	Cardiac death, acute MI, and unstable angina	30 days
Graham 2014 [39]	China	925	68.0	51.7	TIMI	Chest pain in the ED	Death, readmission with MI, ACS not diagnosed at initial ED presentation and coronary revascularization	30 days
Visser 2015 [40]	The Netherlands	255	64.0	56.0	HEART	Chest pain in the ED	MI, or PCI, or CABG, or coronary angiography revealing significant stenosis or death	6 weeks
Boubaker 2015 [41]	Tunisia	3125	57.7	58.3	TIMI, GRACE	Chest pain in the ED	All-cause mortality, acute coronary syndrome, and coronary non-ED planned revascularization	30 days
Wang 2016 [42]	China	986	54.0	55.0	TIMI, HEART, GRACE	Chest pain in the ED	Death, MI, and/or the need for revascularization by CABG or PCI	6 months
Chen 2016 [43]	China	833	65.1	55.3	TIMI, HEART, GRACE	Chest pain in the ED	Coronary revascularization, ventricular arrhythmia needing intervention and	30 days

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							high-degree atrioventricular block needing intervention	
Sakamoto 2016 [44]	Singapore	604	60.8	69.2	TIMI, HEART, GRACE	Chest pain in the ED	Death, acute MI, PCI, CABG	30 days
Leung 2017 [45]	China	602	66.0	48.8	TIMI, HEART	Chest pain in the ED	Death, cardiac arrest, MI, and cardiogenic shock	30 days
Poldervaart 2017 [46]	The Netherlands	1748	62.0	54.0	TIMI, HEART, GRACE	Chest pain in the ED	UA, MI, PCI, CABG, stenosis managed conservatively, and death	3.0 months
McCord 2017 [47]	Europe, Australia, and the US	661	58.3	58.2	HEART	Chest pain in the ED	Death or acute MI	30 days
Reaney 2018 [48]	UK	1000	62.5	57.6	TIMI, HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, cardiac death, cardiogenic shock, and life-threatening arrhythmias requiring emergency intervention.	30 days
Greenslade 2018 [49]	Australia	1760	60.4	59.3	TIMI	Chest pain in the ED	Cardiac death, cardiac arrest, cardiogenic shock, acute MI, UA, emergency or urgent revascularization, high-level atrioventricular block, and ventricular arrhythmias	30 days
Moumneh 2018 [50]	France	641	53.3	53.4	HEART	Non-traumatic chest pain patients	MI, coronary angioplasty, coronary bypass, and sudden unexplained death	6 weeks
Ishak 2018 [51]	The Netherlands	1127	63.8	57.7	HEART	Chest pain in the ED	Acute MI, PCI, CABG, or death	30 days
Wong 2018 [52]	China	1081	48.0	52.3	TIMI, HEART, GRACE	Undifferentiated chest pain	Acute MI, PCI, CABG, and death	30 days

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Al-Zaiti 2019 [53]	USA	750	59.0	58.0	TIMI, HEART, GRACE	Chest pain in the ED	Resuscitated or un-resuscitated sudden cardiac arrest, all-cause death, post-discharge re-infarction requiring cardiac revascularization	30 days
Huang 2020 [54]	China	509	59.8	53.4	HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, cardiac death, cardiogenic shock, or life-threatening arrhythmias requiring emergency intervention or resulting in mortality	30 days
Torralba 2020 [55]	Colombia	519	64.3	56.1	TIMI, HEART, GRACE	Chest pain in the ED	Death from any cause, MI, and surgical or percutaneous myocardial revascularization	30 days
Shin 2020 [56]	Korea	1247	62.0	60.8	TIMI, HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, or death from cardiac causes	30 days

*ACS: acute coronary syndrome; CABG: coronary artery bypass grafting; ECG: electrocardiogram; ED: emergency department; MI: myocardial infarction; PCI: percutaneous coronary intervention; UA: unstable angina.

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Table 2. Systematic comparisons of the sensitivity, specificity, positive and negative likelihood ratio, diagnostic odds ratio, and the area under the receiver operating characteristic curve of risk stratifying measured by TIMI, HEART, and GRACE.

Parameters	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity and 95%CI	0.95 (0.91-0.98)	0.96 (0.91-0.98)	0.78 (0.64-0.87)	0.99 (0.94-1.04)	1.22 (1.04-1.43)	1.23 (1.05-1.44)
Specificity and 95%CI	0.36 (0.24-0.50)	0.50 (0.41-0.60)	0.56 (0.46-0.66)	0.72 (0.48-1.09)	0.64 (0.43-0.97)	0.89 (0.69-1.16)
PLR and 95%CI	1.49 (1.25-1.79)	1.94 (1.61-2.35)	1.77 (1.51-2.08)	0.77 (0.59-1.00)	0.84 (0.66-1.07)	1.10 (0.86-1.40)
NLR and 95%CI	0.13 (0.07-0.21)	0.08 (0.03-0.17)	0.40 (0.27-0.59)	0.46 (0.15-1.39)	0.32 (0.17-0.64)	0.20 (0.08-0.52)
DOR and 95%CI	9.18 (6.22-13.55)	17.92 (9.40-34.18)	4.00 (2.78-5.74)	0.51 (0.24-1.09)	2.29 (1.35-3.91)	4.48 (2.14-9.39)
AUC and 95%CI	0.80 (0.76-0.83)	0.80 (0.77-0.84)	0.70 (0.66-0.74)	1.00 (0.94-1.06)	1.14 (1.06-1.23)	1.14 (1.06-1.23)

PLR: positive likelihood ratio; NLR: negative likelihood ratio; DOR: diagnostic odds ratio; AUC: the area under the receiver operating characteristic curve.

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Table 3. Sensitivity analysis for direct comparisons of the sensitivity, specificity, positive and negative likelihood ratio, diagnostic odds ratio, and the area under the receiver operating characteristic curve of risk stratifying measured by TIMI, HEART, and GRACE.

Parameters	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity and 95%CI	0.81 (0.70-0.89)	0.94 (0.84-0.98)	0.71 (0.58-0.81)	0.86 (0.75-0.99)	1.14 (0.94-1.39)	1.32 (1.11-1.58)
Specificity and 95%CI	0.70 (0.54-0.83)	0.53 (0.39-0.66)	0.59 (0.44-0.73)	1.32 (0.94-1.86)	1.19 (0.85-1.65)	0.90 (0.62-1.29)
PLR and 95%CI	2.72 (1.67-4.41)	1.99 (1.51-2.63)	1.74 (1.33-2.28)	1.37 (0.78-2.39)	1.56 (0.90-2.72)	1.14 (0.78-1.68)
NLR and 95%CI	0.27 (0.16-0.44)	0.11 (0.05-0.28)	0.49 (0.36-0.66)	2.45 (0.90-6.66)	0.55 (0.31-0.99)	0.22 (0.09-0.56)
DOR and 95%CI	9.45 (4.77-18.73)	15.21 (6.45-35.88)	3.49 (2.16-5.63)	0.62 (0.21-1.86)	2.71 (1.17-6.24)	4.36 (1.63-11.64)
AUC and 95%CI	0.83 (0.80-0.86)	0.80 (0.76-0.83)	0.70 (0.66-0.74)	1.04 (0.98-1.10)	1.19 (1.11-1.27)	1.14 (1.06-1.23)

PLR: positive likelihood ratio; NLR: negative likelihood ratio; DOR: diagnostic odds ratio; AUC: the area under the receiver operating characteristic curve.

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Table 4. Subgroup analyses.

Parameters	Factors	Groups	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity	Country	Western	0.97 (0.93-0.99)	0.97 (0.93-0.99)	0.88 (0.72-0.95)	1.00 (0.96-1.05)	1.10 (0.96-1.27)	1.10 (0.96-1.27)
		Eastern	0.86 (0.69-0.95)	0.94 (0.79-0.98)	0.60 (0.49-0.70)	0.91 (0.75-1.11)	1.43 (1.13-1.82)	1.57 (1.27-1.93)
	Mean age (years)	≥ 60.0	0.97 (0.92-0.99)	0.97 (0.94-0.99)	0.83 (0.67-0.92)	1.00 (0.96-1.05)	1.17 (0.99-1.38)	1.17 (1.00-1.37)
		< 60.0	0.93 (0.81-0.97)	0.91 (0.70-0.98)	0.68 (0.44-0.85)	1.02 (0.84-1.24)	1.37 (0.97-1.92)	1.34 (0.92-1.94)
	Percentage male (%)	≥ 60.0	0.98 (0.92-0.99)	-	0.73 (0.50-0.88)	-	1.34 (1.01-1.79)	-
		< 60.0	0.92 (0.85-0.96)	0.96 (0.91-0.99)	0.80 (0.62-0.90)	0.96 (0.89-1.03)	1.15 (0.95-1.40)	1.20 (0.99-1.45)
	Follow-up duration	≤ 30.0 days	0.96 (0.91-0.98)	0.97 (0.92-0.99)	0.76 (0.61-0.87)	0.99 (0.94-1.04)	1.26 (1.05-1.51)	1.28 (1.06-1.53)
		> 30.0 days	-	0.93 (0.78-0.98)	-	-	-	-
	Study quality	High	0.94 (0.83-0.98)	0.97 (0.91-0.99)	0.77 (0.50-0.92)	0.97 (0.88-1.06)	1.22 (0.89-1.67)	1.26 (0.93-1.71)
		Moderate	0.96 (0.90-0.98)	0.95 (0.85-0.98)	0.78 (0.61-0.89)	1.01 (0.93-1.10)	1.23 (1.01-1.49)	1.22 (1.00-1.49)
Specificity	Country	Western	0.33 (0.20-0.50)	0.56 (0.47-0.64)	0.50 (0.34-0.66)	0.59 (0.36-0.96)	0.66 (0.37-1.16)	1.12 (0.78-1.61)
		Eastern	0.46 (0.27-0.67)	0.44 (0.27-0.62)	0.64 (0.55-0.73)	1.05 (0.56-1.94)	0.72 (0.45-1.16)	0.69 (0.44-1.07)
	Mean age	≥ 60.0	0.35 (0.19-0.55)	0.45 (0.34-0.56)	0.51 (0.37-0.64)	0.78 (0.43-1.40)	0.69 (0.38-1.25)	0.88 (0.61-1.28)

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	(years)	< 60.0	0.38 (0.23-0.56)	0.60 (0.44-0.74)	0.63 (0.49-0.76)	0.63 (0.38-1.06)	0.60 (0.37-0.99)	0.95 (0.68-1.34)
	Percentage male (%)	≥ 60.0	0.21 (0.10-0.37)	-	0.56 (0.45-0.66)	-	0.38 (0.19-0.74)	-
		< 60.0	0.45 (0.30-0.61)	0.52 (0.42-0.63)	0.56 (0.42-0.70)	0.87 (0.58-1.30)	0.80 (0.52-1.24)	0.93 (0.67-1.29)
	Follow-up duration	≤ 30.0 days	0.35 (0.23-0.50)	0.46 (0.35-0.57)	0.56 (0.45-0.67)	0.76 (0.48-1.20)	0.63 (0.40-0.97)	0.82 (0.60-1.13)
		> 30.0 days	-	0.60 (0.43-0.75)	-	-	-	-
	Study quality	High	0.41 (0.24-0.62)	0.47 (0.35-0.60)	0.61 (0.36-0.81)	0.87 (0.51-1.51)	0.67 (0.36-1.25)	0.77 (0.47-1.25)
		Moderate	0.33 (0.19-0.51)	0.54 (0.39-0.68)	0.54 (0.44-0.64)	0.61 (0.35-1.08)	0.61 (0.36-1.04)	1.00 (0.72-1.40)
PLR	Country	Western	1.45 (1.16-1.80)	2.20 (1.79-2.70)	1.75 (1.37-2.23)	0.66 (0.49-0.89)	0.83 (0.60-1.15)	1.26 (0.91-1.73)
		Eastern	1.61 (1.24-2.08)	1.66 (1.30-2.13)	1.67 (1.37-2.03)	0.97 (0.68-1.39)	0.96 (0.70-1.33)	0.99 (0.72-1.36)
	Mean age (years)	≥ 60.0	1.49 (1.13-1.96)	1.76 (1.44-2.16)	1.67 (1.34-2.09)	0.85 (0.60-1.19)	0.89 (0.63-1.27)	1.05 (0.78-1.42)
		< 60.0	1.50 (1.23-1.83)	2.26 (1.61-3.18)	1.86 (1.61-2.15)	0.66 (0.45-0.98)	0.81 (0.63-1.03)	1.22 (0.84-1.02)
	Percentage male (%)	≥ 60.0	1.23 (1.06-1.43)	-	1.67 (1.44-1.93)	-	0.74 (0.60-0.91)	-
		< 60.0	1.68 (1.29-2.19)	2.02 (1.64-2.50)	1.81 (1.45-2.28)	0.83 (0.59-1.17)	0.93 (0.66-1.31)	1.12 (0.82-1.52)
	Follow-up duration	≤ 30.0 days	1.47 (1.22-1.79)	1.79 (1.47-2.19)	1.74 (1.45-2.09)	0.82 (0.62-1.08)	0.84 (0.65-1.10)	1.03 (0.78-1.35)
		> 30.0 days	-	2.34 (1.63-3.35)	-	-	-	-
	Study	High	1.61 (1.20-2.17)	1.84 (1.45-2.34)	1.97 (1.32-2.95)	0.88 (0.60-1.28)	0.82 (0.50-1.35)	0.93 (0.58-1.49)

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	quality	Moderate	1.44 (1.15-1.79)	2.04 (1.53-2.72)	1.70 (1.46-1.97)	0.71 (0.49-1.01)	0.85 (0.65-1.11)	1.20 (0.87-1.66)
NLR	Country	Western	0.10 (0.05-0.18)	0.05 (0.02-0.14)	0.25 (0.13-0.49)	2.00 (0.62-6.41)	0.40 (0.16-1.01)	0.20 (0.06-0.65)
		Eastern	0.30 (0.18-0.48)	0.14 (0.06-0.37)	0.63 (0.51-0.77)	2.14 (0.76-6.02)	0.48 (0.28-0.81)	0.22 (0.09-0.56)
	Mean age (years)	≥ 60.0	0.08 (0.03-0.20)	0.06 (0.02-0.14)	0.34 (0.19-0.60)	1.33 (0.34-5.19)	0.24 (0.08-0.71)	0.18 (0.06-0.55)
		< 60.0	0.19 (0.11-0.35)	0.15 (0.04-0.51)	0.50 (0.31-0.81)	1.27 (0.31-5.13)	0.38 (0.18-0.81)	0.30 (0.08-1.17)
	Percentage male (%)	≥ 60.0	0.11 (0.05-0.24)	-	0.48 (0.27-0.87)	-	0.23 (0.09-0.61)	-
		< 60.0	0.17 (0.10-0.28)	0.07 (0.03-0.17)	0.36 (0.22-0.60)	2.43 (0.89-6.66)	0.47 (0.23-0.97)	0.19 (0.07-0.53)
	Follow-up duration	≤ 30.0 days	0.12 (0.07-0.22)	0.07 (0.02-0.18)	0.42 (0.27-0.65)	1.71 (0.50-5.92)	0.29 (0.14-0.59)	0.17 (0.05-0.54)
		> 30.0 days	-	0.12 (0.04-0.35)	-	-	-	-
	Study quality	High	0.13 (0.05-0.33)	0.06 (0.02-0.20)	0.38 (0.20-0.71)	2.17 (0.49-9.60)	0.34 (0.11-1.07)	0.16 (0.04-0.59)
		Moderate	0.12 (0.06-0.24)	0.10 (0.04-0.26)	0.40 (0.24-0.69)	1.20 (0.37-3.85)	0.30 (0.13-0.72)	0.25 (0.09-0.73)
DOR	Country	Western	12.68 (7.19-22.38)	30.41 (10.41-88.82)	6.32 (3.35-11.92)	0.42 (0.12-1.40)	2.01 (0.86-4.70)	4.81 (1.38-16.72)
		Eastern	4.56 (3.67-5.65)	10.05 (5.30-19.06)	2.59 (1.83-3.67)	0.45 (0.23-0.89)	1.76 (1.17-2.65)	3.88 (1.87-8.04)
	Mean age (years)	≥ 60.0	12.86 (6.11-27.07)	23.01 (10.84-48.85)	4.55 (2.50-8.28)	0.56 (0.19-1.61)	2.83 (1.09-7.35)	5.06 (1.93-13.23)
		< 60.0	6.24 (4.46-8.75)	11.52 (3.44-38.55)	3.41 (2.25-5.16)	0.54 (0.15-1.90)	1.83 (1.07-3.12)	3.38 (0.94-12.12)
	Percentage	≥ 60.0	8.06 (4.08-15.91)	12.32 (1.74-87.48)	3.08 (1.76-5.37)	0.65 (0.08-5.20)	2.62 (1.09-6.31)	4.00 (0.52-30.66)

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	male (%)	< 60.0	9.16 (5.69-14.75)	19.66 (9.31-41.51)	4.51 (2.82-7.22)	0.47 (0.19-1.13)	2.03 (1.04-3.97)	4.36 (1.80-10.54)
	Follow-up duration	≤ 30.0 days	9.45 (6.19-14.41)	19.21 (8.02-45.99)	3.70 (2.46-5.55)	0.49 (0.19-1.30)	2.55 (1.42-4.59)	5.19 (1.98-13.61)
		> 30.0 days	8.42 (5.14-13.80)	15.90 (8.10-31.21)	5.53 (2.34-13.03)	0.53 (0.23-1.22)	1.52 (0.57-4.10)	2.88 (0.96-8.57)
	Study quality	High	7.96 (4.90-12.91)	21.21 (7.10-63.34)	5.17 (2.51-10.64)	0.38 (0.11-1.24)	1.54 (0.65-3.67)	4.10 (1.11-15.22)
		Moderate	9.44 (5.17-17.23)	16.25 (6.88-38.41)	3.33 (2.22-5.01)	0.58 (0.20-1.66)	2.83 (1.37-5.86)	4.88 (1.88-12.63)
AUC	Country	Western	0.85 (0.82-0.88)	0.82 (0.78-0.85)	0.74 (0.70-0.78)	1.04 (0.98-1.10)	1.15 (1.08-1.23)	1.11 (1.03-1.19)
		Eastern	0.73 (0.69-0.77)	0.74 (0.70-0.78)	0.66 (0.62-0.70)	0.99 (0.91-1.07)	1.11 (1.02-1.20)	1.12 (1.03-1.22)
	Mean age (years)	≥ 60.0	0.85 (0.82-0.88)	0.83 (0.80-0.86)	0.70 (0.66-0.74)	1.02 (0.97-1.08)	1.21 (1.14-1.30)	1.19 (1.11-1.27)
		< 60.0	0.73 (0.69-0.77)	0.79 (0.75-0.82)	0.70 (0.65-0.73)	0.92 (0.86-0.99)	1.04 (0.96-1.13)	1.13 (1.05-1.21)
	Percentage male (%)	≥ 60.0	0.74 (0.70-0.77)	-	0.65 (0.60-0.69)	-	1.14 (1.05-1.24)	-
		< 60.0	0.82 (0.79-0.85)	0.81 (0.78-0.85)	0.72 (0.68-0.76)	1.01 (0.96-1.07)	1.14 (1.07-1.22)	1.13 (1.05-1.21)
	Follow-up duration	≤ 30.0 days	0.81 (0.77-0.84)	0.79 (0.75-0.82)	0.69 (0.65-0.73)	1.03 (0.96-1.09)	1.17 (1.09-1.26)	1.14 (1.06-1.23)
		> 30.0 days	-	0.83 (0.79-0.86)	-	-	-	-
	Study quality	High	0.79 (0.75-0.83)	0.80 (0.76-0.83)	0.74 (0.70-0.78)	0.99 (0.92-1.06)	1.07 (0.99-1.15)	1.08 (1.01-1.16)
		Moderate	0.80 (0.76-0.83)	0.81 (0.77-0.84)	0.67 (0.62-0.71)	0.99 (0.93-1.05)	1.19 (1.10-1.29)	1.21 (1.05-1.14)

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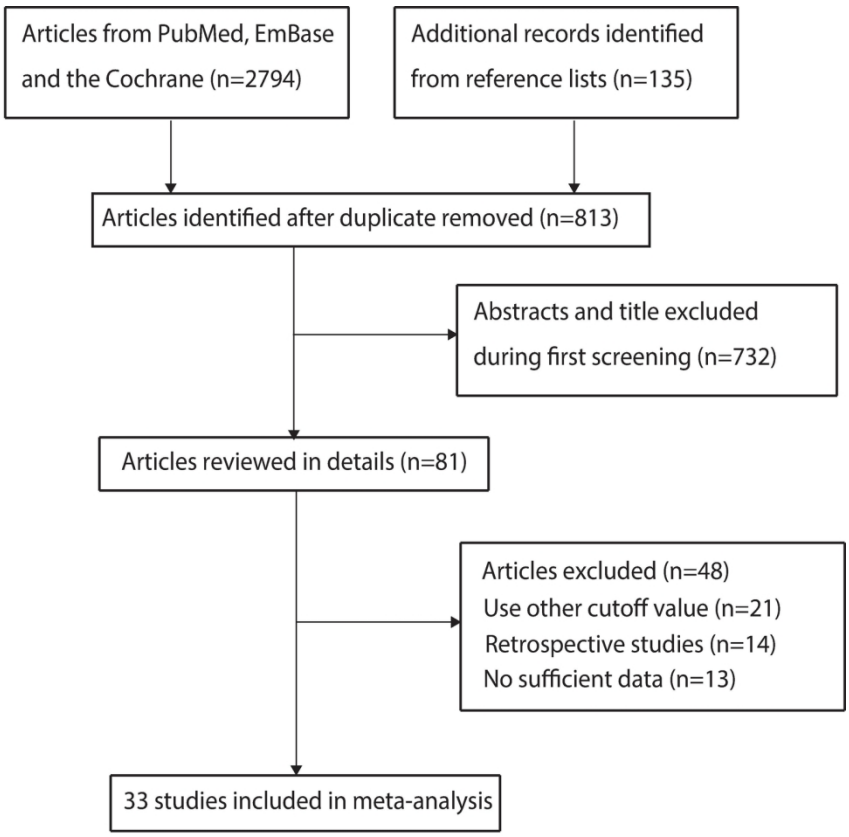


Figure 1. Flow diagram of the literature search and study selection process.

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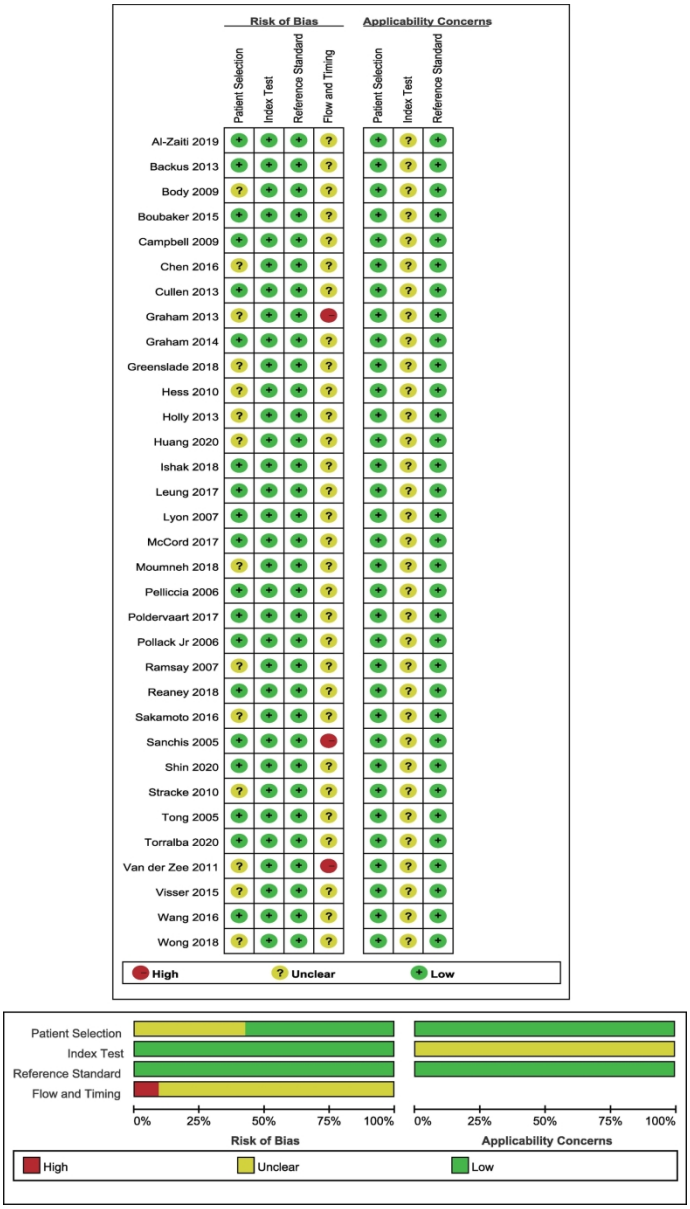


Figure 2. QUADAS-2 scoring of included studies.

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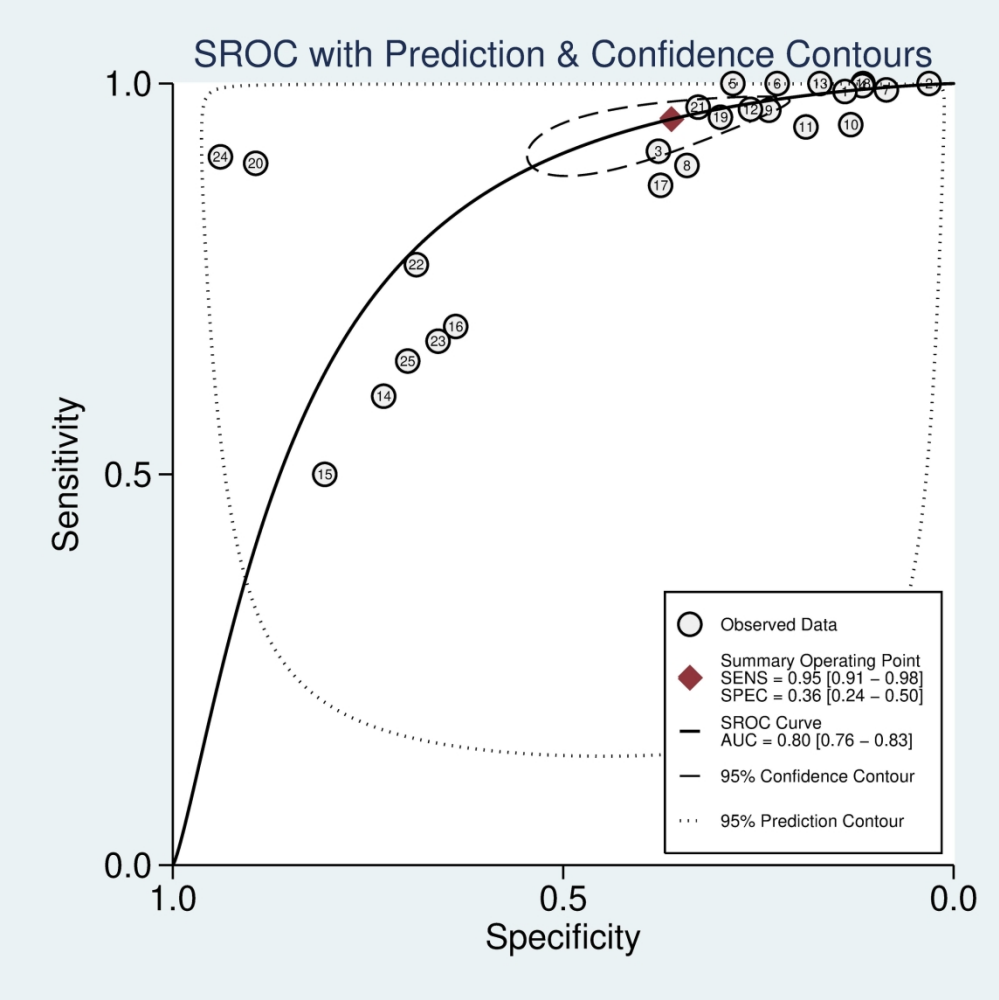


Figure 3. The summary receiver operating characteristic curve of risk stratification assessed by the TIMI score.

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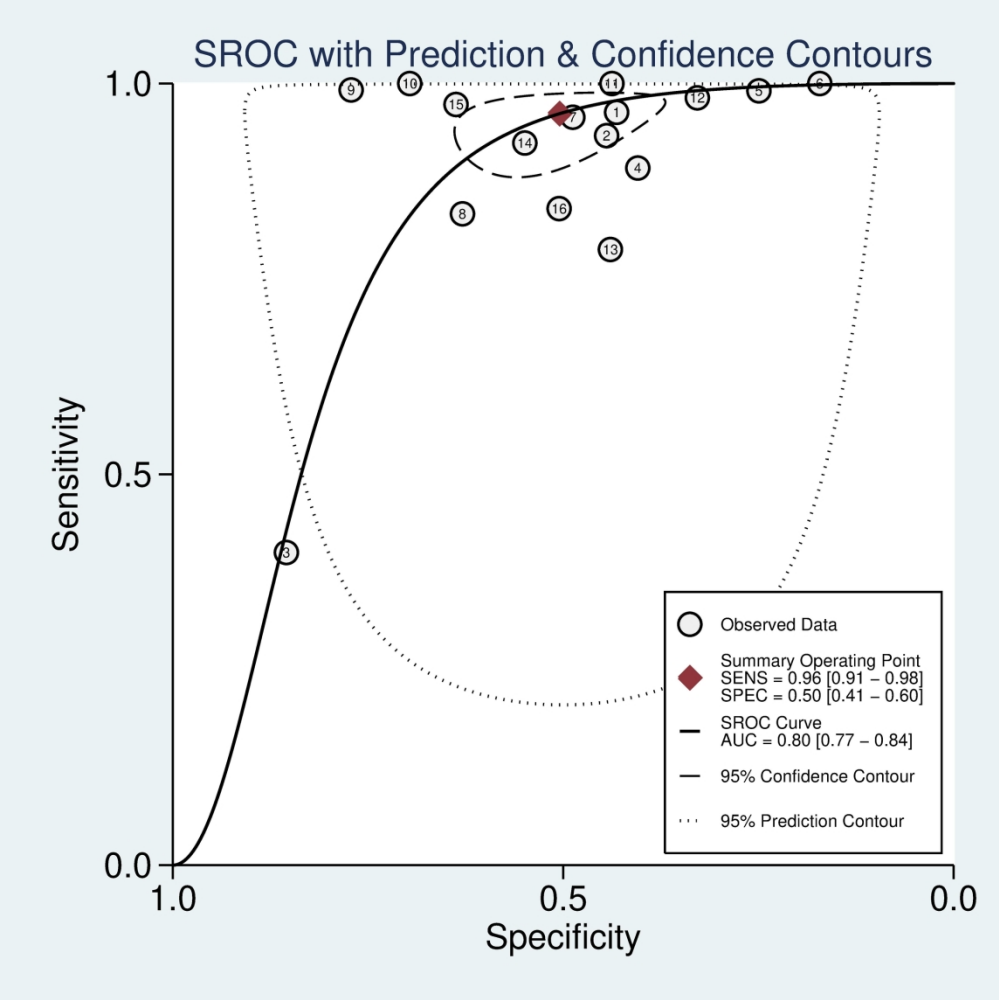


Figure 4. The summary receiver operating characteristic curve of risk stratification assessed by the HEART score.

203x203mm (300 x 300 DPI)

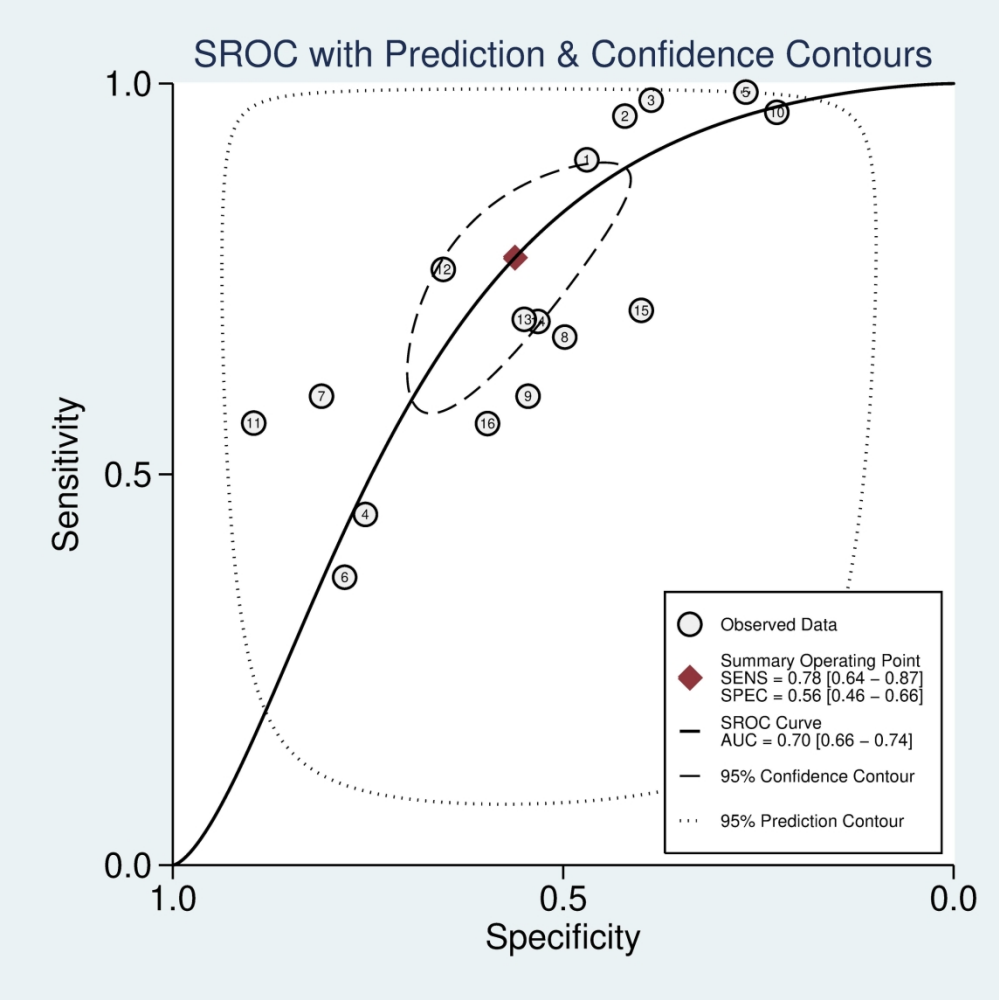


Figure 5. The summary receiver operating characteristic curve of risk stratification assessed by the GRACE score.

203x203mm (300 x 300 DPI)

1. Search strategies in PubMed:

#1 ("Chest Pain"[Mesh] OR "Chest Pain"[Title/Abstract])

#2 ("Emergency Service, Hospital/classification"[Mesh] OR "Emergency Service, Hospital/organization and administration"[Mesh] OR "emergency department"[Title/Abstract])

#3 ("TIMI"[Title/Abstract] OR "HEART"[Title/Abstract] OR "GRACE"[Title/Abstract])

#4 registr*[Title/Abstract] OR Registries[Mesh] OR cohort*[Title/Abstract] OR Follow-Up Stud*[Title/Abstract] OR Follow-Up*[Title/Abstract] OR Longitudinal Stud*[Title/Abstract] OR Prospective*[Title/Abstract] OR Prospective Stud*[Title/Abstract] OR Concurrent Stud*[Title/Abstract] OR Incidence Stud*[Title/Abstract] OR "Cohort Studies"[Mesh]

#5 AND/#1-#4

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2. Sensitivity and Specificity

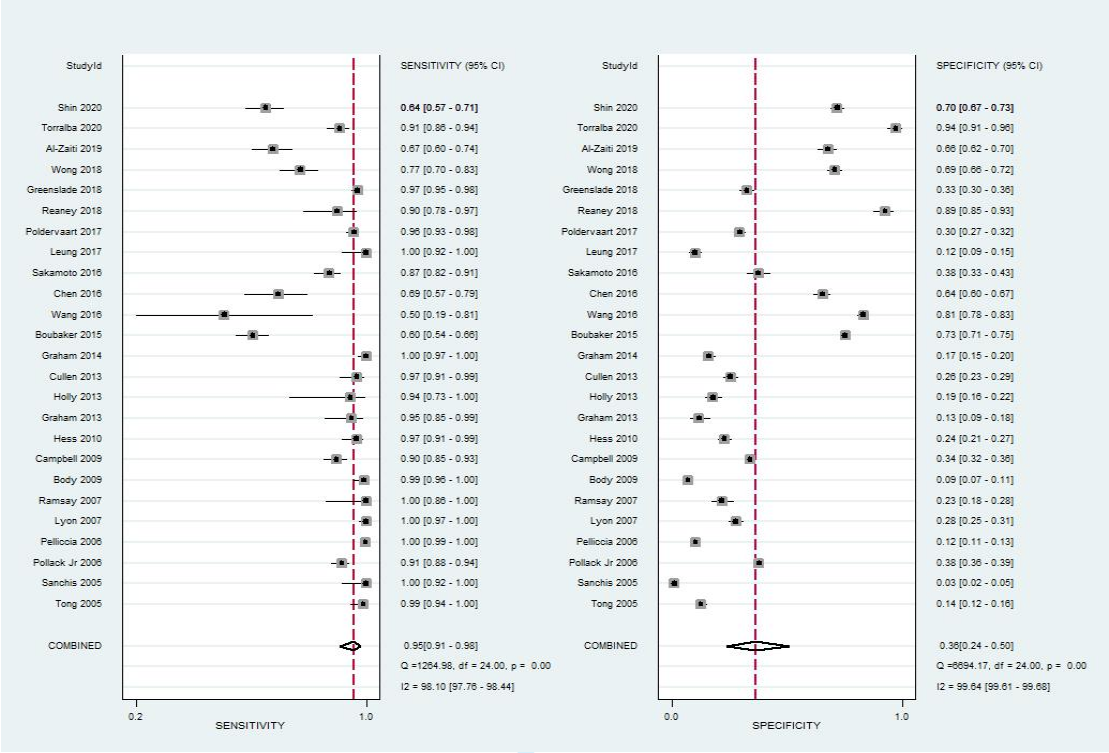


Figure S1. Sensitivity and specificity for TIMI

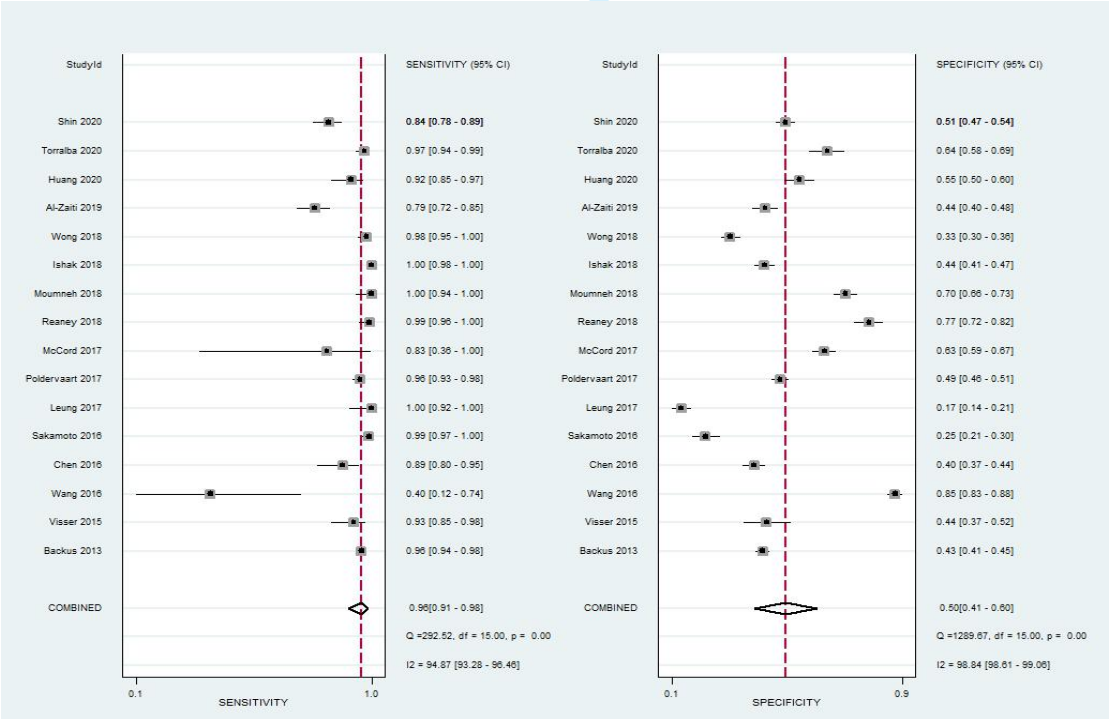


Figure S2. Sensitivity and specificity for HEART

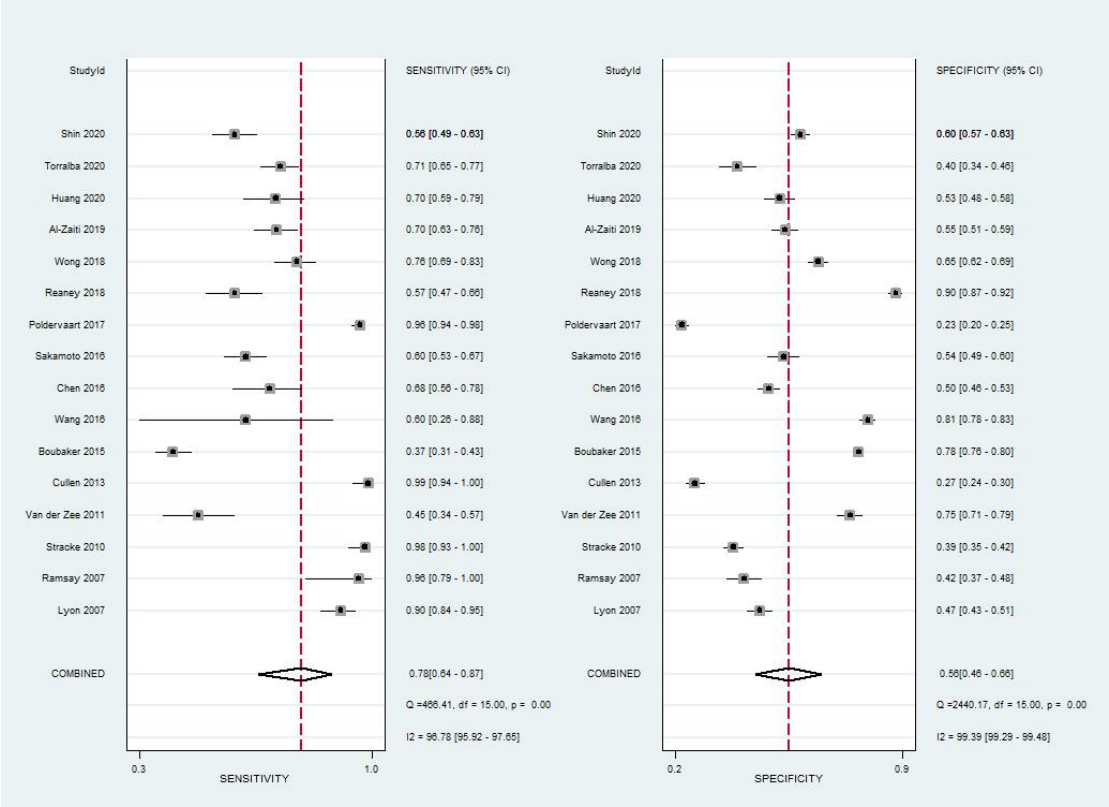


Figure S3. Sensitivity and specificity for GRACE

3. PLR and NLR

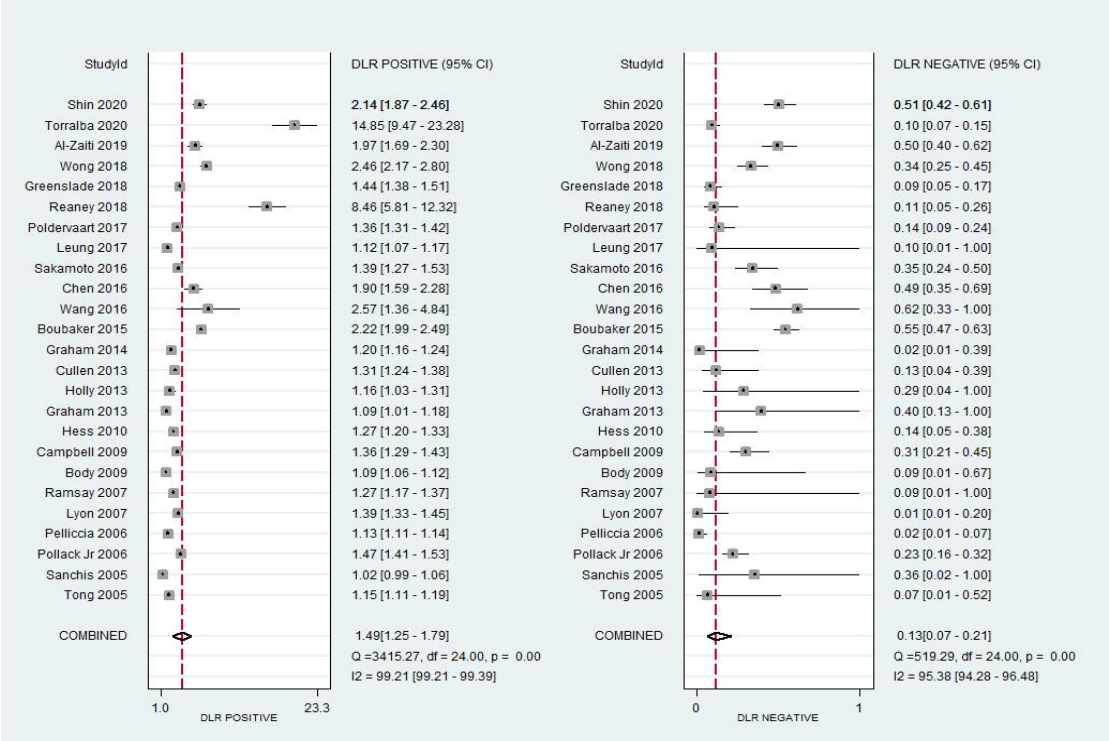


Figure S1. PLR and NLR for TIMI

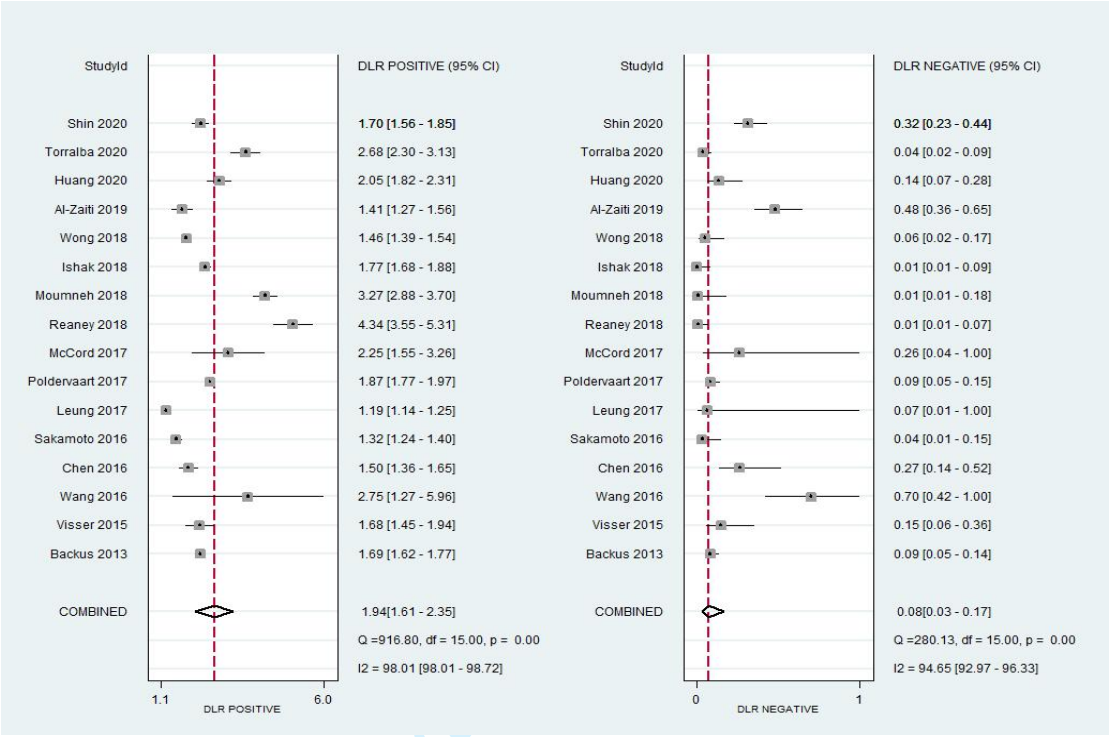


Figure S2. PLR and NLR for HEART

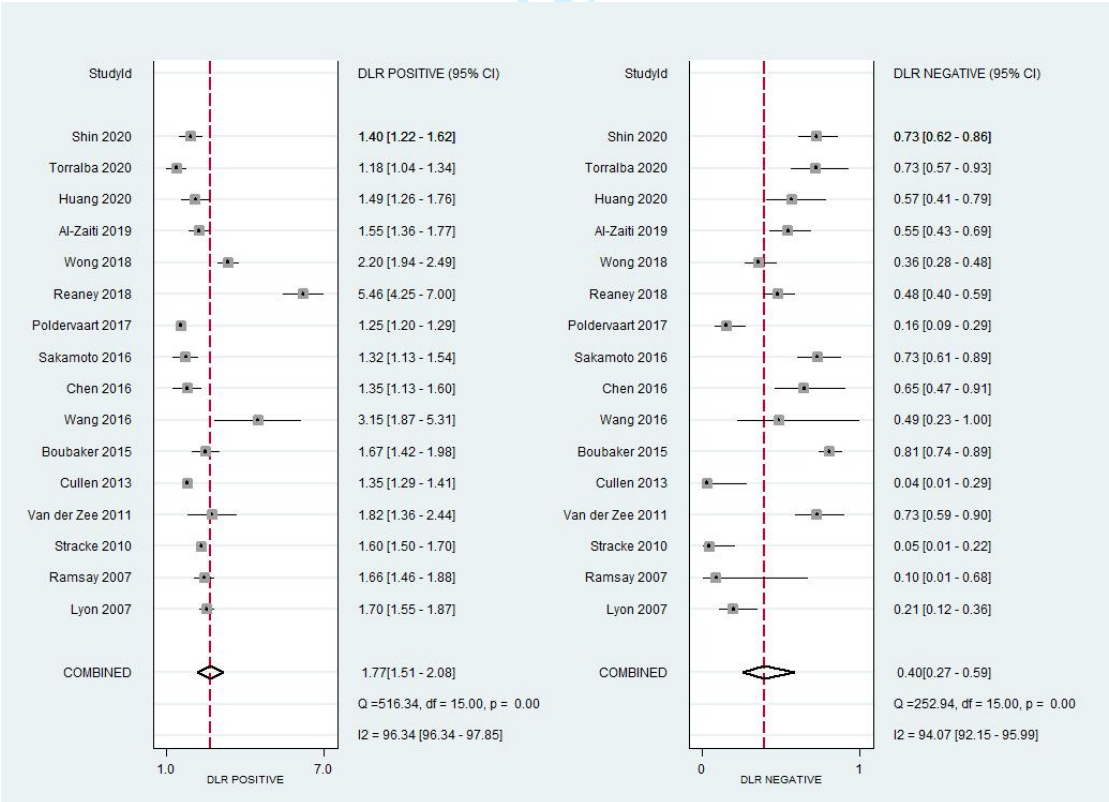


Figure S3. PLR and NLR for GRACE

4. DOR

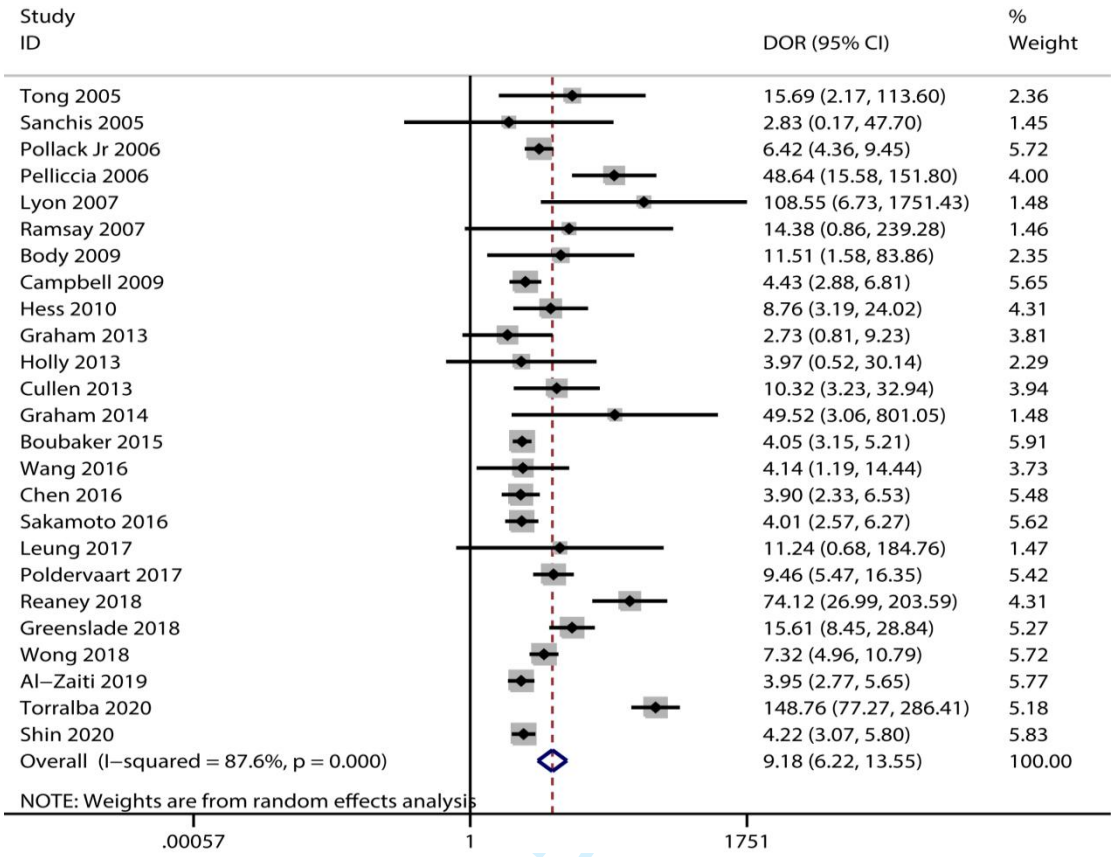


Figure S1. The summary DOR for TIMI

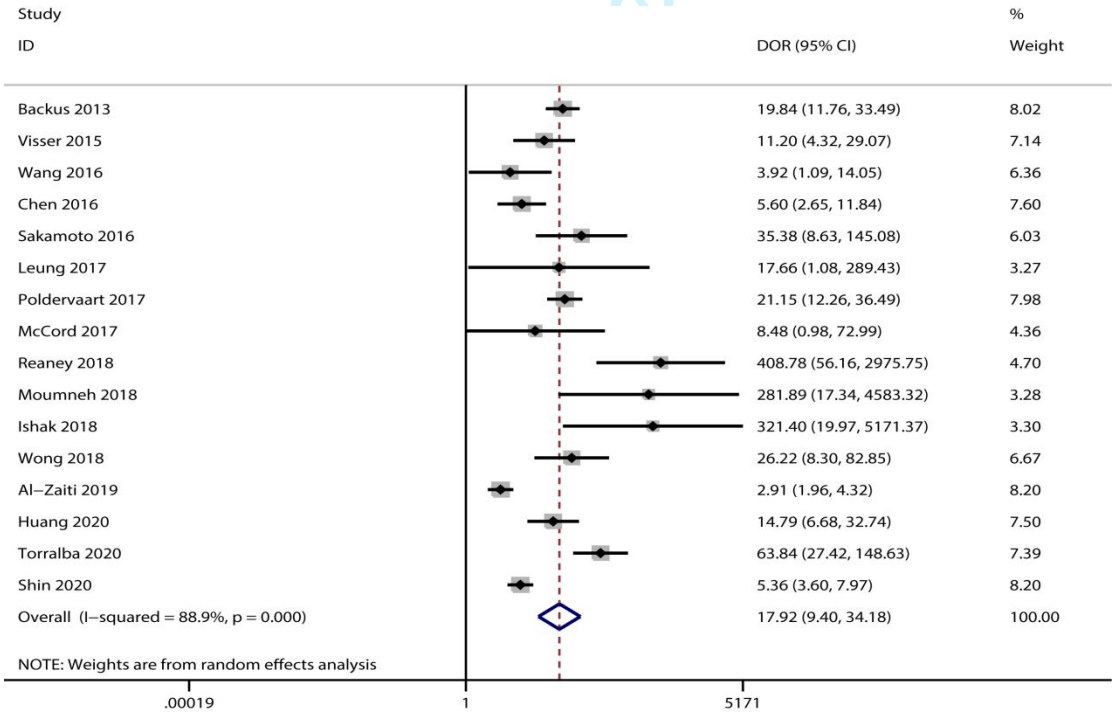


Figure S2. The summary DOR for HEART

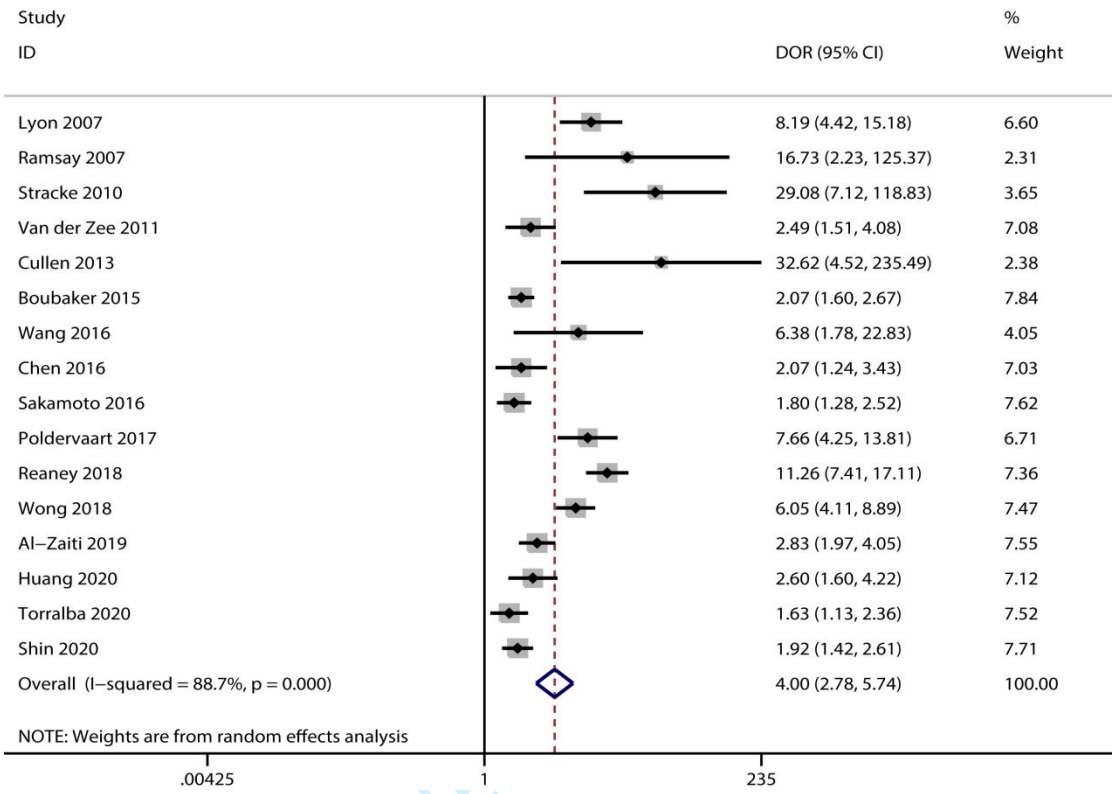


Figure S3. The summary DOR for GRACE

5. Publication bias

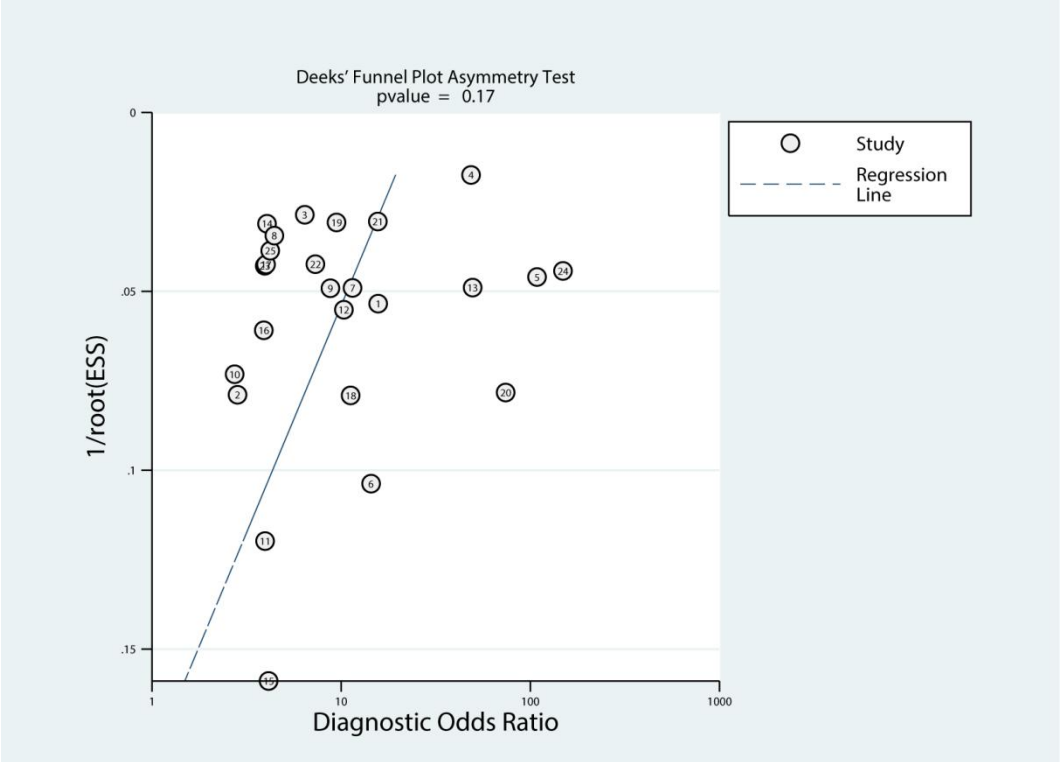


Figure S1. Publication bias for TIMI

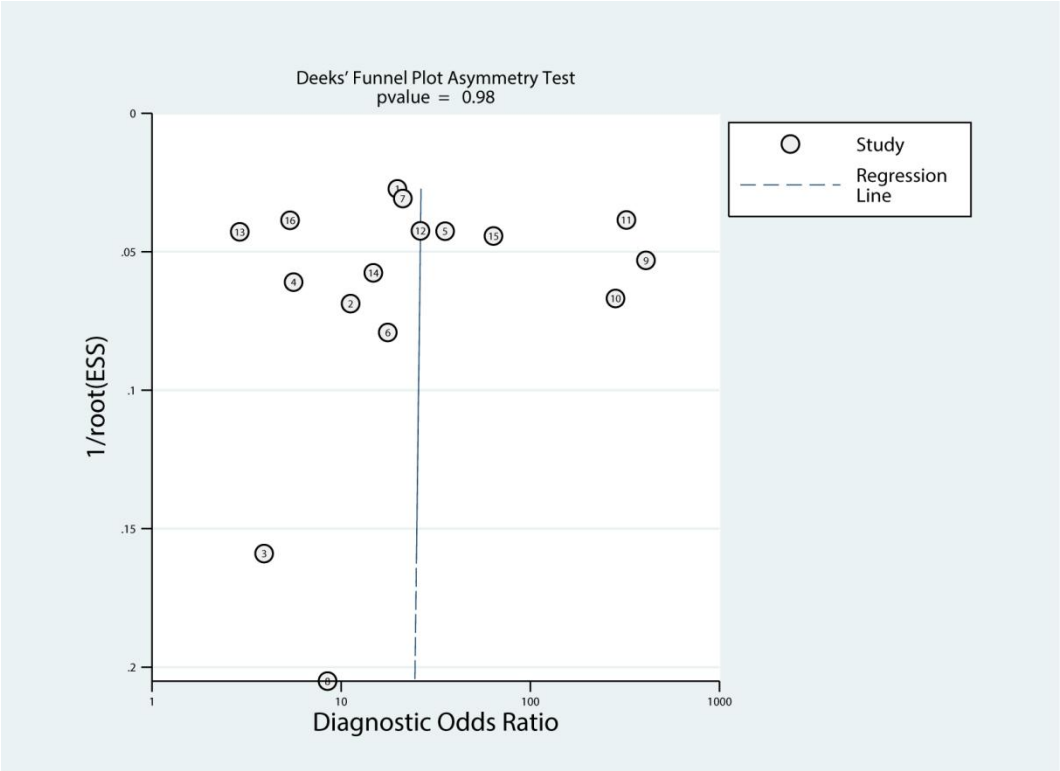


Figure S2. Publication bias for HEART

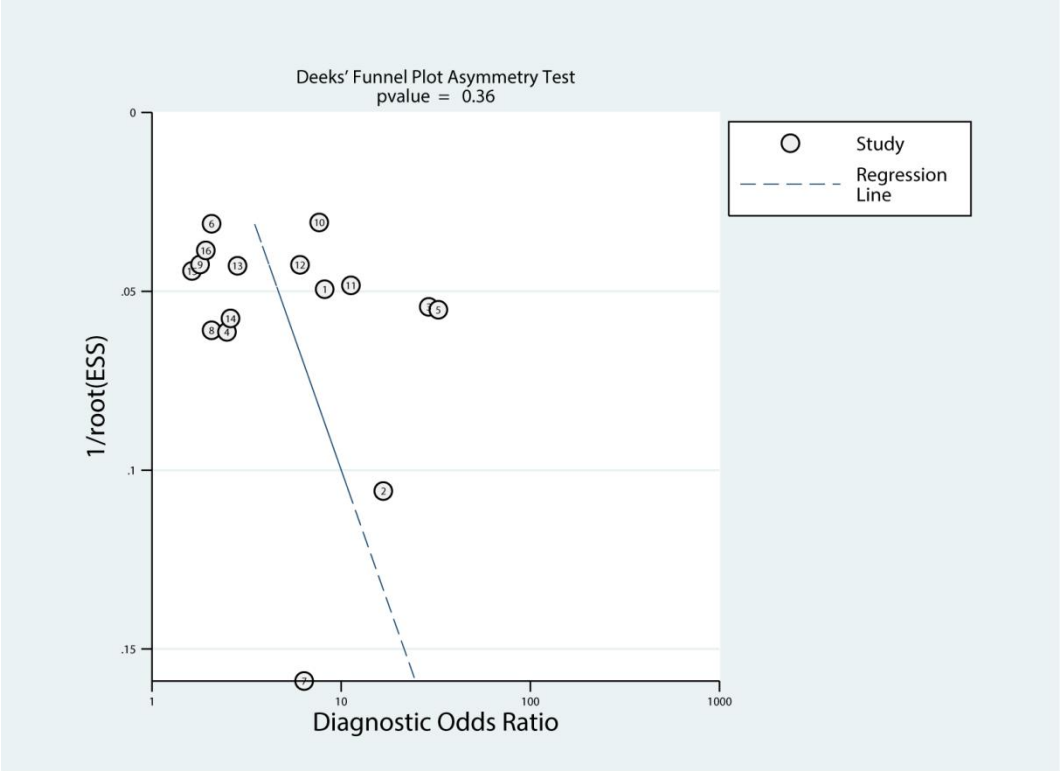


Figure S3. Publication bias for GRACE

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N/A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	9-10
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10-13
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	13
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	14

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Indirect comparison of TIMI, HEART, and GRACE for predicting major cardiovascular events in patients admitted to the emergency department with acute chest pain: A systematic review and meta-analysis

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Indirect comparison of TIMI, HEART, and GRACE for predicting major cardiovascular events in patients admitted to the emergency department with acute chest pain: A systematic review and meta-analysis

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ABSTRACT

Background: The study aimed to compare the predictive values of the Thrombolysis in Myocardial Infarction (TIMI); History, Electrocardiography, Age, Risk factors, and Troponin (HEART); and Global Registry in Acute Coronary Events (GRACE) scoring systems for major adverse cardiovascular events (MACEs) in acute chest pain (ACP) patients admitted to the emergency department (ED).

Methods: We systematically searched PubMed, Embase, and the Cochrane Library from their inception until June 2020; we compared the following parameters: sensitivity, specificity, positive and negative likelihood ratios (PLR and NLR), diagnostic odds ratio (DOR), and area under the receiver operating characteristic curves (AUC).

Results: The pooled sensitivity and specificity for TIMI, HEART, and GRACE were 0.95 and 0.36, 0.96 and 0.50, and 0.78 and 0.56, respectively. The pooled PLR and NLR for TIMI, HEART, and GRACE were 1.49 and 0.13, 1.94 and 0.08, and 1.77 and 0.40, respectively. The pooled DOR for TIMI, HEART, and GRACE was 9.18, 17.92, and 4.00, respectively. The AUC for TIMI, HEART, and GRACE was 0.80, 0.80, and 0.70, respectively. Finally, the results of indirect comparison suggested the superiority of values of TIMI and HEART to those of GRACE for predicting MACEs, while there were no significant differences between TIMI and HEART for predicting MACEs.

Conclusions: TIMI and HEART were superior to GRACE for predicting MACE risk in ACP patients admitted to the ED.

Keywords: TIMI; HEART; GRACE; cardiovascular events; meta-analysis

Strengths and limitations of this study

(1) The analysis is based on prospective studies and used consistent cut-off values.

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(2) The pooled results were stable owing to a large sample size.

(3) The indirect comparisons among TIMI, HEART, and GRACE scoring systems were provided.

(4) The analysis is based on crude data; the predictive values could be affected by covariates.

(5) Substantial heterogeneity was not fully explained.

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INTRODUCTION

Acute chest pain (ACP) is a common symptom accounting for a significant proportion

of attendance and burden in the emergency department (ED) [1]. ACP patients require effective risk stratification to ensure timely initiation of proper treatment in high-risk cases to achieve better prognoses. The early identification of cardiovascular disease (CVD) in ACP patients is important although CVD accounts for only a small proportion of ACP patients with electrocardiograms (ECGs) on presentation [2-4]. Moreover, patients diagnosed with acute coronary syndrome (ACS) should remain hospitalized, whereas non-ACS patients are unnecessarily admitted to hospitals due to the heavy burden on resource constraints [5]. Therefore, accurate risk stratification for ACP patients is essential to improve hospital efficacy by administering timely interventions to high-risk patients, avoiding unnecessary tests, and minimizing admissions for low-risk patients.

Currently, the Thrombolysis in Myocardial Infarction (TIMI); History, ECG, Age, Risk factors, and Troponin (HEART); and Global Registry in Acute Coronary Events (GRACE) scores are widely used for the risk stratification of ACP patients; however, the predictive values using these methods on major adverse cardiovascular events (MACEs) have not been elucidated. The TIMI scoring system was established in 2000 for evaluating patients with unstable angina or non-ST-segment elevation myocardial infarction [6-10]. The HEART score, developed in 2008, aims to improve the accuracy of diagnosing ACS for patients with undifferentiated chest pain [11]. The GRACE score was developed in 2001 for adults with symptoms of ACS; it comprises the following factors: age, vital signs, kidney function, ECG, and troponin levels [12,13]. However, the predictive values of risk stratification measured by the TIMI, HEART, and GRACE scoring systems on MACEs have not been fully compared. Therefore, this study was conducted based on prospective cohort studies to provide comprehensive results regarding the risk stratification assessed by the TIMI, HEART, and GRACE scoring systems on MACEs in ACP patients admitted to the ED. Furthermore, the predictive values of risk stratification assessed by the TIMI, HEART, and GRACE scoring systems on MACEs were compared through an indirect analytic approach.

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Methods

Data Sources, Search Strategy, and Selection Criteria

This study was conducted and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Statement issued in 2009 [14]. Any prospective cohort studies investigating the predictive value of TIMI, HEART, and GRACE on MACEs in ACP patients were eligible for inclusion in this study. No restrictions were placed on publishing language and status. The electronic databases of PubMed, Embase, and the Cochrane library were systematically searched for studies from their inception up to June 2020, and the search strategy was performed using the following terms with Medical Subject Heading and free words: ("TIMI" or "HEART" or "GRACE") and "emergency department" and "chest pain" and ("prospective" or "cohort"). The search strategy details are summarized in the Supplementary file. The reference lists of retrieved studies were also searched manually to find new eligible studies.

Two authors independently performed the literature search and study selection; any conflicts were resolved through group discussion until a consensus was reached. A study was included if they met the following inclusion criteria: (1) study design: the study had a prospective design; (2) patients: ACP patients admitted to the ED; (3) risk stratifying tools: TIMI, HEART, or GRACE; (4) outcomes: the study had to report the incidence of MACEs and provided clear definitions of MACEs; (5) data abstracted: true and false positives or negatives, or data could transform into the above information must be reported; and (6) cut-off value: the cut-off value of TIMI and HEART was 0-3, and the cut-off value of GRACE was 55-110. Retrospective studies were excluded due to various confounding factors. Additionally, studies that used other cut-off values were excluded.

Data Collection and Quality Assessment

Two authors independently abstracted data items and assessed the quality of the included studies, and any disagreement was settled by an additional author reviewing the original article. The collected information from retrieved studies including the first author's name, publication year, country, sample size, age at baseline, percentage of

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males, risk stratifying tools, patients' status, MACE definition, follow-up duration, and true and false positives/negatives. Study quality was assessed using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2), which is based on patient selection, index test, reference standard, risk of bias, and concerns about applicability [15].

Statistical Analysis

The pooled sensitivity, specificity, positive likelihood ratio (PLR), negative likelihood ratio (NLR), and the area under the receiver operating characteristic curves (AUC) for the risk stratification by TIMI, HEART, and GRACE on MACEs were calculated using a bivariate generalized linear mixed model [16,17], while the pooled diagnostic odds ratio (DOR) was calculated using a random-effects model [18]. The I^2 and Q statistic were used for assessing the heterogeneity across included studies, and P<0.10 was considered as significant heterogeneity [19,20]. The robustness of pooled results was also assessed by sensitivity analyses, which were also performed for studies using all three scoring systems and the endpoint MACE [21]. TIMI-, HEART-, and GRACE-based risk stratification were assessed using an indirect comparison analysis, and the ratios among these scoring systems were calculated [22]. Furthermore, subgroup analyses for the predictive values of TIMI, HEART, and GRACE on subsequent MACE risk were also estimated based on country, mean age, percentage of males, follow-up duration, and study quality. The funnel plots and Deeks' asymmetry tests were used to assess publication bias [23,24]. The inspection level for pooled diagnostic parameters was two-sided, and P<0.05 was considered statistically significant. All statistical analyses were performed using Stata software (version 10.0; Stata Corporation, College Station, TX, USA).

Patient and Public Involvement

There was no patient or public involvement in the design or conduct of this study.

RESULTS

Literature Search

The details regarding the literature search and study selection of eligible studies are presented in Figure 1. A total of 2,794 articles were identified through the electronic search from PubMed, Embase, and the Cochrane library, and 1,981 were excluded because of term duplications. Subsequently, the remaining 813 studies were selected through title and abstract review; 732 were excluded based on irrelevance. A total of 81 full texts were retrieved for further evaluation, and 48 studies were excluded due to the following reasons: used other cut-off values (n=21), retrospective study design (n=14), and insufficient data (n=13). An additional 135 potential studies identified from the reference lists of retrieved studies were excluded because of duplication with the electronic search. Subsequently, 33 prospective cohort studies that recruited 40,262 ACP patients were selected for final quantitative meta-analysis [13,25-56].

Study Characteristics

The baseline characteristics of included studies are summarized in Table 1. The retrieved studies were published from 2005 to 2020, and 255-4,333 ACP patients were included in each study. Nine studies were conducted in Eastern countries, and the remaining 24 studies were conducted in Western countries. The mean age of enrolled patients ranged from 48.0 to 69.0 years, and the percentage of males ranged from 40.0 to 68.8%. Risk stratification by the TIMI score was available in 25 studies published between 2005 and 2020, 16 studies used the HEART score and were published between 2013 and 2020, and 16 studies employed GRACE and were published between 2007 and 2020. The definition of MACEs across included studies contained all-cause death, cardiac death, myocardial infarction, revascularization, cardiac arrest, cardiogenic shock, unstable angina, ACS, percutaneous coronary intervention, coronary artery bypass grafting, coronary angiography revealing procedurally correctable stenosis managed conservatively, ventricular arrhythmia needing intervention, high-degree atrioventricular block needing intervention, and life-threatening arrhythmias requiring emergency intervention. The study quality of the included studies was assessed by QUADAS-2 (Figure 2).

TIMI

The predictive value of risk stratification by the TIMI score on MACEs in ACP patients

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was available in 25 studies. The pooled sensitivity and specificity of the TIMI score for predicting MACEs were 0.95 (95% CI: 0.91-0.98; I^2 = 98.10%) and 0.36 (95% CI: 0.24-0.50; I^2 = 99.64%), respectively (Supplementary file). Moreover, the pooled PLR and NLR of the TIMI score for predicting MACEs were 1.49 (95% CI: 1.25-1.79; I^2 = 99.21%) and 0.13 (95% CI: 0.07-0.21; I^2 = 95.38%), respectively (Supplementary file). The pooled DOR of the TIMI score for predicting MACEs was 9.18 (95% CI: 6.22-13.55; P<0.001) with significant heterogeneity across the included studies (I^2 = 87.6%; P<0.001) (Supplementary file). The AUC of the TIMI score for predicting MACEs was 0.80 (95% CI: 0.76-0.83; Figure 3). No significant publication bias for the TIMI score was detected (P=0.17; Supplementary file).

HEART

The predictive value of risk stratification by the HEART score on MACEs in ACP patients was available in 16 studies. The pooled sensitivity and specificity of the HEART score for predicting MACEs were 0.96 (95% CI: 0.91-0.98; I^2 = 94.87%) and 0.50 (95% CI: 0.41-0.60; I^2 = 98.84%), respectively (Supplementary file). The pooled PLR and NLR of the HEART score for predicting MACEs were 1.94 (95% CI: 1.61-2.35; I^2 = 98.01%) and 0.08 (95% CI: 0.03-0.17; I^2 = 94.65%), respectively (Supplementary file). The pooled DOR for the HEART score was 17.92 (95% CI: 9.40-34.18; P<0.001) with significant heterogeneity across the included studies (I^2 = 88.9%; P<0.001) (Supplementary file). The AUC of the HEART score for predicting MACEs was 0.80 (95% CI: 0.77-0.84; Figure 4). There was no significant publication bias for the HEART score (P=0.98; Supplementary file).

GRACE

The predictive value of risk stratification by the GRACE score on MACEs in ACP patients was available in 16 studies. The pooled sensitivity and specificity of the GRACE score for predicting MACEs were 0.78 (95% CI: 0.64-0.87; I^2 = 96.78%) and 0.56 (95% CI: 0.46-0.66; I^2 = 99.39%), respectively (Supplementary file). The pooled PLR and NLR of the GRACE score for predicting MACEs were 1.77 (95% CI: 1.51-2.08; I^2 = 96.34%) and 0.12 (95% CI: 0.06-0.26; I^2 = 94.07%), respectively (Supplementary file). The DOR of the GRACE score for predicting MACEs was 4.00

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(95% CI: 2.78-5.74; P<0.001) with significant heterogeneity across the included studies (I^2 = 88.7%; P<0.001) (Supplementary file). The AUC of the GRACE score for predicting MACEs was 0.70 (95% CI: 0.66-0.74; Figure 5). No significant publication bias for the GRACE score was observed (P=0.36; Supplementary file).

Indirect Comparisons

Indirect comparisons of the diagnostic parameters (sensitivity, specificity, PLR, NLR, DOR, and AUC) among the TIMI, HEART, and GRACE scoring systems for predicting MACEs are summarized in Table 2. First, the sensitivity of TIMI (ratio: 1.22; 95% CI: 1.04-1.43) and HEART (ratio: 1.23; 95% CI: 1.05-1.44) was significantly higher than that of GRACE for predicting MACEs. Second, the specificity of TIMI was significantly lower than that of GRACE for predicting MACEs (ratio: 0.64; 95% CI: 0.43-0.97). Third, the PLR of TIMI was lower than that of HEART for predicting MACEs (ratio: 0.77; 95% CI: 0.59-1.00). Fourth, TIMI (ratio: 0.32; 95% CI: 0.17-0.64) and HEART (ratio: 0.20; 95% CI: 0.08-0.52) were associated with a lower NLR than GRACE for predicting MACEs. Fifth, the DOR of TIMI (ratio: 2.29; 95% CI: 1.35-3.91) and HEART (ratio: 4.48; 95% CI: 2.14-9.39) was significantly higher than that of GRACE for predicting MACEs. Finally, the AUC of TIMI (ratio: 1.14; 95% CI: 1.06-1.23) and HEART (ratio: 1.14; 95% CI: 1.06-1.23) was significantly higher than that of GRACE for predicting MACEs.

Sensitivity and Subgroup Analysis

The results of sensitivity analyses found that the predictive values of TIMI, HEART, and GRACE for predicting MACEs were stable and unaltered by sequential removal of one study from the overall analysis (data not shown). Sensitivity analyses were also performed after removing studies not using all 3 scoring systems (Table 3). We noted that TIMI had lower sensitivity that that of HEART (ratio: 0.86; 95% CI: 0.75-0.99). Moreover, TIMI had lower NLR (ratio: 0.55; 95% CI: 0.31-0.99), higher DOR (ratio: 2.71; 95% CI: 1.17-6.24), and higher AUC (ratio: 1.19; 95% CI: 1.11-1.27), compared to GRACE. Furthermore, HEART had higher sensitivity (ratio: 1.32; 95% CI: 1.11-1.58), DOR (ratio: 4.36; 95% CI: 1.63-11.64), and AUC (ratio: 1.14; 95% CI: 1.06-1.23) and lower NLR (ratio: 0.22; 95% CI: 0.09-0.56) as compared with GRACE.

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Additionally, Table 4 presents the subgroup analysis results for the predictive values of TIMI, HEART, and GRACE. First, TIMI has lower specificity than HEART in the pooled studies from Western countries. Furthermore, TIMI versus HEART showed lower PLR in the pooled studies of Western countries (mean age < 60.0 years). Moreover, the DOR of TIMI was lower than of HEART in the pooled studies of Eastern countries, while TIMI had a lower AUC than HEART for mean age <60.0 years. Second, TIMI with higher sensitivity than GRACE when pooled studies conducted in Eastern countries, percentage of males ≥60.0%, follow-up duration ≤30.0 days, and studies of moderate quality. Moreover, TIMI versus GRACE showed lower specificity if mean age <60.0 years, percentage of males ≥60.0%, and follow-up duration ≤30.0 days. Furthermore, TIMI has lower PLR than GRACE if percentage of males ≥60.0%, while TIMI has lower NLR than GRACE in the pooled studies of Eastern countries, irrespective of the mean age or percentage male status, follow-up duration ≤30.0 days, and studies of moderate quality. In addition, TIMI versus GRACE showed higher DOR and AUC in most subgroups. Third, HEART versus GRACE showed higher sensitivity in the pooled studies of Eastern countries, mean age ≥60.0 years, follow-up duration ≤30.0 days, and studies of moderate quality. Moreover, HEART has lower NLR than GRACE in most subgroups, except that of mean age <60.0 years. Furthermore, HEART versus GRACE showed higher DOR and AUC in most subgroups.

Discussion

This study was the first meta-analysis to conduct indirect comparisons of the predictive values of risk stratification assessed by the TIMI, HEART, and GRACE scores on MACEs in ACP patients. The current study included a total of 40,262 ACP patients from 33 prospective cohort studies and across a wide range of patient characteristics. The findings of this study suggest that the predictive values of TIMI, HEART, and GRACE scoring systems were better for MACEs in ACP patients admitted to the ED. Moreover, an indirect analysis indicated that the predictive value of TIMI and HEART was superior to that of GRACE for predicting MACEs, while there were no significant

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differences between TIMI and HEART for predicting MACEs. The results of sensitivity analyses for studies using all 3 scoring systems were consistent with those of the overall analysis. Meanwhile, we noted that the sensitivity of TIMI was lower than HEART for predicting MACEs.

Several systematic reviews and meta-analyses have illustrated the predictive values of the TIMI, HEART, and GRACE scoring systems on MACEs in ACP patients [57-59]. Hess et al. included 8 prospective studies and found that the TIMI score provided effective risk stratification for predicting MACEs in potential ACS patients, whereas it should not be used as the sole means for determining patient disposition [57]. Van Den Berg et al. identified 2 prospective and 10 retrospective cohort studies and suggested that the HEART score could be used to identify MACEs in patients with a suspected diagnosis of ACS [58]. Roche et al. included 11 studies and found that using 100 as the cut-off value of the GRACE score could predict the discharge of nearly 70% of presentations, while the predictive value for subsequent MACE risk was not obtained [59]. However, the above studies only reported the diagnostic value of a single scoring system for predicting MACEs in ACS patients. Therefore, we performed the current meta-analysis of prospective studies to evaluate the predictive values of the TIMI, HEART, and GRACE scoring systems on the risk of MACEs in ACP patients and systematically compared the predictive values among them.

The predictive value of the TIMI score for MACEs in ACP patients was statistically significant, whereas several studies reported inconsistent results. Sanchis et al. found that the TIMI score was not associated with the risk of MACEs when 0 was used as the cut-off value [26]. Graham et al. found that a low TIMI score could not rule out cardiac causes of chest pain [35]. Holly et al. suggested that the TIMI score was not associated with the risk of MACE at 30 days when 0 was used as the cut-off value [36]. Leung et al. indicated that a modified TIMI score of 0 could not rule out 30-day MACEs in ACP patients admitted to the ED [45]. The potential reasons for this could be that the TIMI score was designed for risk stratification in patients with non-ST-segment elevation ACS, which is mainly based on appropriate ECG changes or elevations of biomarkers of necrosis. Moreover, the presentation characteristics in ACP patients were not entered

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into the TIMI score. Finally, the prevalence of MACEs during follow-up in these studies was lower than expected, resulting in broad 95% CIs, i.e., no statistically significant difference [26,35,36,45].

The predictive value of the HEART score for predicting MACEs in ACP patients was statistically significant. Nearly all included studies reported a similar conclusion, whereas the study conducted by McCord et al. suggested that the HEART score after 4 hours of the presentation was associated with marginal predictive values for the risk of MACEs [47]. The potential reason for this may be that this study used a modified HEART score, and the original HEART score only considered the initial cTn value, without taking serial sampling into account, which is associated with a lower prevalence of MACEs [60].

The predictive value of the GRACE score for predicting MACE in ACP patients was statistically significant, and all included studies reported similar conclusions for the predictive value of the GRACE score on the risk of MACEs. Especially, the GRACE score was initially designed for post-ACS risk stratification, including unstable angina and non-ST-elevation ACS [11]. The American Heart Association suggested the use of the GRACE score for admission and discharge of ACS patients. Moreover, the risk assessment for patients evaluated outside the hospital should be recommended to use GRACE [61]. Therefore, the predictive value of the GRACE score in low-risk individuals was restricted, which should be addressed in clinical practice.

We noted that the predictive value of the GRACE score was inferior to that of TIMI and HEART scores. Sensitivity analyses were performed for studies reporting all 3 scoring systems, which included 9 studies [42-44,46,48,52,53,55,56], and the results indicated TIMI and HEART having higher predictive values for MACEs than GRACE. The results of the sensitivity analyses were more reliable owing to the analysis of three scoring systems based on direct comparisons. Moreover, the GRACE score was initially developed for ACS patients but not for ACP patients, and the potential risk factors for the progression of MACEs were not considered in the GRACE score. Interestingly, we noted no significant difference between TIMI and HEART for predicting MACEs, while several studies reported that the predictive value of HEART

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for MACEs was superior to that of TIMI [44,46,48,52,53,55,56]. Subgroup analysis found TIMI with lower AUC compared with HEART if the mean age of patients was <60.0 years. The potential explanation could be the use of HEART score in the absence of exact definitions for medical history across included studies [11], and the predictive value of HEART was more suitable in low-risk individuals.

Three strengths of this quantitative meta-analysis should be highlighted: (1) the study was based on prospective cohort studies and used relatively uniform cut-off values, which were associated with lower selective and informative biases; (2) the analysis of this study was based on a large sample size, and the findings in our study were more robust than any individual study; and (3) the predictive values of the TIMI, HEART, and GRACE scoring systems on the risk of MACEs in ACP patients were compared through an indirect analytic approach.

Despite the abovementioned findings, the predictive values of the TIMI, HEART, and GRACE could be affected by the definitions of MACEs and the ranges from a single endpoint (death or myocardial infarction) to a composite endpoint. Subgroup analysis based on MACE definition were not performed because of the definition of MACE across included studies are various. Therefore, MACE definition could affect the predictive value and follow-up durations owing to these factors that attribute to the weight of pooled conclusion. Moreover, this meta-analysis was based on crude data, and the adjusted results were not available. Furthermore, substantial heterogeneity was detected across the included studies, and the heterogeneity was not fully explained by sensitivity and subgroup analyses. In addition, the analysis was conducted on published articles, which causes inevitable publication biases. Finally, the current study was based on indirect comparisons between the predictive values of the TIMI, HEART, and GRACE scoring systems as direct comparisons were not available.

The findings of this study indicated that risk stratification assessed by the TIMI, HEART, and GRACE scores provides relatively appropriate predictive values for MACEs in ACP patients. The results of indirect comparison analysis indicated that TIMI and HEART had relatively relative better predictive values than GRACE on subsequent MACE risk. Further prospective cohort studies should be conducted to

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directly compare the predictive values of TIMI, HEART, and GRACE on MACEs in ACP patients.

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Authors' Contributions

JK: study design and manuscript drafting

YC: study supervision, data analysis, and result interpretation

XW: literature search and data collection

ZW: data curation

FC: manuscript revising and final approval

Data Availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Competing Interests

The authors declare no competing interests.

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Ethics Approval

Neither ethics approval nor participant consent was required as this study was based solely on the summary results of previously published articles. Individual patient data were not obtained or accessed.

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Figure Legends

Figure 1. Flow diagram of the literature search and study selection process.

Figure 2. QUADAS-2 scoring of included studies.

Figure 3. The summary receiver operating characteristic curve of risk stratification assessed by the TIMI score.

Figure 4. The summary receiver operating characteristic curve of risk stratification assessed by the HEART score.

Figure 5. The summary receiver operating characteristic curve of risk stratification assessed by the GRACE score.

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Table 1. Baseline characteristics of studies included in the systematic review and meta-analysis.

Study	Country	Sample size	Mean age (years)	Percentage male (%)	Risk stratifying tools	Patients' status	MACE definition	Follow-up
Tong 2005 [25]	US	957	60.0	52.0	TIMI	Chest pain and a nondiagnostic ECG	Death and MI	30 days
Sanchis 2005 [26]	Spain	646	64.0	65.8	TIMI	Acute chest pain	Death, MI, or urgent revascularization	14 days
Pollack Jr 2006 [27]	US	3929	51.6	40.0	TIMI	Chest pain in the ED	Death, acute MI, and revascularization	30 days
Pelliccia 2006 [28]	Italy	4333	58.4	68.8	TIMI	Acute chest pain	MI	In-hospital
Lyon 2007 [29]	UK	954	60.0	62.0	TIMI, GRACE	Undifferentiated chest pain	MI, cardiac arrest, revascularization, unstable angina with myocardial damage and death	30 days
Ramsay 2007 [13]	UK	347	65.2	62.3	TIMI, GRACE	Suspected cardiac pain	Death, non-fatal MI, and emergency revascularization	3.0 months
Body 2009 [30]	UK	796	58.9	60.4	TIMI	Chest pain in the ED	Death, acute MI or urgent coronary revascularization	30 days
Campbell 2009 [31]	US	3169	53.6	45.0	TIMI	Chest pain in the ED	Death, MI, or revascularization	30 days
Hess 2010 [32]	Canada	1017	59.3	60.6	TIMI	ED patients with chest pain and possible ACS	Acute MI, revascularization, or death	30 days
Stracke 2010 [33]	Germany	1014	66.0	55.0	GRACE	Chest pain in the ED	Death	In-hospital

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Van der Zee 2011 [34]	The Netherlands	524	57.7	60.5	GRACE	Chest pain in the ED	Death	9.4 years
Graham 2013 [35]	China	315	69.0	54.9	TIMI	Chest pain in the ED	Death, MI, troponin positive ACS and PCI	30 days
Holly 2013 [36]	US	552	54.1	46.0	TIMI	Chest pain in the ED	MI, revascularization, or death	30 days
Backus 2013 [37]	The Netherlands	2388	60.6	57.5	HEART	Chest pain in the ED	Acute MI, PCI, CABG, coronary angiography revealing procedurally correctable stenosis managed conservatively, and death	6 weeks
Cullen 2013 [38]	Australia	948	54.0	59.9	TIMI, and GRACE	Chest pain in the ED	Cardiac death, acute MI, and unstable angina	30 days
Graham 2014 [39]	China	925	68.0	51.7	TIMI	Chest pain in the ED	Death, readmission with MI, ACS not diagnosed at initial ED presentation and coronary revascularization	30 days
Visser 2015 [40]	The Netherlands	255	64.0	56.0	HEART	Chest pain in the ED	MI, or PCI, or CABG, or coronary angiography revealing significant stenosis or death	6 weeks
Boubaker 2015 [41]	Tunisia	3125	57.7	58.3	TIMI, GRACE	Chest pain in the ED	All-cause mortality, acute coronary syndrome, and coronary non-ED planned revascularization	30 days
Wang 2016 [42]	China	986	54.0	55.0	TIMI, HEART, GRACE	Chest pain in the ED	Death, MI, and/or the need for revascularization by CABG or PCI	6 months
Chen 2016 [43]	China	833	65.1	55.3	TIMI, HEART, GRACE	Chest pain in the ED	Coronary revascularization, ventricular arrhythmia needing intervention and	30 days

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							high-degree atrioventricular block needing intervention	
Sakamoto 2016 [44]	Singapore	604	60.8	69.2	TIMI, HEART, GRACE	Chest pain in the ED	Death, acute MI, PCI, CABG	30 days
Leung 2017 [45]	China	602	66.0	48.8	TIMI, HEART	Chest pain in the ED	Death, cardiac arrest, MI, and cardiogenic shock	30 days
Poldervaart 2017 [46]	The Netherlands	1748	62.0	54.0	TIMI, HEART, GRACE	Chest pain in the ED	UA, MI, PCI, CABG, stenosis managed conservatively, and death	3.0 months
McCord 2017 [47]	Europe, Australia, and the US	661	58.3	58.2	HEART	Chest pain in the ED	Death or acute MI	30 days
Reaney 2018 [48]	UK	1000	62.5	57.6	TIMI, HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, cardiac death, cardiogenic shock, and life-threatening arrhythmias requiring emergency intervention.	30 days
Greenslade 2018 [49]	Australia	1760	60.4	59.3	TIMI	Chest pain in the ED	Cardiac death, cardiac arrest, cardiogenic shock, acute MI, UA, emergency or urgent revascularization, high-level atrioventricular block, and ventricular arrhythmias	30 days
Moumneh 2018 [50]	France	641	53.3	53.4	HEART	Non-traumatic chest pain patients	MI, coronary angioplasty, coronary bypass, and sudden unexplained death	6 weeks
Ishak 2018 [51]	The Netherlands	1127	63.8	57.7	HEART	Chest pain in the ED	Acute MI, PCI, CABG, or death	30 days
Wong 2018 [52]	China	1081	48.0	52.3	TIMI, HEART, GRACE	Undifferentiated chest pain	Acute MI, PCI, CABG, and death	30 days

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Al-Zaiti 2019 [53]	USA	750	59.0	58.0	TIMI, HEART, GRACE	Chest pain in the ED	Resuscitated or un-resuscitated sudden cardiac arrest, all-cause death, post-discharge re-infarction requiring cardiac revascularization	30 days
Huang 2020 [54]	China	509	59.8	53.4	HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, cardiac death, cardiogenic shock, or life-threatening arrhythmias requiring emergency intervention or resulting in mortality	30 days
Torralba 2020 [55]	Colombia	519	64.3	56.1	TIMI, HEART, GRACE	Chest pain in the ED	Death from any cause, MI, and surgical or percutaneous myocardial revascularization	30 days
Shin 2020 [56]	Korea	1247	62.0	60.8	TIMI, HEART, GRACE	Chest pain in the ED	Acute MI, PCI, CABG, or death from cardiac causes	30 days

*ACS: acute coronary syndrome; CABG: coronary artery bypass grafting; ECG: electrocardiogram; ED: emergency department; MI: myocardial infarction; PCI: percutaneous coronary intervention; UA: unstable angina.

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Table 2. Systematic comparisons of the sensitivity, specificity, positive and negative likelihood ratio, diagnostic odds ratio, and the area under the receiver operating characteristic curve of risk stratifying measured by TIMI, HEART, and GRACE.

Parameters	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity and 95%CI	0.95 (0.91-0.98)	0.96 (0.91-0.98)	0.78 (0.64-0.87)	0.99 (0.94-1.04)	1.22 (1.04-1.43)	1.23 (1.05-1.44)
Specificity and 95%CI	0.36 (0.24-0.50)	0.50 (0.41-0.60)	0.56 (0.46-0.66)	0.72 (0.48-1.09)	0.64 (0.43-0.97)	0.89 (0.69-1.16)
PLR and 95%CI	1.49 (1.25-1.79)	1.94 (1.61-2.35)	1.77 (1.51-2.08)	0.77 (0.59-1.00)	0.84 (0.66-1.07)	1.10 (0.86-1.40)
NLR and 95%CI	0.13 (0.07-0.21)	0.08 (0.03-0.17)	0.40 (0.27-0.59)	0.46 (0.15-1.39)	0.32 (0.17-0.64)	0.20 (0.08-0.52)
DOR and 95%CI	9.18 (6.22-13.55)	17.92 (9.40-34.18)	4.00 (2.78-5.74)	0.51 (0.24-1.09)	2.29 (1.35-3.91)	4.48 (2.14-9.39)
AUC and 95%CI	0.80 (0.76-0.83)	0.80 (0.77-0.84)	0.70 (0.66-0.74)	1.00 (0.94-1.06)	1.14 (1.06-1.23)	1.14 (1.06-1.23)

PLR: positive likelihood ratio; NLR: negative likelihood ratio; DOR: diagnostic odds ratio; AUC: the area under the receiver operating characteristic curve.

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Table 3. Sensitivity analysis for direct comparisons of the sensitivity, specificity, positive and negative likelihood ratio, diagnostic odds ratio, and the area under the receiver operating characteristic curve of risk stratifying measured by TIMI, HEART, and GRACE.

Parameters	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity and 95%CI	0.81 (0.70-0.89)	0.94 (0.84-0.98)	0.71 (0.58-0.81)	0.86 (0.75-0.99)	1.14 (0.94-1.39)	1.32 (1.11-1.58)
Specificity and 95%CI	0.70 (0.54-0.83)	0.53 (0.39-0.66)	0.59 (0.44-0.73)	1.32 (0.94-1.86)	1.19 (0.85-1.65)	0.90 (0.62-1.29)
PLR and 95%CI	2.72 (1.67-4.41)	1.99 (1.51-2.63)	1.74 (1.33-2.28)	1.37 (0.78-2.39)	1.56 (0.90-2.72)	1.14 (0.78-1.68)
NLR and 95%CI	0.27 (0.16-0.44)	0.11 (0.05-0.28)	0.49 (0.36-0.66)	2.45 (0.90-6.66)	0.55 (0.31-0.99)	0.22 (0.09-0.56)
DOR and 95%CI	9.45 (4.77-18.73)	15.21 (6.45-35.88)	3.49 (2.16-5.63)	0.62 (0.21-1.86)	2.71 (1.17-6.24)	4.36 (1.63-11.64)
AUC and 95%CI	0.83 (0.80-0.86)	0.80 (0.76-0.83)	0.70 (0.66-0.74)	1.04 (0.98-1.10)	1.19 (1.11-1.27)	1.14 (1.06-1.23)

PLR: positive likelihood ratio; NLR: negative likelihood ratio; DOR: diagnostic odds ratio; AUC: the area under the receiver operating characteristic curve.

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Table 4. Subgroup analyses.

Parameters	Factors	Groups	TIMI	HEART	GRACE	TIMI versus HEART	TIMI versus GRACE	HEART versus GRACE
Sensitivity	Country	Western	0.97 (0.93-0.99)	0.97 (0.93-0.99)	0.88 (0.72-0.95)	1.00 (0.96-1.05)	1.10 (0.96-1.27)	1.10 (0.96-1.27)
		Eastern	0.86 (0.69-0.95)	0.94 (0.79-0.98)	0.60 (0.49-0.70)	0.91 (0.75-1.11)	1.43 (1.13-1.82)	1.57 (1.27-1.93)
	Mean age (years)	≥ 60.0	0.97 (0.92-0.99)	0.97 (0.94-0.99)	0.83 (0.67-0.92)	1.00 (0.96-1.05)	1.17 (0.99-1.38)	1.17 (1.00-1.37)
		< 60.0	0.93 (0.81-0.97)	0.91 (0.70-0.98)	0.68 (0.44-0.85)	1.02 (0.84-1.24)	1.37 (0.97-1.92)	1.34 (0.92-1.94)
	Percentage male (%)	≥ 60.0	0.98 (0.92-0.99)	-	0.73 (0.50-0.88)	-	1.34 (1.01-1.79)	-
		< 60.0	0.92 (0.85-0.96)	0.96 (0.91-0.99)	0.80 (0.62-0.90)	0.96 (0.89-1.03)	1.15 (0.95-1.40)	1.20 (0.99-1.45)
	Follow-up duration	≤ 30.0 days	0.96 (0.91-0.98)	0.97 (0.92-0.99)	0.76 (0.61-0.87)	0.99 (0.94-1.04)	1.26 (1.05-1.51)	1.28 (1.06-1.53)
		> 30.0 days	-	0.93 (0.78-0.98)	-	-	-	-
	Study quality	High	0.94 (0.83-0.98)	0.97 (0.91-0.99)	0.77 (0.50-0.92)	0.97 (0.88-1.06)	1.22 (0.89-1.67)	1.26 (0.93-1.71)
		Moderate	0.96 (0.90-0.98)	0.95 (0.85-0.98)	0.78 (0.61-0.89)	1.01 (0.93-1.10)	1.23 (1.01-1.49)	1.22 (1.00-1.49)
Specificity	Country	Western	0.33 (0.20-0.50)	0.56 (0.47-0.64)	0.50 (0.34-0.66)	0.59 (0.36-0.96)	0.66 (0.37-1.16)	1.12 (0.78-1.61)
		Eastern	0.46 (0.27-0.67)	0.44 (0.27-0.62)	0.64 (0.55-0.73)	1.05 (0.56-1.94)	0.72 (0.45-1.16)	0.69 (0.44-1.07)
	Mean age	≥ 60.0	0.35 (0.19-0.55)	0.45 (0.34-0.56)	0.51 (0.37-0.64)	0.78 (0.43-1.40)	0.69 (0.38-1.25)	0.88 (0.61-1.28)

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	(years)	< 60.0	0.38 (0.23-0.56)	0.60 (0.44-0.74)	0.63 (0.49-0.76)	0.63 (0.38-1.06)	0.60 (0.37-0.99)	0.95 (0.68-1.34)
	Percentage male (%)	≥ 60.0	0.21 (0.10-0.37)	-	0.56 (0.45-0.66)	-	0.38 (0.19-0.74)	-
		< 60.0	0.45 (0.30-0.61)	0.52 (0.42-0.63)	0.56 (0.42-0.70)	0.87 (0.58-1.30)	0.80 (0.52-1.24)	0.93 (0.67-1.29)
	Follow-up duration	≤ 30.0 days	0.35 (0.23-0.50)	0.46 (0.35-0.57)	0.56 (0.45-0.67)	0.76 (0.48-1.20)	0.63 (0.40-0.97)	0.82 (0.60-1.13)
		> 30.0 days	-	0.60 (0.43-0.75)	-	-	-	-
	Study quality	High	0.41 (0.24-0.62)	0.47 (0.35-0.60)	0.61 (0.36-0.81)	0.87 (0.51-1.51)	0.67 (0.36-1.25)	0.77 (0.47-1.25)
		Moderate	0.33 (0.19-0.51)	0.54 (0.39-0.68)	0.54 (0.44-0.64)	0.61 (0.35-1.08)	0.61 (0.36-1.04)	1.00 (0.72-1.40)
PLR	Country	Western	1.45 (1.16-1.80)	2.20 (1.79-2.70)	1.75 (1.37-2.23)	0.66 (0.49-0.89)	0.83 (0.60-1.15)	1.26 (0.91-1.73)
		Eastern	1.61 (1.24-2.08)	1.66 (1.30-2.13)	1.67 (1.37-2.03)	0.97 (0.68-1.39)	0.96 (0.70-1.33)	0.99 (0.72-1.36)
	Mean age (years)	≥ 60.0	1.49 (1.13-1.96)	1.76 (1.44-2.16)	1.67 (1.34-2.09)	0.85 (0.60-1.19)	0.89 (0.63-1.27)	1.05 (0.78-1.42)
		< 60.0	1.50 (1.23-1.83)	2.26 (1.61-3.18)	1.86 (1.61-2.15)	0.66 (0.45-0.98)	0.81 (0.63-1.03)	1.22 (0.84-1.02)
	Percentage male (%)	≥ 60.0	1.23 (1.06-1.43)	-	1.67 (1.44-1.93)	-	0.74 (0.60-0.91)	-
		< 60.0	1.68 (1.29-2.19)	2.02 (1.64-2.50)	1.81 (1.45-2.28)	0.83 (0.59-1.17)	0.93 (0.66-1.31)	1.12 (0.82-1.52)
	Follow-up duration	≤ 30.0 days	1.47 (1.22-1.79)	1.79 (1.47-2.19)	1.74 (1.45-2.09)	0.82 (0.62-1.08)	0.84 (0.65-1.10)	1.03 (0.78-1.35)
		> 30.0 days	-	2.34 (1.63-3.35)	-	-	-	-
	Study	High	1.61 (1.20-2.17)	1.84 (1.45-2.34)	1.97 (1.32-2.95)	0.88 (0.60-1.28)	0.82 (0.50-1.35)	0.93 (0.58-1.49)

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	quality	Moderate	1.44 (1.15-1.79)	2.04 (1.53-2.72)	1.70 (1.46-1.97)	0.71 (0.49-1.01)	0.85 (0.65-1.11)	1.20 (0.87-1.66)
NLR	Country	Western	0.10 (0.05-0.18)	0.05 (0.02-0.14)	0.25 (0.13-0.49)	2.00 (0.62-6.41)	0.40 (0.16-1.01)	0.20 (0.06-0.65)
		Eastern	0.30 (0.18-0.48)	0.14 (0.06-0.37)	0.63 (0.51-0.77)	2.14 (0.76-6.02)	0.48 (0.28-0.81)	0.22 (0.09-0.56)
	Mean age (years)	≥ 60.0	0.08 (0.03-0.20)	0.06 (0.02-0.14)	0.34 (0.19-0.60)	1.33 (0.34-5.19)	0.24 (0.08-0.71)	0.18 (0.06-0.55)
		< 60.0	0.19 (0.11-0.35)	0.15 (0.04-0.51)	0.50 (0.31-0.81)	1.27 (0.31-5.13)	0.38 (0.18-0.81)	0.30 (0.08-1.17)
	Percentage male (%)	≥ 60.0	0.11 (0.05-0.24)	-	0.48 (0.27-0.87)	-	0.23 (0.09-0.61)	-
		< 60.0	0.17 (0.10-0.28)	0.07 (0.03-0.17)	0.36 (0.22-0.60)	2.43 (0.89-6.66)	0.47 (0.23-0.97)	0.19 (0.07-0.53)
	Follow-up duration	≤ 30.0 days	0.12 (0.07-0.22)	0.07 (0.02-0.18)	0.42 (0.27-0.65)	1.71 (0.50-5.92)	0.29 (0.14-0.59)	0.17 (0.05-0.54)
		> 30.0 days	-	0.12 (0.04-0.35)	-	-	-	-
	Study quality	High	0.13 (0.05-0.33)	0.06 (0.02-0.20)	0.38 (0.20-0.71)	2.17 (0.49-9.60)	0.34 (0.11-1.07)	0.16 (0.04-0.59)
		Moderate	0.12 (0.06-0.24)	0.10 (0.04-0.26)	0.40 (0.24-0.69)	1.20 (0.37-3.85)	0.30 (0.13-0.72)	0.25 (0.09-0.73)
DOR	Country	Western	12.68 (7.19-22.38)	30.41 (10.41-88.82)	6.32 (3.35-11.92)	0.42 (0.12-1.40)	2.01 (0.86-4.70)	4.81 (1.38-16.72)
		Eastern	4.56 (3.67-5.65)	10.05 (5.30-19.06)	2.59 (1.83-3.67)	0.45 (0.23-0.89)	1.76 (1.17-2.65)	3.88 (1.87-8.04)
	Mean age (years)	≥ 60.0	12.86 (6.11-27.07)	23.01 (10.84-48.85)	4.55 (2.50-8.28)	0.56 (0.19-1.61)	2.83 (1.09-7.35)	5.06 (1.93-13.23)
		< 60.0	6.24 (4.46-8.75)	11.52 (3.44-38.55)	3.41 (2.25-5.16)	0.54 (0.15-1.90)	1.83 (1.07-3.12)	3.38 (0.94-12.12)
	Percentage	≥ 60.0	8.06 (4.08-15.91)	12.32 (1.74-87.48)	3.08 (1.76-5.37)	0.65 (0.08-5.20)	2.62 (1.09-6.31)	4.00 (0.52-30.66)

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	male (%)	< 60.0	9.16 (5.69-14.75)	19.66 (9.31-41.51)	4.51 (2.82-7.22)	0.47 (0.19-1.13)	2.03 (1.04-3.97)	4.36 (1.80-10.54)
	Follow-up duration	≤ 30.0 days	9.45 (6.19-14.41)	19.21 (8.02-45.99)	3.70 (2.46-5.55)	0.49 (0.19-1.30)	2.55 (1.42-4.59)	5.19 (1.98-13.61)
		> 30.0 days	8.42 (5.14-13.80)	15.90 (8.10-31.21)	5.53 (2.34-13.03)	0.53 (0.23-1.22)	1.52 (0.57-4.10)	2.88 (0.96-8.57)
	Study quality	High	7.96 (4.90-12.91)	21.21 (7.10-63.34)	5.17 (2.51-10.64)	0.38 (0.11-1.24)	1.54 (0.65-3.67)	4.10 (1.11-15.22)
		Moderate	9.44 (5.17-17.23)	16.25 (6.88-38.41)	3.33 (2.22-5.01)	0.58 (0.20-1.66)	2.83 (1.37-5.86)	4.88 (1.88-12.63)
AUC	Country	Western	0.85 (0.82-0.88)	0.82 (0.78-0.85)	0.74 (0.70-0.78)	1.04 (0.98-1.10)	1.15 (1.08-1.23)	1.11 (1.03-1.19)
		Eastern	0.73 (0.69-0.77)	0.74 (0.70-0.78)	0.66 (0.62-0.70)	0.99 (0.91-1.07)	1.11 (1.02-1.20)	1.12 (1.03-1.22)
	Mean age (years)	≥ 60.0	0.85 (0.82-0.88)	0.83 (0.80-0.86)	0.70 (0.66-0.74)	1.02 (0.97-1.08)	1.21 (1.14-1.30)	1.19 (1.11-1.27)
		< 60.0	0.73 (0.69-0.77)	0.79 (0.75-0.82)	0.70 (0.65-0.73)	0.92 (0.86-0.99)	1.04 (0.96-1.13)	1.13 (1.05-1.21)
	Percentage male (%)	≥ 60.0	0.74 (0.70-0.77)	-	0.65 (0.60-0.69)	-	1.14 (1.05-1.24)	-
		< 60.0	0.82 (0.79-0.85)	0.81 (0.78-0.85)	0.72 (0.68-0.76)	1.01 (0.96-1.07)	1.14 (1.07-1.22)	1.13 (1.05-1.21)
	Follow-up duration	≤ 30.0 days	0.81 (0.77-0.84)	0.79 (0.75-0.82)	0.69 (0.65-0.73)	1.03 (0.96-1.09)	1.17 (1.09-1.26)	1.14 (1.06-1.23)
		> 30.0 days	-	0.83 (0.79-0.86)	-	-	-	-
	Study quality	High	0.79 (0.75-0.83)	0.80 (0.76-0.83)	0.74 (0.70-0.78)	0.99 (0.92-1.06)	1.07 (0.99-1.15)	1.08 (1.01-1.16)
		Moderate	0.80 (0.76-0.83)	0.81 (0.77-0.84)	0.67 (0.62-0.71)	0.99 (0.93-1.05)	1.19 (1.10-1.29)	1.21 (1.05-1.14)

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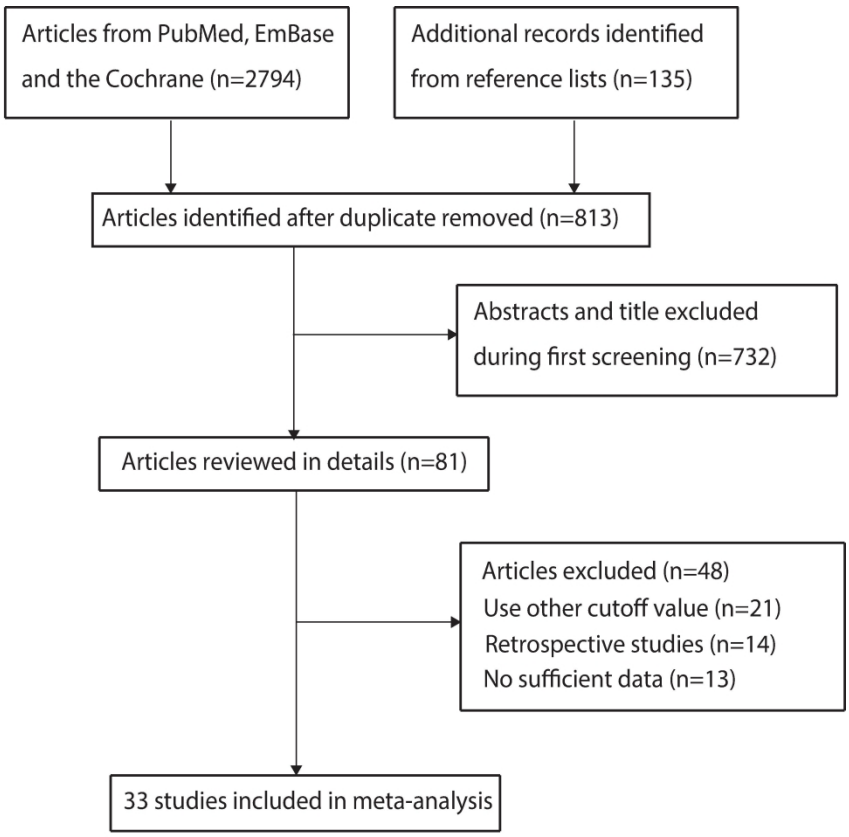


Figure 1. Flow diagram of the literature search and study selection process.

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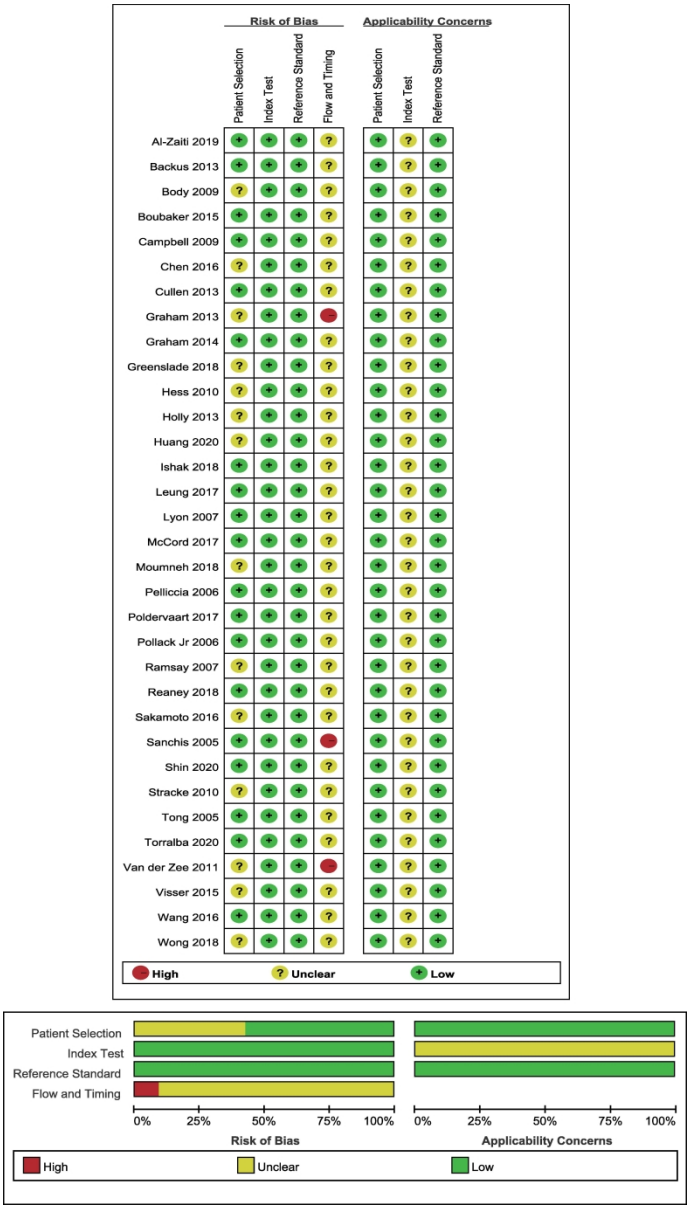


Figure 2. QUADAS-2 scoring of included studies.

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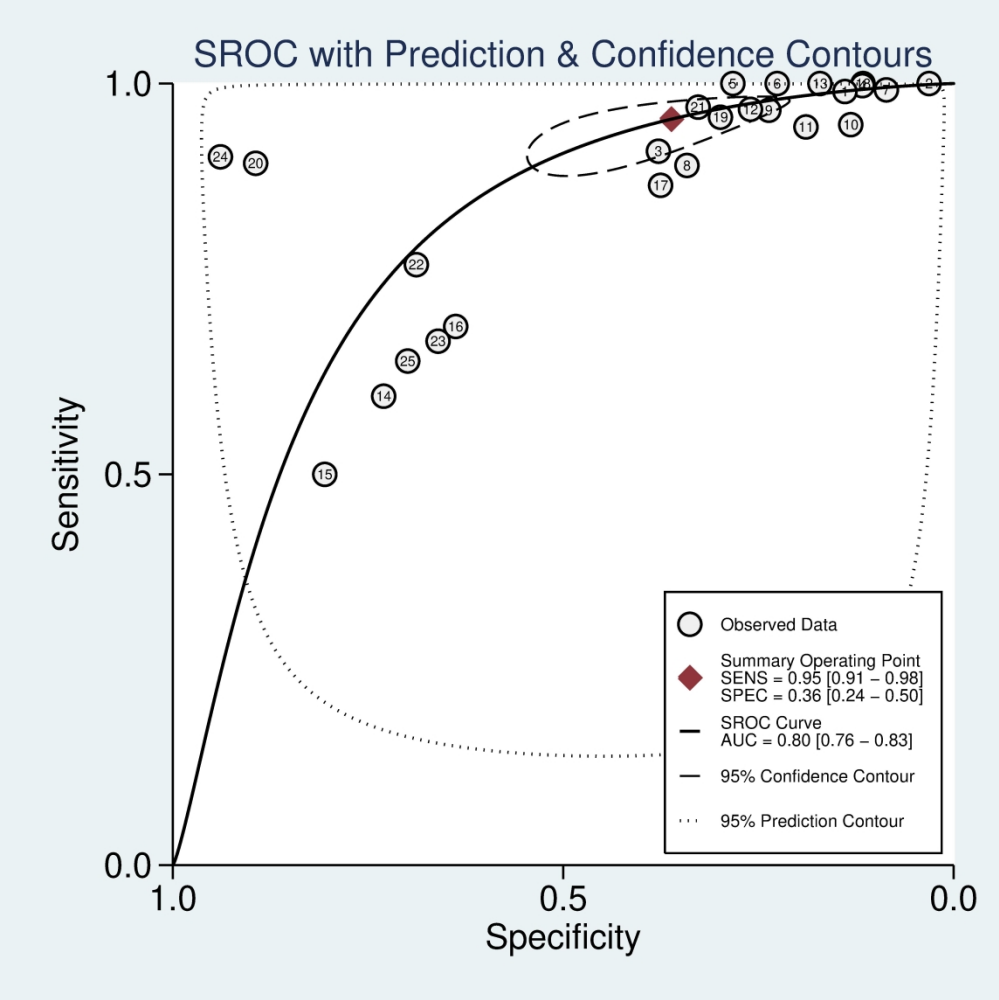


Figure 3. The summary receiver operating characteristic curve of risk stratification assessed by the TIMI score.

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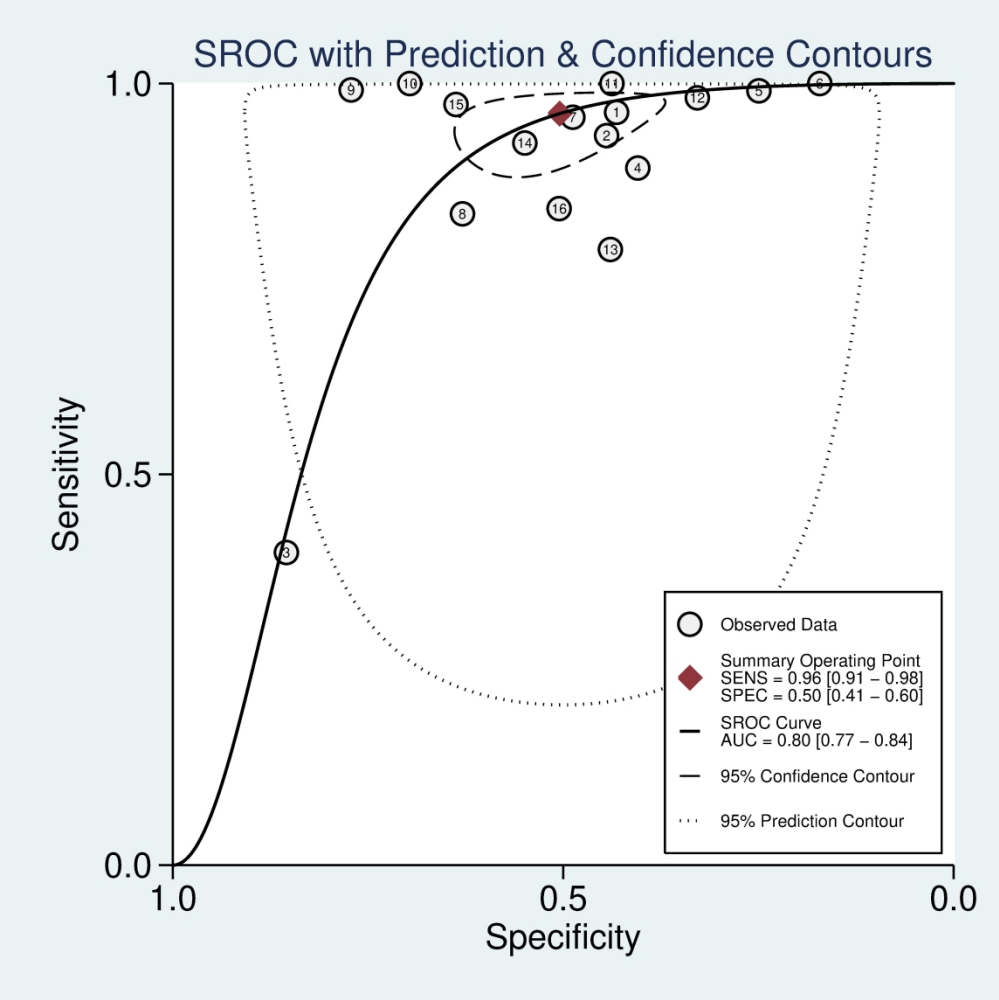


Figure 4. The summary receiver operating characteristic curve of risk stratification assessed by the HEART score.

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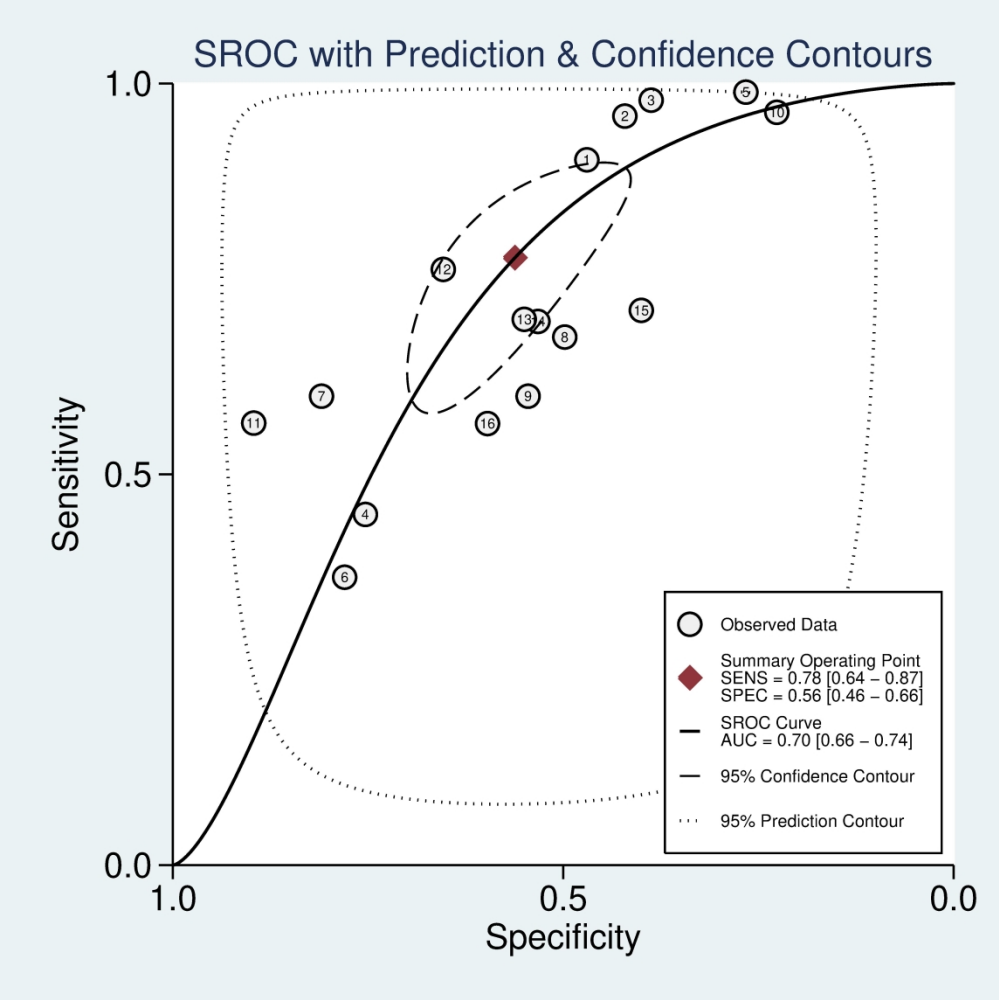


Figure 5. The summary receiver operating characteristic curve of risk stratification assessed by the GRACE score.

203x203mm (600 x 600 DPI)

1. Search strategies in PubMed and Cochrane library:

#1 ("Chest Pain"[Mesh] OR "Chest Pain"[Title/Abstract])

#2 ("Emergency Service, Hospital/classification"[Mesh] OR "Emergency Service, Hospital/organization and administration"[Mesh] OR "emergency department"[Title/Abstract])

#3 ("TIMI"[Title/Abstract] OR "HEART"[Title/Abstract] OR "GRACE"[Title/Abstract])

#4 registr*[Title/Abstract] OR Registries[Mesh] OR cohort*[Title/Abstract] OR Follow-Up Stud*[Title/Abstract] OR Follow-Up*[Title/Abstract] OR Longitudinal Stud*[Title/Abstract] OR Prospective*[Title/Abstract] OR Prospective Stud*[Title/Abstract] OR Concurrent Stud*[Title/Abstract] OR Incidence Stud*[Title/Abstract] OR "Cohort Studies"[Mesh]

#5 AND/#1-#4

Search strategies in EmBase:

#1 Chest Pain.mp.

#2 Chest Pain.tw.

#3 #1 OR #2

#4 TIMI.mp.

#5 TIMI.tw.

#6 HEART.mp.

#7 HEART.tw.

#8 GRACE.mp.

#9 GRACE.tw.

#10 #4-#9/OR

#11 Cohort studies/exp

#12 Follow-up studies/exp

#13 Longitudinal Studies/exp

#14 Prospective Studies/exp

#15 Registry/exp

#16 #11-#15/OR

#17 #3 AND #10 AND #16

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2. Sensitivity and Specificity

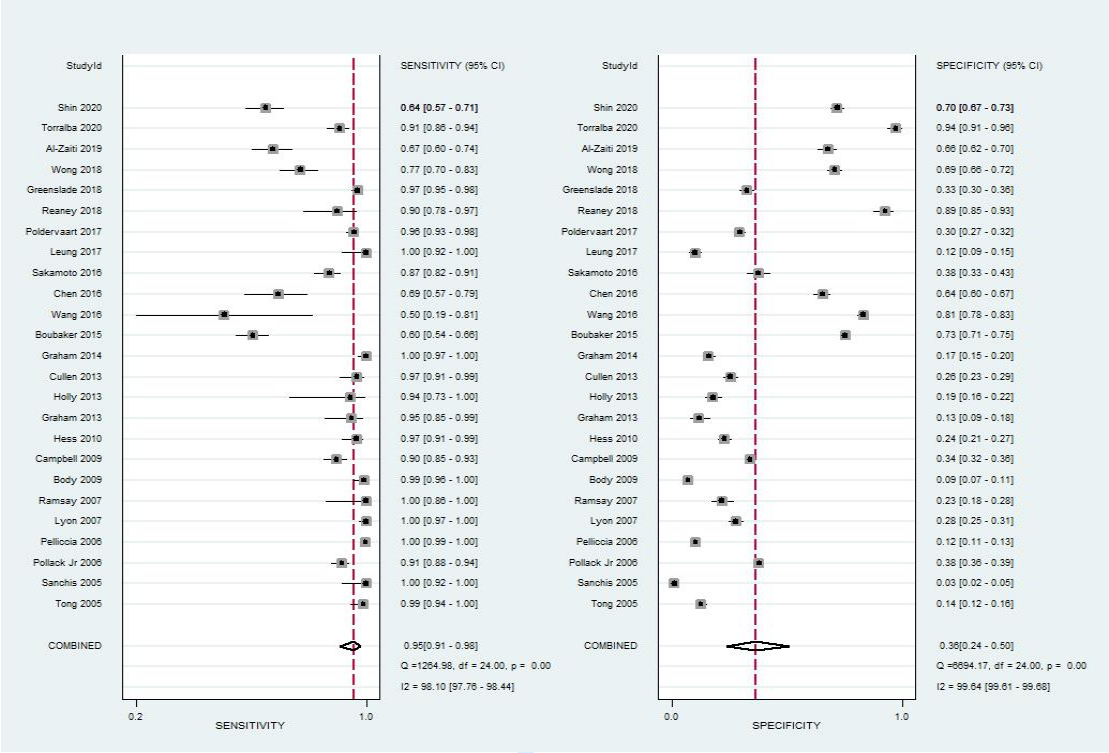


Figure S1. Sensitivity and specificity for TIMI

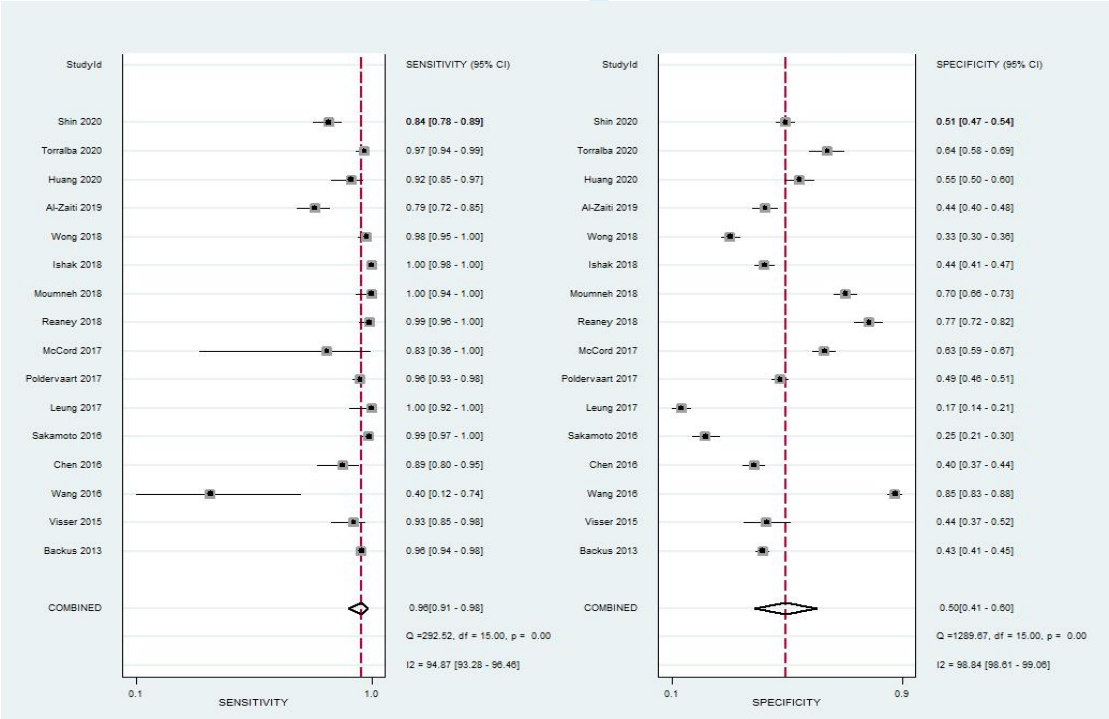


Figure S2. Sensitivity and specificity for HEART

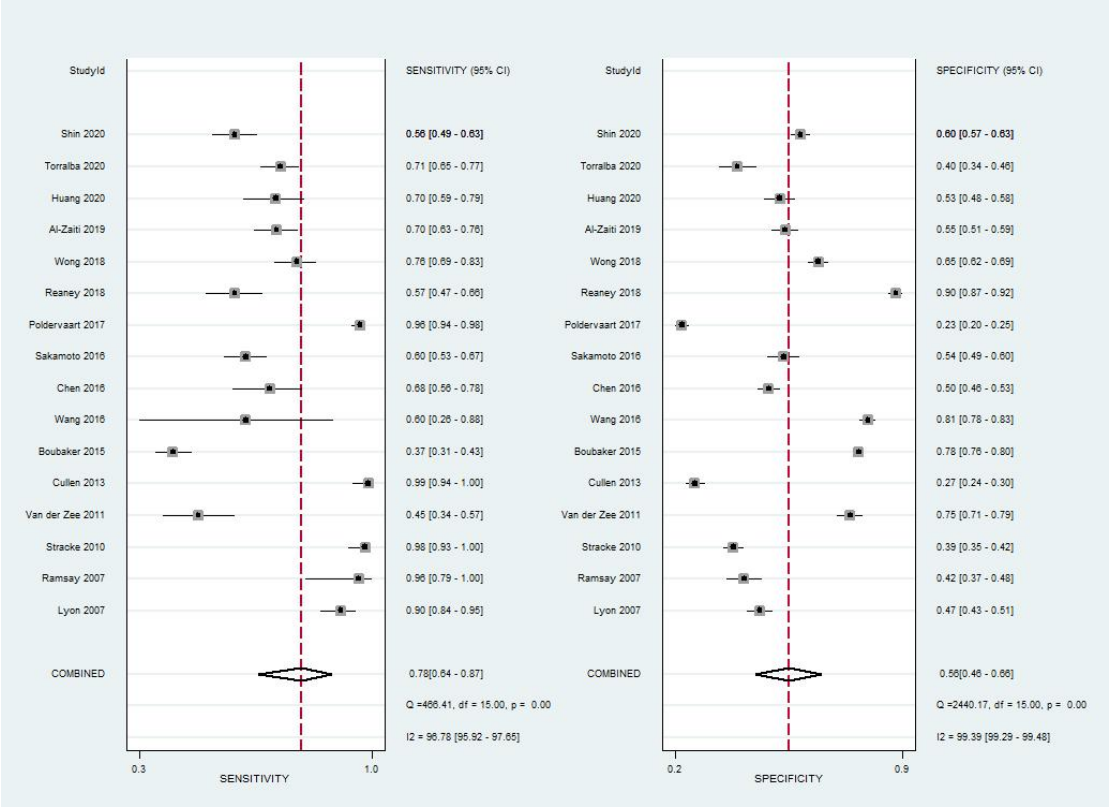


Figure S3. Sensitivity and specificity for GRACE

3. PLR and NLR

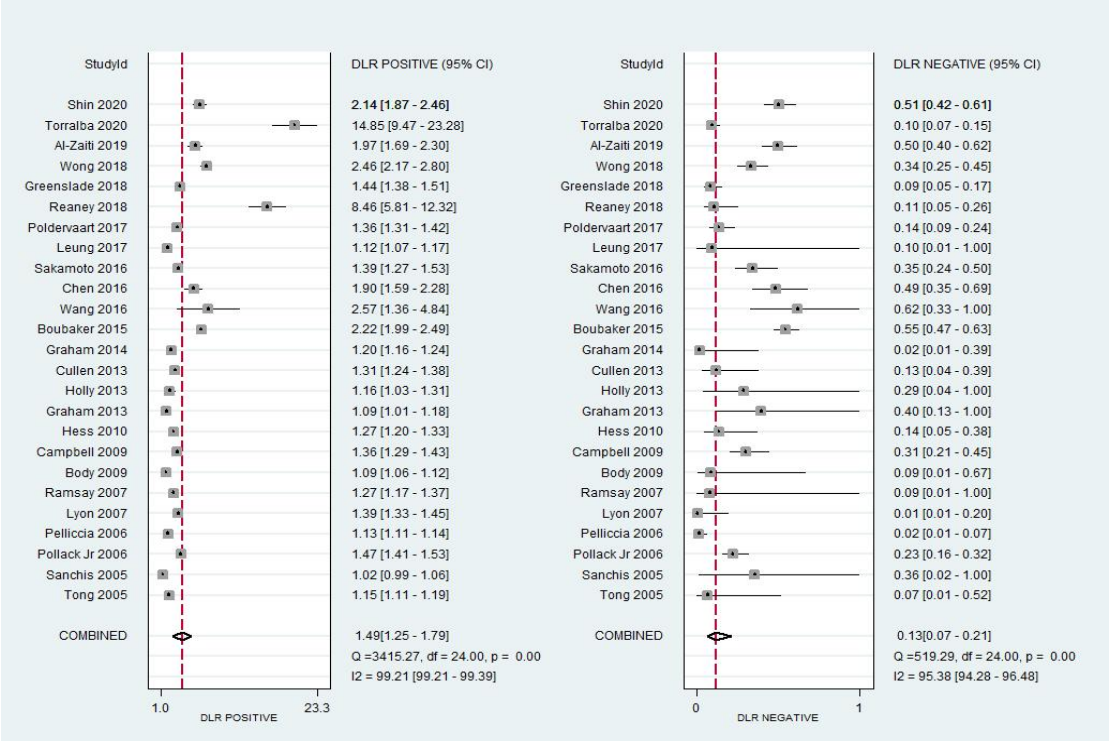


Figure S1. PLR and NLR for TIMI

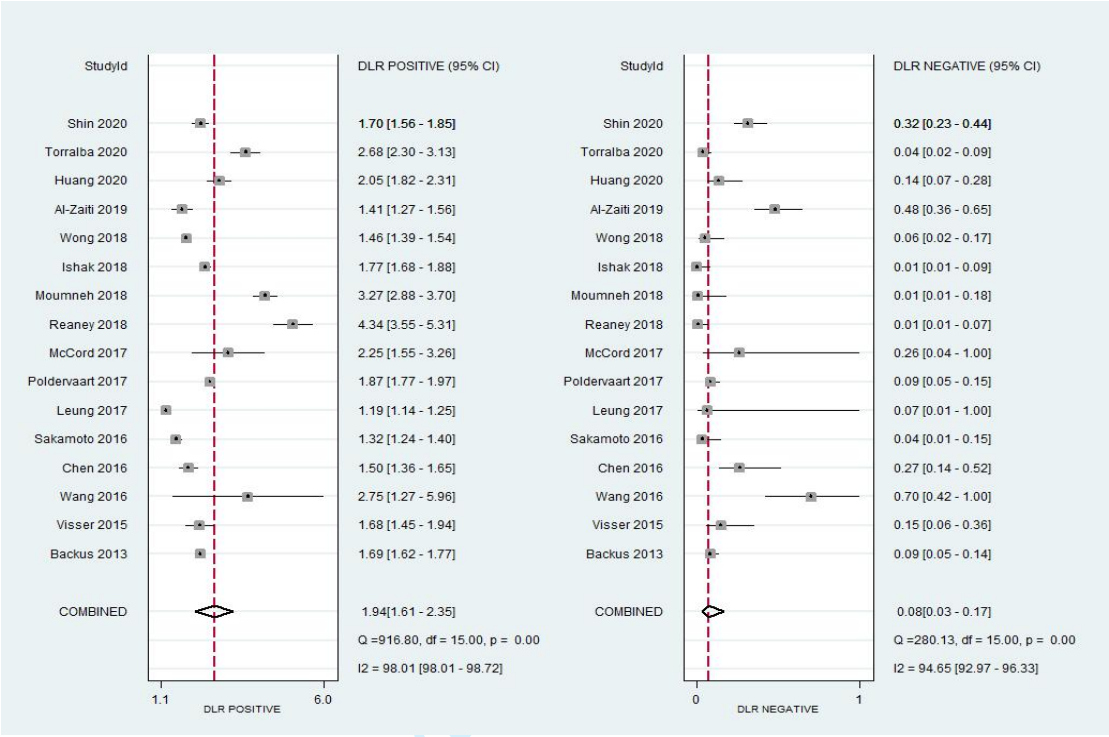


Figure S2. PLR and NLR for HEART

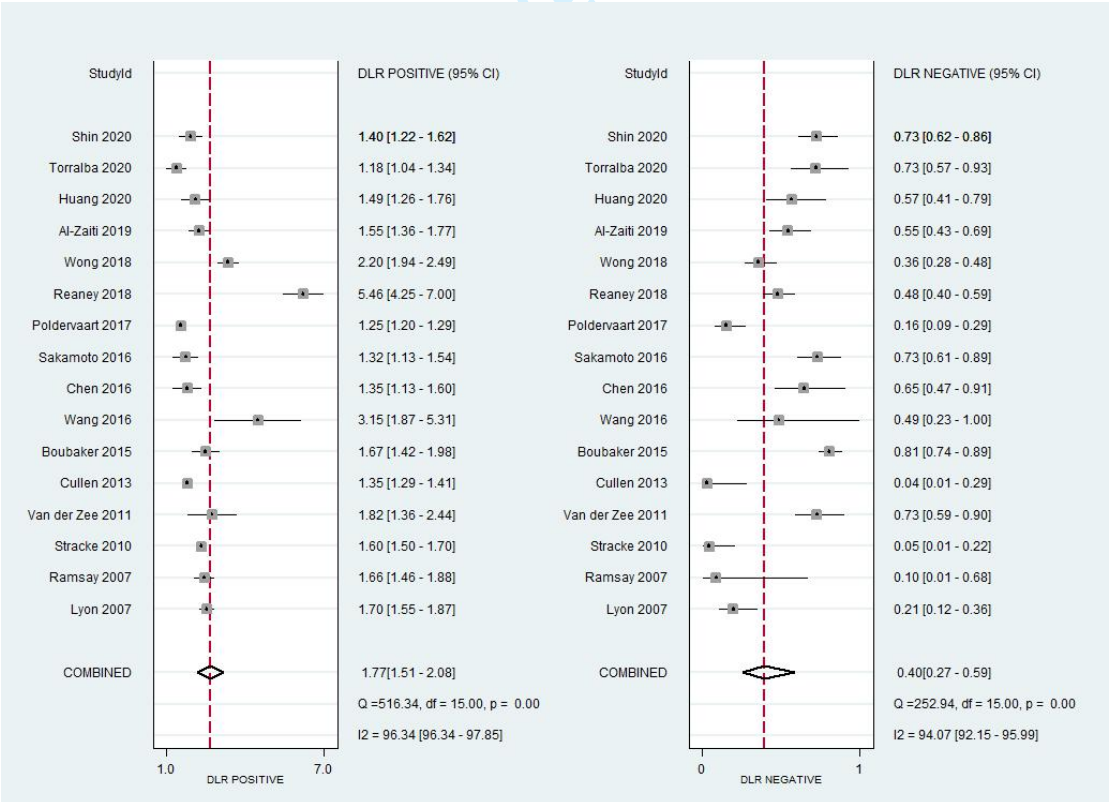


Figure S3. PLR and NLR for GRACE

4. DOR

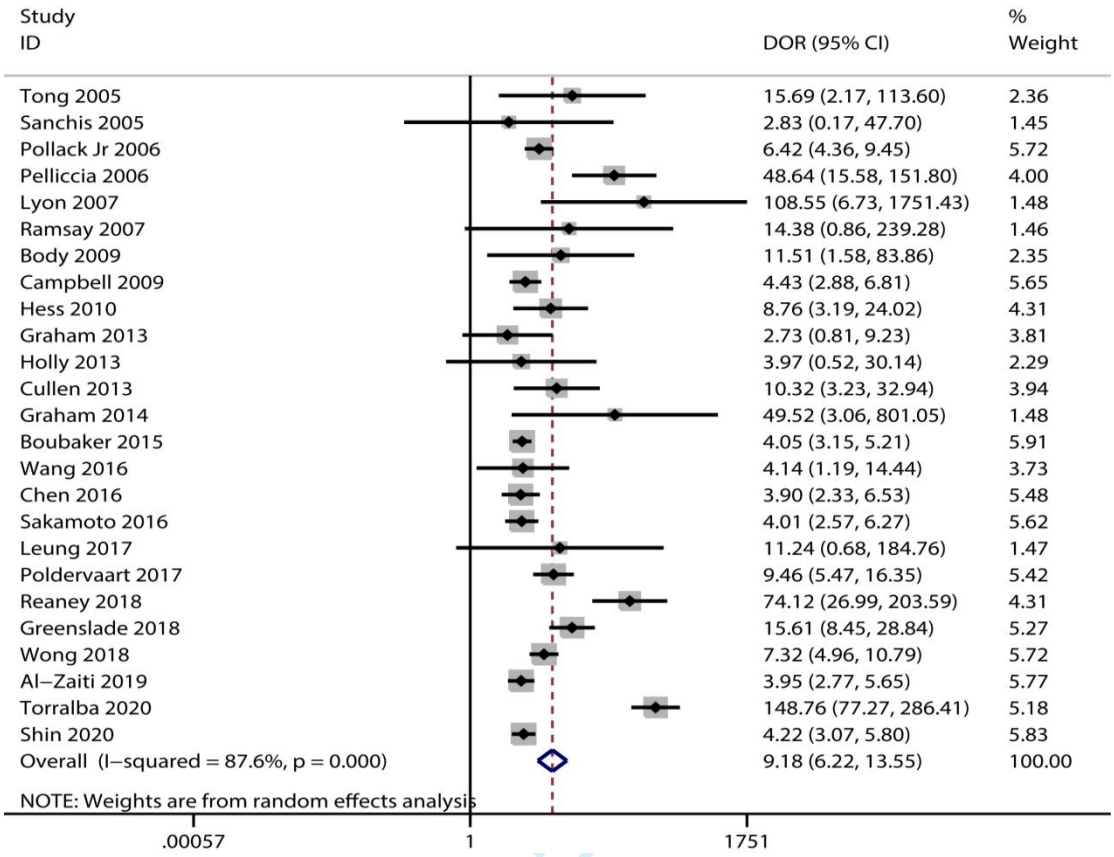


Figure S1. The summary DOR for TIMI

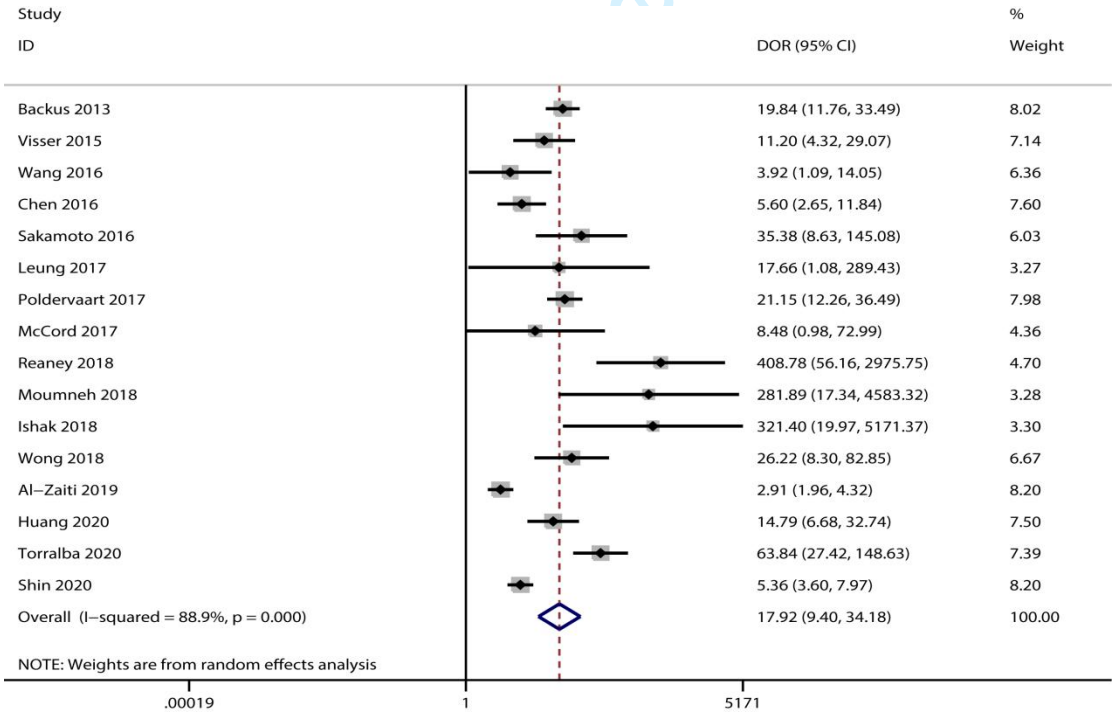


Figure S2. The summary DOR for HEART

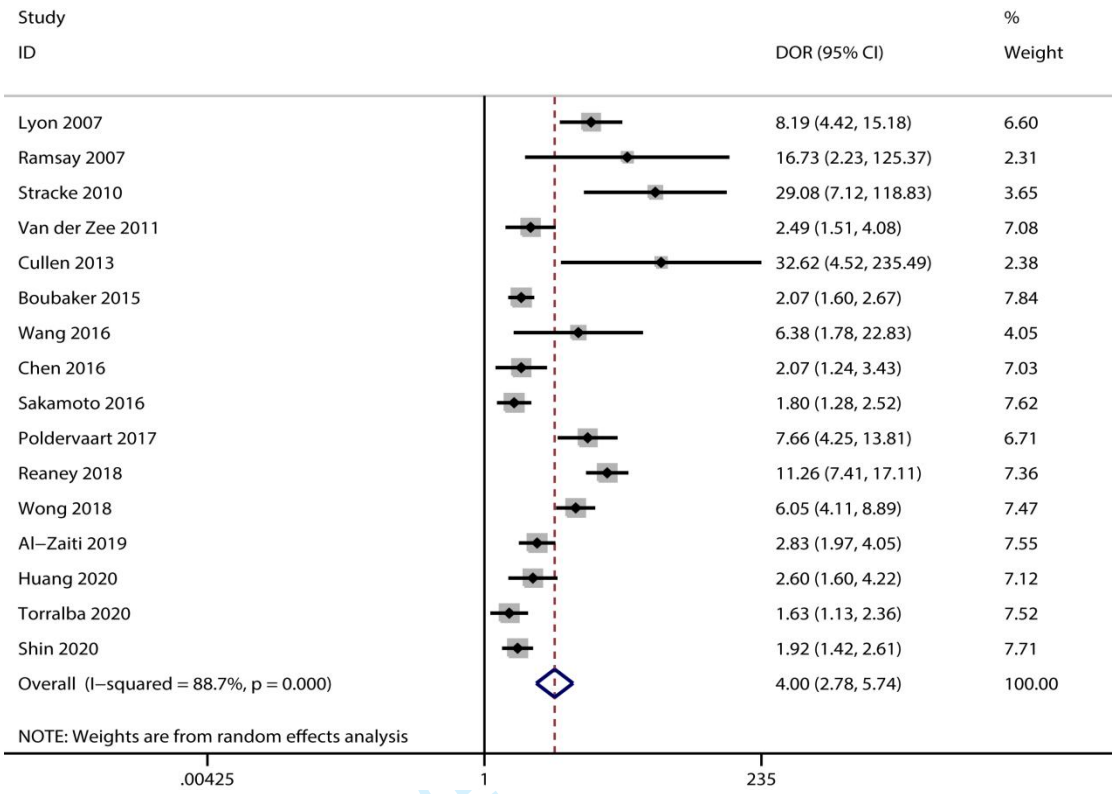


Figure S3. The summary DOR for GRACE

5. Publication bias

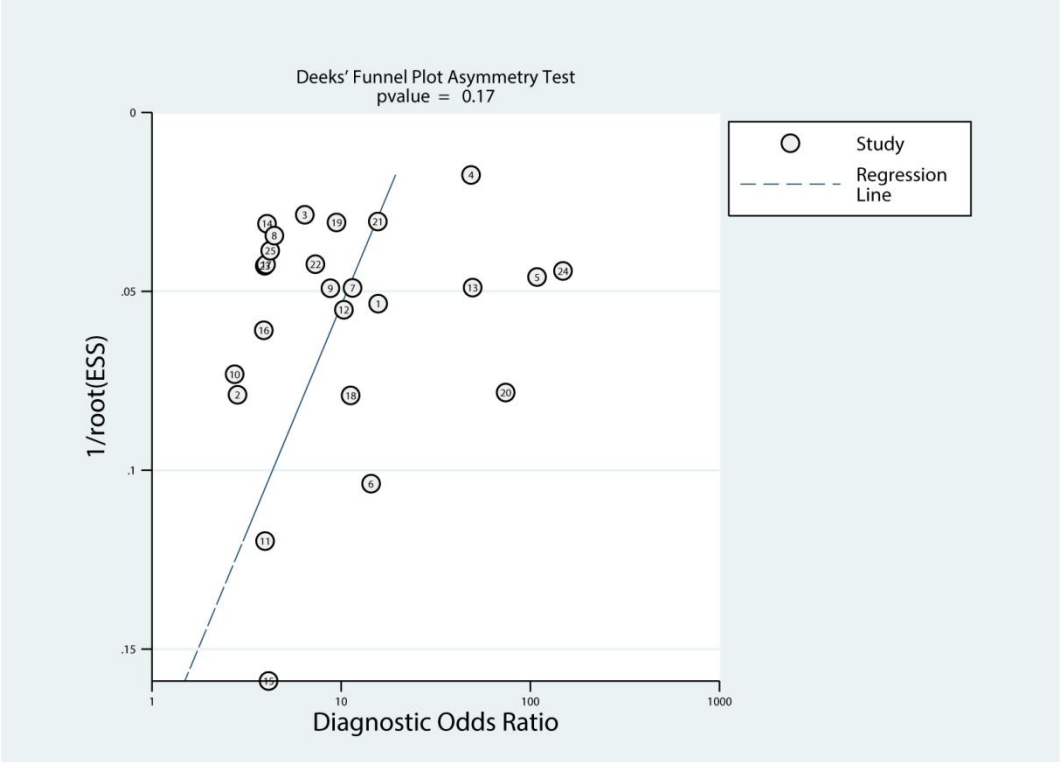


Figure S1. Publication bias for TIMI

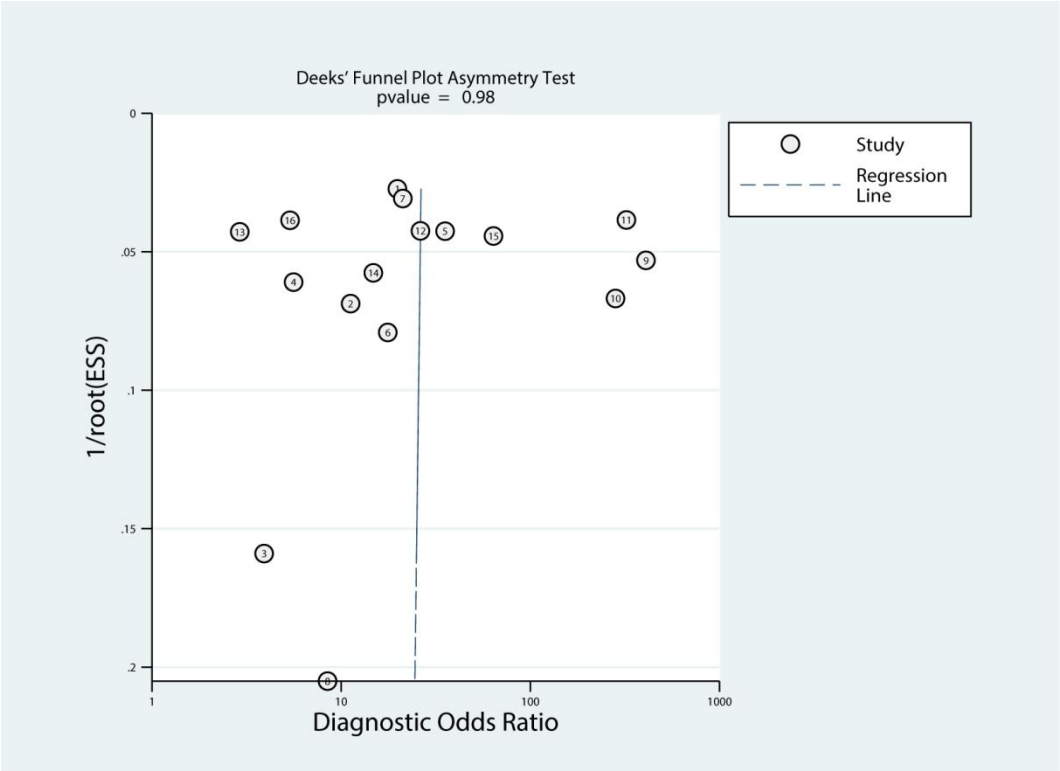


Figure S2. Publication bias for HEART

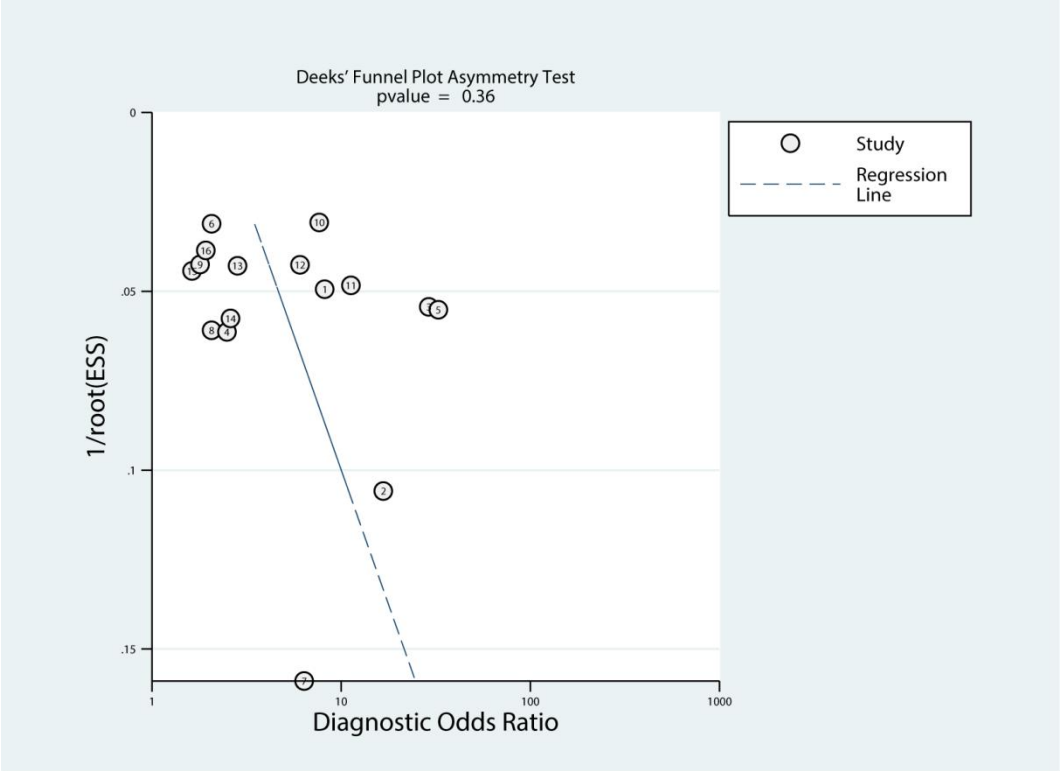


Figure S3. Publication bias for GRACE

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N/A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	9-10
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10-13
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	13
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	14

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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